Service

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Warning

Joule for Service

Joule, SAP's generative AI copilot, offers dedicated use cases for Service. To find out how exactly Joule can support you when working in Service in SAP S/4HANA Cloud Private Edition, see the Joule Capabilities Guide for Service.

Related Information

Joule in SAP S/4HANA Cloud Private Edition

Cross Functions

Cross functions include generic tools, components, and features that are available in more than one area or throughout all of Service, such as business transactions, actions, pricing, or data archiving.

Worklist

The Worklist is a central entry point for processing alerts, workflow tasks, and business objects. It includes inboxes for alerts, workflows, and business objects, each with various functions and navigation options.

Business Transaction

A business transaction provides business structures and functions that can be used in the various processes of a company.

Communication Management Software

Integration with Communication Management Software enables use of CMS functions in business roles in Service and allows you to achieve simplified interactions, faster response times, and improved end-to-end processes.

Partner Processing

Date Management

Availability Check

During the availability check (ATP check), the system checks whether a product contained in a service order can be confirmed as available (sufficient stock is available or can be produced or purchased on time). The required quantity of the product is reserved, and the ATP requirements are transferred to production or procurement.

Product Substitution, Listing, and Exclusion in Service Transactions

Text Management

Attachments

Attachments in service transactions may include documents, images, or other files, such as inspection reports, photographs, or product manuals, which provide additional context or detailed instructions throughout the service delivery process. The attachments are important for record-keeping, ensuring the accuracy of data, and enhancing the overall service process by providing comprehensive information.

Advanced Variant Configuration

Output Management

Actions

Survey Tool

Pricing

Billing

Get an overview of how Service and Sales Billing work together to facilitate the billing of service transactions.

Multilevel Categorization

Rule Modeler

Rule-Based Dispatching of Service Transactions

External References for Service Transactions

You can use references to synchronize your service transactions with external systems.

Plant and Storage Location

Learn about plant and storage location in service transactions.

Worklist

The Worklist is a central entry point for processing alerts, workflow tasks, and business objects. It includes inboxes for alerts, workflows, and business objects, each with various functions and navigation options.

Use

The Worklist is used to personalize alert and workflow delivery, execute follow-up activities, and process business objects.

Prerequisites

- You have configured alert management (ALM) in Customizing for **Service**, under **Basic Functions** > **Alert Management**.

  For more information, search for *Alert Management (ALM)* in the relevant version of SAP NetWeaver on SAP Help Portal.

- You have configured SAP Business Workflow in Customizing for **Service**.

- You have defined the appropriate profiles for the substitute function of the workflow inbox, in Customizing for **SAP NetWeaver** at **Application Server** > **Business Management** > **SAP Business Workflow** > **Basic Settings** > **Substitute Profile**.

- You have configured the alert inbox, the workflow inbox, and the business object inbox of the worklist, in Customizing for **Service** at **Basic Functions** > **Worklist**.

Features

Alert Inbox

- Displaying the alerts in a list

  The alerts are automatically displayed in a list.

  You can use the different columns of the list to sort the entries so that you see an overview of the priority and due date of the various alerts.

- Personalizing the alert delivery

  You can specify how the alert is displayed, whether a substitute can read the alert, and how/when the alert should be delivered. In addition to this, you can cancel the default subscription to alert categories.

  > **Note**
  > The substitute will only receive alerts generated after they have been designated as a substitute, not alerts delivered prior to this.

- Executing follow-up activities

  You can execute follow-up activities from the alert list or from the details page.

- Reserving, completing, rejecting, and forwarding alerts

  On the details page, you can reserve, complete, reject, or forward an alert:

  - When you reserve an alert, it is removed from the worklists of other recipients and you can process it at a later time, if appropriate.

   - When you reject an alert, it is removed from your worklist and you can no longer process it.
   - When you have finished processing an alert, you must manually set it to **Completed**.
   - When you forward an alert, it appears in the worklist of the recipient, for whom it is simultaneously reserved. This alert is therefore removed from your worklist and from the worklists of other users.

- More Information

  There is other information available on the details page, such as a list of the recipients and details about the possible escalation of the alert.

Workflow Inbox

- Displaying the workflow tasks in a list

  The workflow tasks are displayed in a list.

  You can sort the entries in the different columns of the list. For example, you can display all overdue workflow tasks together, or you can see an overview of the priority and due date of the various workflow tasks.

- Search criteria are available for the workflow tasks.

  > **i Note**
  >
  > If you search using the search criterion **Due Date**, the search result list is automatically sorted by due date, and this can no longer be changed.

- Using the new field **Maximum Number of Results**, you can limit the number of workflow tasks displayed.

  > **i Note**
  >
  > In the background, the system loads all data records from the database, but only displays a subset of these on the WebClient UI.

- Using the field **Save Search As**, you can save your search.
- On the page **Home**, in the screen area **Workflow Tasks**, there is a new link called **More than 5 workflow tasks found**, which you can use to access the **Workflow Tasks** assignment block for the **Worklist**.
- Executing workflow steps simultaneously for multiple workflow tasks

  In the result list, you can simultaneously execute workflow steps such as **Confirm** or **Reject**, for multiple workflow tasks that require a decision.

- Processing decision tasks

  For workflow tasks that require a decision from you, the system presents decision options that influence the rest of the business process.

- Personalizing the delivery of workflow tasks
   - You can define substitutes, who process your workflow tasks in your absence.
   - In the case of an unplanned absence you can activate yourself as the substitute for an absent user, if the user has specified you as such but has not activated the substitution. When the substitution is no longer required, you can deactivate your assignment as substitute.

- Executing, reserving, confirming, and forwarding workflow tasks

  On the details page, you can execute, reserve, confirm, and forward a workflow task:

    - When you execute a workflow task, it is deleted from the worklists of other recipients.
    - If a workflow task was sent to multiple processors, you can also reserve it. The task is then deleted from the worklists of the other recipients and you can process it at a later time, if appropriate.
    - If a workflow task requires confirmation, you must manually confirm the execution of the task on the details page.
    - When you forward a workflow task, it appears in the worklist of the recipient, for whom it is simultaneously reserved. This workflow task is therefore removed from your worklist and from the worklists of other users.

- More Information

  There is other information available on the details page, such as a list of the associated objects, attachments, and recipients.

- Adding attachments

  You can add attachments to a workflow task.

  You can also enter notes as attachments. The notes are stored as plain text files, and are not deleted during workflow processing.

- Integrating other SAP systems

  In the workflow inbox of the worklist in service solution, you can integrate workflow tasks from other SAP systems.

  For more information, see Integration of Other SAP Systems.

i Note

Notifications, about a rejected lead for example, go to SAP Business Workplace. You can set up the system so that it automatically forwards notifications to an e-mail address. For more information, see Setting Up Automatic Forwarding.

Business Object Inbox

The business object inbox is a central entry point for calling the business objects that you need to process. The inbox offers a comprehensive search function for business objects, and displays a corresponding result list. You can access the business objects from the result list.

You can use the search criterion **Assigned To** for example, to search for all business objects to which you are assigned as employee responsible. In addition, you can use the criterion **Category** to search for business objects in a certain category, such as sales orders.

i Note

In the standard system, the business object inbox is set up exclusively for the processing of **business transactions**. In the Customizing for the worklist however, you can configure the inbox so that it supports other types of business objects, such as cases, fax messages, or e-mails.

Integration of Other SAP Systems

Use

In the workflow inbox of the worklist in Service, you can integrate other SAP systems.

At runtime, service solution checks whether there are workflow tasks in integrated SAP systems, for the user who is currently logged on to service solution.

You can define which of the SAP systems that are integrated in the workflow inbox of the worklist are relevant for each user. This means that each user sees only the relevant workflow tasks, in their workflow inbox in the worklist.

In addition, the settings for workflow substitutes that the worklist users make using the workflow task personalization, are transferred to the integrated systems at runtime. This makes the substitute function in the worklist in service solution available for workflow tasks that come from integrated SAP systems.

Prerequisites

You have made the necessary settings in Customizing for **Service**, at **Basic Functions** > **Worklist** > **Define Alert Inbox and Workflow Inbox**:

- You have defined the SAP systems that you want to integrate in the workflow inbox. You do this in the Customizing activity **Integrate Other SAP Systems In Workflow Inbox**.

  You must also meet all the prerequisites listed in the documentation for this Customizing activity.

- At user level, you can define which of the external SAP systems that are integrated in the workflow inbox of the worklist are relevant for each user at runtime. You can use the Business Add-In (BAdI) **Integrate Other SAP Systems In Workflow Inbox** to do this.

- When defining the navigation options for object types in the system in service solution, you can also define the navigation options for object types from other logical SAP systems that are integrated in the workflow inbox.

  You do this in the Customizing activity **Define Object-Specific Navigation**.

- When defining the navigation options for workflow tasks in the system in service solution, you can also define the navigation options for workflow tasks from other logical SAP systems that are integrated in the workflow inbox.

  You do this in the Customizing activity **Define Task-Specific Navigation**.

Features

If you use the worklist as an entry point for the processing of workflows, you have the following options for workflow tasks that come from other SAP systems:

- You can view details about workflow tasks in the workflow inbox of the worklist in service solution.
- You can view details about workflow tasks in the original system.
- You can trigger all workflow task-specific actions, such as the execution or reservation of a workflow task, in the workflow inbox of the worklist in service solution.
- You can trigger workflow task-specific actions in the original system.
- You can navigate to associated business objects in the original system.

Setting Up Automatic Forwarding

Context

Notifications do not arrive in the workflow inbox of the worklist, but in SAP Business Workplace instead. You can set up the system so that it automatically forwards notifications to a specific e-mail address.

Procedure

1. In the user maintenance (transaction `SU01`), check whether an e-mail address is defined for the specified user.
2. Define the automatic forwarding to e-mail addresses.

   You can set up the automatic forwarding for particular users, or across the whole system:

   - For specific users only:

     Call transaction `SO36` and define the automatic forwarding for the relevant user.

   - System-wide:

     a. Call transaction `SO16`.

     b. Select **Send to Home Addresses of Users** on the **Mail sy.grp** (Mail System Group) tab page.
3. Check whether e-mails can be sent using SAPconnect (transaction `SCOT`).

Business Transaction

A business transaction provides business structures and functions that can be used in the various processes of a company.

Definition

Business object that depicts the following possibilities:

- A business interaction with a business partner at a given time
- The result of an interaction, for example, a contract
- The initiation of or request for an interaction, for example, an opportunity
- The collection of information in preparation for a possible business interaction, for example, an activity
- The object of a (potential) interaction, for example, a service transaction

Each business transaction is stored as a separate database record.

Use

In Service, some examples of business transactions are service orders, service requests, service confirmations.

Structure

Business transactions are structured as follows:

- Header

  The header of a business transaction contains information that is relevant for the entire transaction, for example, the business partner, status of the transaction, and certain dates.
- Items (not for all business objects)

  Items contain the details of individual products to which the business transaction relates, and information such as the item status. Items can have subitems.

You can combine items from different business contexts in one transaction.

Business Transaction Customizing

Definition

This section provides information about the model of business transactions in service solution, which is the basis for business transaction Customizing. You define business transactions in Customizing for **Service** under **Transactions** > **Basic Settings**. For information about where the individual elements are defined, see “Overview of Transaction Customizing” below.

Concept

Transaction Types

Transaction types define the characteristics and attributes of a business transaction (for example, whether it is a service order or a service request), and determine how the transaction is processed (for example, profiles and procedures used to determine texts, business partners, statuses, and organizational data).

Assignment of Business Transaction Categories

You assign a transaction type to one or more business transaction categories. Only specific combinations of business transaction categories are allowed. In addition to the general settings for a transaction type, you need to make Customizing settings specifically for each business transaction category assigned to a transaction type.

In the definition of the transaction type, you define one business transaction category as the leading business transaction category. This indicates that the transaction category has the greatest relevance for the transaction type (and not that the category is related hierarchically to the categories).

Example

A service order with business activity data, for example, has **Service** as a leading transaction category, but also has the transaction category **Business Activity** assigned. The transaction is considered a service transaction rather than as an activity for the purpose of processing on the user interface.

The leading transaction category of a transaction type determines which Customizing options are available for this transaction type, such as which item categories can belong to the transaction type or which further transaction categories can be assigned. The assigned transaction categories define which additional functionality is available for this transaction type (for example, whether it can be used in service processes).

The assigned transaction category can be the same category as the leading transaction category, if additional Customizing is required for the leading transaction category.

Assignment of Item Categories

You specify the default item category for a transaction type, and you can also define alternative item categories that are allowed for a transaction type.

Item Categories

Item categories define the characteristics and attributes of a transaction item (for example, whether it is a sales item or service material item) and determine how the item is processed (for example, profiles and procedures used to determine texts, business partners, statuses, and organizational data).

Assignment of Business Transaction Categories

An item category is assigned to one or more business transaction categories (see “Business Transaction Categories” below). Only specific combinations of business transaction categories are allowed. In addition to the general settings for each item category, you need to make Customizing settings specifically for each business transaction category assigned to an item category.

Assignment of Item Object Type

Each item category is assigned an item object type, which defines the context in which the item can be used (for example service processes). This context is defined by the transaction categories that are assigned to the item.

Business Transaction Categories

Business transaction categories are defined by SAP and correspond to business object types in the business object repository, for example, Service Process (`BUS2000116`), Business Activity (`BUS2000126`).

A business transaction category determines the business context in which a business transaction or item can be used. It determines the following:

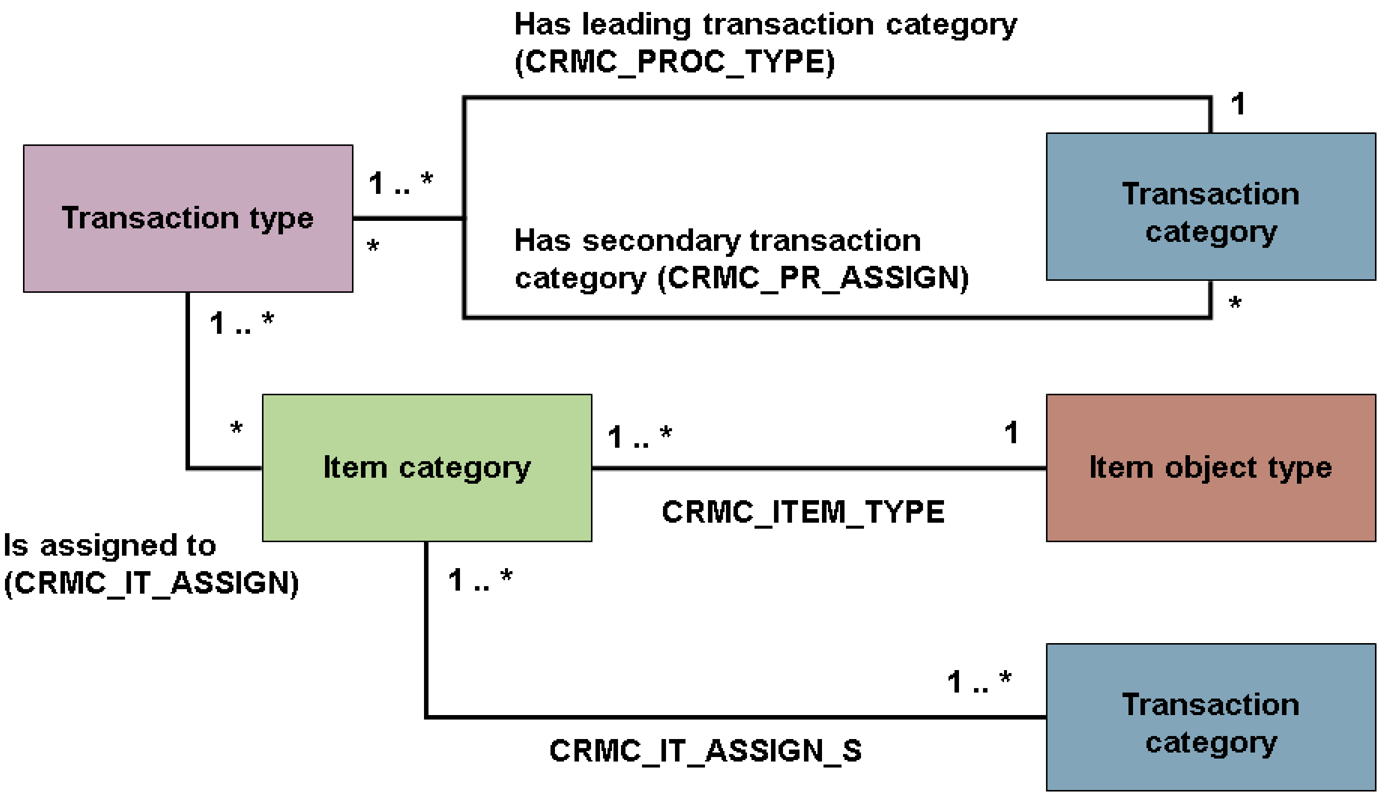
- The maximum allowed structure of a certain type of transaction
- The objects of which the transaction category consists
- Secondary transaction categories with which the transaction category can be combined
- The item object types of which the transaction category consists, and hence the objects of which the item object type consists

Overview of Transaction Customizing

The following table shows the Customizing activities and corresponding tables in which the elements of business transactions are defined. All activities are in Customizing for Service under Transactions > Basic Settings.

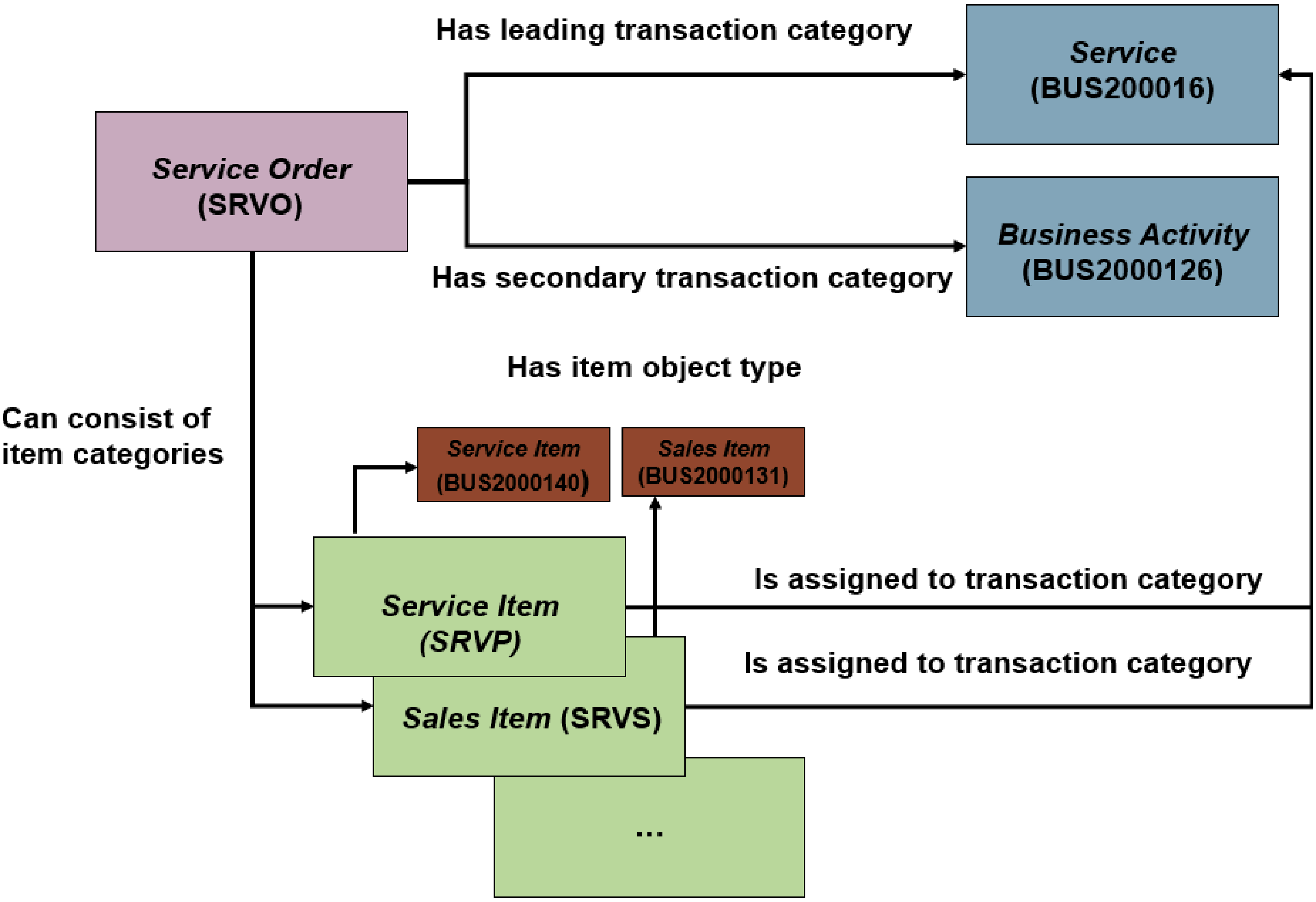
Element	Customizing	Table	Note
Transaction type	Define Transaction Types (view Definition of Transaction Types)	`CRMC_PROC_TYPE`	The leading transaction category is also specified in this table.
Secondary transaction categories for transaction type	Define Transaction Types (view Assignment of Business Transaction Categories)	`CRMC_PR_ASSIGN`	The allowed transaction categories are predefined by SAP.
Item category	Define Item Categories	`CRMC_ITEM_TYPE`	The item object type is also defined in this table.
Item categories belonging to transaction type	Define Item Category Determination	`CRMC_IT_ASSIGN`	Not applicable
Transaction categories for item categories	Define Item Categories (view Assignment of Business Transaction Categories)	`CRMC_IT_ASSIGN_S`	The allowed transaction categories are predefined by SAP.

The elements of business transaction Customizing described above is shown in the following figure:



Example

The following figure shows how the **Service Order** transaction type is modeled, and is followed by an explanation:



The transaction type Service Order (SRVO) has the leading transaction category Service (BUS2000116) and one secondary transaction category, namely Business Activity (BUS2000126). It can consist of several item categories such as the Service Item (SRVP) with item object type Service Item (BUS2000140) or the Sales Item (SRVS) with item object type Sales Item (BUS2000131).

Business Transaction Processing

Use

Business transactions in Service are based on a generic, modular framework with a uniform processing concept.

Integration

Basic functions within Service are available for business transaction processing, depending on the Customizing settings that have been made. These basic functions include functions such as availability check, pricing, partner determination, and actions.

Prerequisites

You have defined the basic elements of your business transactions in Customizing for Service under Transactions > Basic Settings.

Features

- Product entry

You can enter products (for example, in service orders and quotations) by product ID. You can additionally enable product entry by product description in Customizing for **Service** under **Transactions** > **Basic Settings** > **Define Transaction Types**.

- Change history

  All changes that have been made to the business transaction at header and item level since its creation can be displayed in the **Change History** assignment block. The information displayed here includes the type of change, the old and new value, the change date and time, and the user who made the change.

  The system displays changes in individual fields on the user interface (for example, changes to organizational data), and changes to a business transaction or to an item (for example, the item was canceled or released).

  If you want to use the change history, you need to specify in Customizing that change documents should be created. In transaction type Customizing, ensure that the **No Change Documents** checkbox is not selected.

- Follow-up transactions

  When you create a new transaction that relates to an existing transaction, you can create the new transaction as a follow-up transaction, thereby creating a relationship between the two transactions that provides additional contextual information to business users in the **Transaction History** assignment block.

  For more information, see Follow-Up Transactions.

- Copying of business transactions

  To save time creating a new transaction that is not a follow-up to an existing transaction, you can copy an existing transaction that contains data relevant to your new transaction. No link is created between the existing and new business transactions.

  You can prevent the copying of business transactions by using a Business Add-In in Customizing for **Service** under **Transactions** > **Basic Settings** > **Copying Control for Business Transactions** > **BAdI: Enhancements for Copying Control** (method `COPY`).

- Authorization checks

  You can control permitted access to business transactions by defining authorization checks on various levels (for example, partner function and business transaction type). For more information, see Authorization Check in Business Transactions.

- Incompleteness check

  You can define the data required for a transaction to be complete. For more information, see Incompleteness Check in Business Transactions.

- Status management

  The status of a business transaction is reflected in a system status and user status. For more information, see Status Management in Business Transactions.

- Modifications to the transaction data structure (enhancements)

  You can add fields to business transactions (see Business Transaction Enhancements).

- Modifications to default processing of transactions

  - There are various Business Add-Ins available in Customizing for **Service** under **Transactions** > **Basic Settings** in the context-specific sections.

  - You can flexibly implement additional processing functions for business transactions by adding callbacks (function modules) to the technical definition of business transaction processing. You can do this in Customizing for**Service** under **Transactions** > **Basic Settings** > **Edit Event Handler Table**.

Note

You require an understanding of the processing logic to be able to register callbacks. For more information, see the documentation for the Customizing activity.

More Information

Specifics for Interaction Center: Business Transactions

Business Transaction Model

This section describes those elements of the business transaction in Service that are predefined by SAP and cannot be modified by customers. This information is useful to SAP developers and development consultants who want to get a better understanding of the technical model of the business transaction.

Business Transaction Categories

Concept

Business transaction categories are defined by SAP and correspond to business object types in the business object repository, for example, **Service Process** (`BUS2000116`), **Business Activity** (`BUS2000126`).

A business transaction category determines the business context in which a business transaction or item can be used. It determines the following:

- The maximum allowed structure of a certain type of transaction (the actual structure is defined by the transaction type)
- The subobjects of which the transaction category consists

  For more information about subobjects, see Business Transaction Subobjects.
- Secondary transaction categories with which the transaction category can be combined (the actual combination is defined in the transaction type)
- The item object types of which the transaction category consists, and hence the subobjects of which the item object type consists

A transaction type is assigned to one or more business transaction categories. Only specific combinations of business transaction categories are allowed. In addition to the general settings for a transaction type, you need to make Customizing settings specifically for each business transaction category assigned to a transaction type.

In the definition of the transaction type, you define one business transaction category as the leading business transaction category. This indicates that the transaction category has the greatest relevance for the transaction type (and not that the category is related hierarchically to the categories).

Example

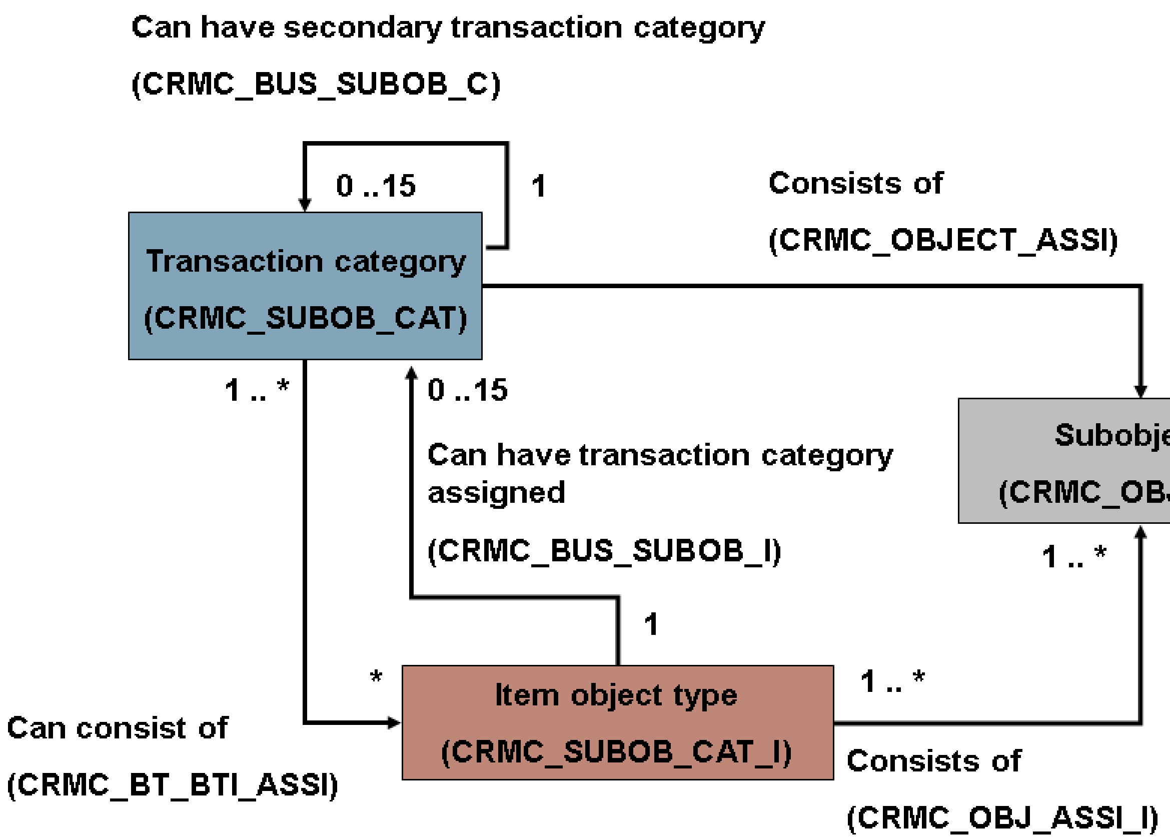
A service order with business activity data, for example, has **Service** as a leading transaction category, but also has the transaction category **Business Activity** assigned. The transaction is considered a service transaction rather than as an activity for the purpose of processing on the user interface.

The leading transaction category of a transaction type determines which Customizing options are available for this transaction type, such as which item categories can belong to the transaction type or which secondary transaction categories can be assigned. The assigned transaction categories define which additional functionality is available for this transaction type (for example, whether it can be used in service processes).

The assigned transaction category can be the same category as the leading transaction category, if additional Customizing is required for the leading transaction category.

Technical Model

The figure below shows the technical modelling of transaction categories and is followed by an explanation:



The following table explains the above figure by showing the individual definitions and assignments that make up the definition of a transaction category:

Definition/Assignment	System Table in Which Definition/Assignment is Stored
Transaction category	CRMC_SUBOB_CAT
Allowed item object types for a transaction category	CRMC_BT_BTI_ASSI
Subobject	CRMC_OBJECTS
Assignment of subobjects to a transaction category	CRMC_OBJECT_ASSI
Allowed combination of transaction categories	CRMC_BUS_SUBOB_C

Definition/Assignment	System Table in Which Definition/Assignment is Stored
Item object type	CRMC_SUBOB_CAT_I
Subobjects of which item object types consist	CRMC_OBJ_ASSI_I
Allowed transaction categories for item object type	CRMC_BUS_SUBOB_I

Example

The **Service** transaction category consists of several subobjects, including header, item, and status. It consists of several item object types, such as the service item. These item object types can, in turn, have several transaction categories assigned to them.

The **Service** transaction category can be combined with the **Activity** transaction category, which also consists of several subobjects.

Business Transaction Subobjects

Business transaction subobjects encapsulate functions, such as status handling, that can be reused by several transactions. These subobjects are defined by SAP in the table `CRMC_OBJECTS`. The object name (name of the subobject) is used by the services such as the link handler and the event handler (see Event Handler and Callback Registration), to identify the initiator.

The following types of subobject are available:

- Header and item

  Each transaction consists of at least one header. Several items can be linked to this header.

- Header and item extension

  Extensions store additional data belonging either to the header or item. Extensions are, therefore, either header extensions or item extensions. Extensions can have multiple entries.

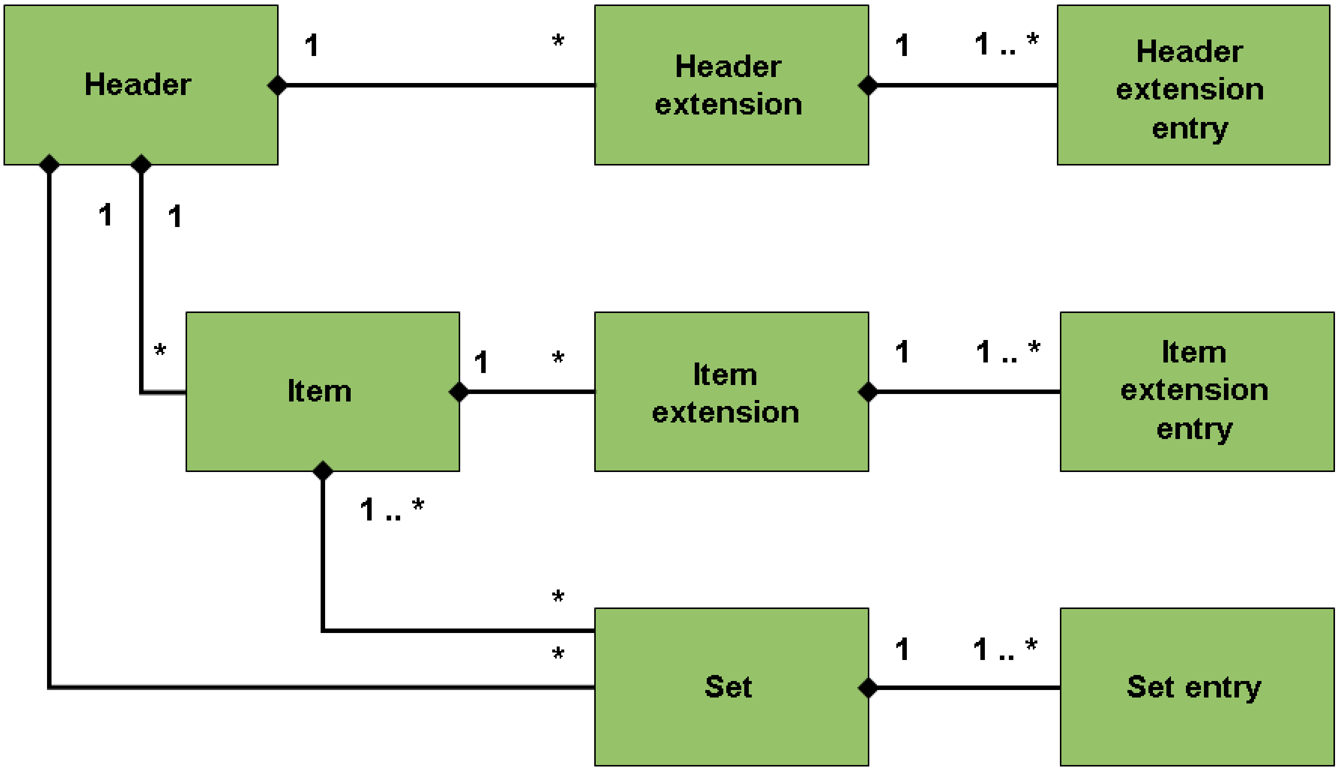
- Set

  Sets store additional data that can belong to a header and/or item. Each set can have multiple entries, assigned to the header or item by a link. A single entry can be assigned to the both the header and/or multiple items by a link. Typically, sets are assigned to both header and item, and the values are inherited from header to item.

  Example

  The partners in a business transaction are modelled as a set on header and item level. By default the partners are inherited to the items, however, you can also change the partner data on item level.

The relationship model between subobject types, as described above, is depicted in the following figure:



Event Handler and Callback Registration

Concept

Business transaction processing is controlled by the event handler, which enables communication between interrelated subobjects (see Business Transaction Subobjects). Subobjects are decoupled and, therefore, do not communicate directly with each other. Instead they use a publish/subscribe communication pattern. Each subobject publishes certain events, and callbacks (function modules) can be registered to react to these events.

When an event is triggered by a subobject, the event handler finds the callbacks that are registered for this event, and executes the corresponding function modules (callbacks) for event handling at the defined time and with the defined priority. The event handler is called when business transactions are edited or saved, but not when they are displayed.

Features

- Predefined events and callbacks

  SAP predefines the events that are published by subobjects, and the callbacks for these events. This is done in transaction `CRMV_EVENT`. The system table is `CRMC_EVENT_CALL`.

  For detailed information about the various parameters involved in callback registration, for example, the execution time, see the documentation linked to this transaction.

- Customer-defined events and callbacks

Customers can register additional callbacks in Customizing for Service, under Transactions > Basic Settings > Edit Event Handler Table. The Customizing table is CRMV_EVENT_CUST.

For more information, see the documentation for the above Customizing activity.

During processing, the event handler searches for entries in tables CRMC_EVENT_CALL and CRMV_EVENT_CUST. If entries exist in the tables, the entries in both tables are analyzed and sorted according to the priority and name of the function module.

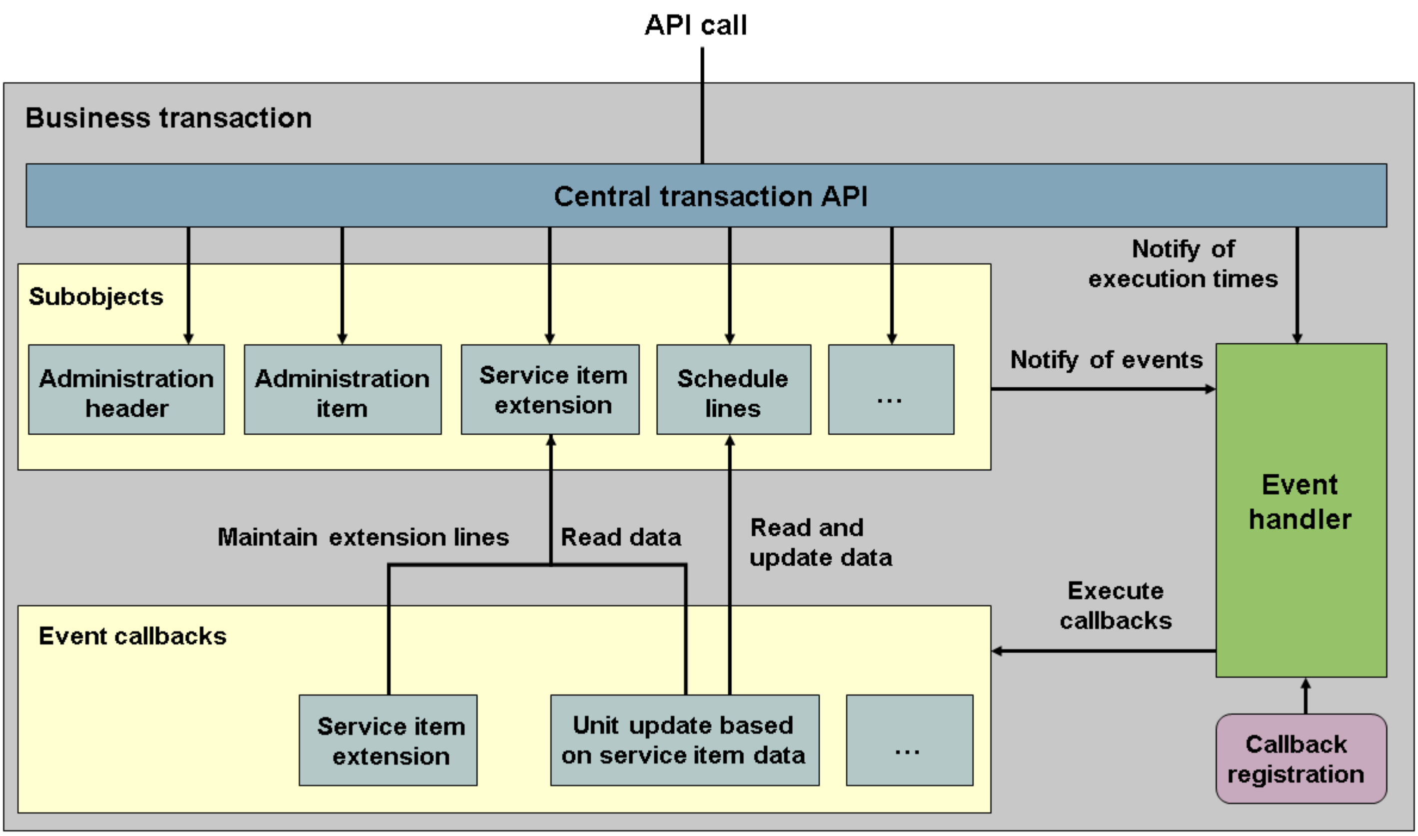
Activities

During the processing of API calls (such as CRM_ORDER_MAINTAIN), the subobjects are accessed (read), and requests (such as create or change) are processed. During the processing of a request, each subobject decides whether events should be triggered, and which events, for example, AFTER_CHANGE.

To trigger an event, the subobject notifies the event handler via function module CRM_EVENT_PUBLISH_OW. The event handler then determines all callbacks that are registered for this event. Depending on the execution time, the callbacks are executed immediately or planned in for a later execution time. The execution times are set by the API at certain times during processing, via the function module CRM_EVENT_SET_EXETIME_OW.

Within callbacks, other subobjects can be accessed and new requests (such as create or change) can be triggered. These requests, in turn, can trigger additional events. For these events, callbacks are also determined and need to be executed depending on the execution time.

The determination and execution of callbacks by the event handler described above is depicted in the following figure, using certain subobjects and callbacks as an example:



Example

A callback for the generation of service item extensions is registered for the administration item creation event. This callback creates a service item extension entry. This triggers the event AFTER_CREATE for the service item extension. For this event, the callback to update the unit in the schedule lines is registered. This updates the schedule lines, which also triggers an event for which other callbacks might be registered.

⚠ Caution

In order to avoid side effects and ensure good system performance, consider carefully the impact of any callbacks that you want to register.

- Keep in mind that any subobject processed by a callback might trigger a chain of additional events. Therefore, carefully check the effects of new entries, for example, by using the event trace transaction described (see the documentation for the transaction `CRMV_EVENT`).
- Consider whether it is absolutely necessary to set the execution time **Immediately**. The other execution times are better for performance since several events might trigger the same callback. If you need to react immediately to changes consider using the following callbacks **(both must used)**:
  - A callback that is registered for several events, which collects all changes and prepares the execution
  - A callback that is registered to execute the desired processing, which is registered less frequently

  This is already set up for pricing, for example. Collection is done using function module `CRM_PRIDOC_COM_DET_PRCTYPE_EC` and execution is done using function module `CRM_PRIDOC_UPDATE_EC`.
- Consider carefully which value you set for the parameter **Call Callback** and restrict the number of times the callback is planned in for a certain execution time to a minimum. For more information on the parameters, see the documentation for the transaction `CRMV_EVENT`.
- Register your callback for individual transaction categories instead of for the generic transaction category `BUS20001` so that your callbacks do not run unnecessarily for other transactions.
- If the subobject you are interested in transfers certain attributes and you are only interested in a certain value, register your callback only for these specific attribute changes, especially when working with partner, status, and appointments.

Business Transaction Enhancements

In addition to defining the structure of your business transactions in Customizing, you can modify business transaction structures using the enhancement options described in this section.

Business Add-Ins (BAdIs)

You implement BAdIs in Customizing for **Service** under **Transactions** > **Basic Settings** in the context-specific sections.

For performance information related to BAdIs, see Performance Considerations: Business Add-Ins for Business Transactions.

Extensibility Tool: Custom Fields and Logic App

You can use the **Custom Fields** and **Custom Logic** apps to extend your master data by creating custom fields and implementing them in your business applications. For more information, see Key User Extensibility

Performance Considerations: Business Add-Ins for Business Transactions

When enhancing business transactions, always keep in mind in which context and how often your enhancement is called. We recommend that you carry out performance measurements for these enhancements and ensure that performance is optimized, especially if you have transactions with a large number of items or a large number of transactions in your database.

Performance recommendations are available for the following BAdIs, in the documentation for the corresponding Customizing activity:

- BAdI: Authorization Check in Business Transaction
- BAdI: Field Selection
- BAdI: Enhancements for Copying Control

  This BAdI is available in Customizing for Service under Transactions > Basic Settings > Copying Control for Business Transactions.

- BAdIs `CRM_<OBJECT>_BADI` / `COM_<OBJECT>_BADI`

  These BAdIs are available in Customizing for Service under Transactions, in the context-specific sections.

  → Recommendation

  If you use one of these BAdIs, ensure that you keep system performance in mind.

  These BAdIs have check methods and merge methods that are called in the corresponding processing step during maintenance of the corresponding object. With the merge method the data set can be completed and the check is done on the completed data. Therefore the check is, in most cases, called after the merge.

Follow-Up Transactions

Use

When you create a new transaction that relates to an existing transaction, you can create the new transaction as a follow-up transaction. When you do this, data from the original transaction is copied to the new transaction and the transactions become linked. This linking provides you with information about the history of a particular business transaction. You can view this history in the Transaction History assignment block in each business transaction.

Example

You receive information that a sales customer has accepted a quotation. You search for the original quotation and create a service order as a follow-up transaction. From the service order, you create an activity as a follow-up transaction, with a task to call the customer. In this example, the service order is a preceding transaction of the activity, and the quotation is a preceding transaction of the service order.

Prerequisites

Follow-Up Transactions

You have set up copying control in Customizing for Service, under Transactions > Basic Settings > Copying Control for Business Transactions. Here you define the allowed combinations of source and target (preceding and follow-up) transactions.

You can only create a follow-up transaction if the preceding transaction does not contain errors. However, this does not apply if you are creating an activity or if you only copy individual items.

Subsequent Assignment of Transactions

You determine which transactions can be assigned to which by defining object reference profiles in Customizing for **Service**, under **Transactions** > **Basic Settings** > **Define Object Reference Profile**. You assign the profiles to the appropriate transaction types or item categories in Customizing.

Features

- Creation of follow-up transactions

  When you create a follow-up transaction, you can select the items that you want to copy from the original transaction. You can only copy subitems together with the higher-level item.

- Subsequent assignment of follow-up transaction

  You can link a transaction to another transaction after both transactions have been created. You do this in the **Transaction History** assignment block.

- Transaction history

  In addition to the transaction history at header level, you can also display the **Transaction History** assignment block at item level.

  In some cases, transaction links are not shown in the **Transaction History** assignment block.

Status Management in Business Transactions

Use

You can use the general SAP status management when processing business transactions.

You can document the current processing status of an object (for example, a service order), using status management. The current object state can consist of a combination of individual statuses. Any number of statuses can be set for an object.

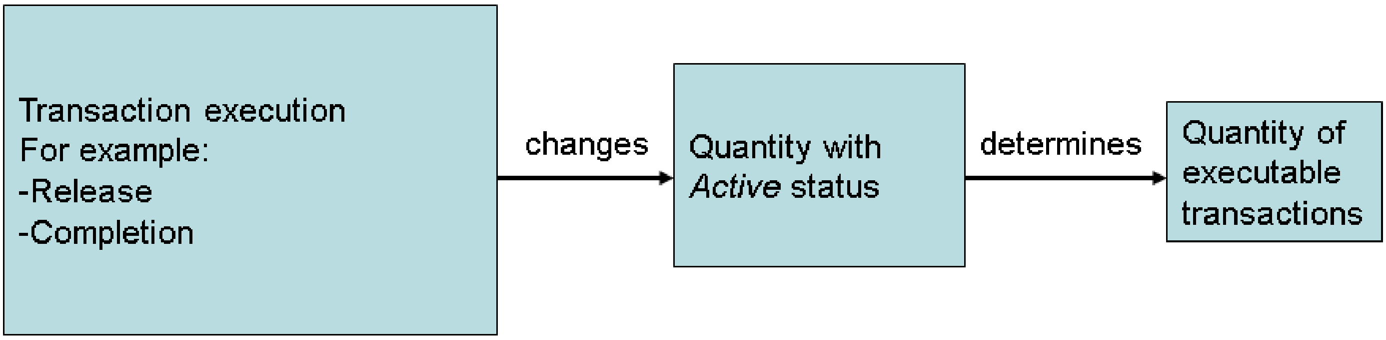
Example

The system statuses **Open** and **Contains Errors** and the user statuses **Quotation created** and **Do not print** can be active simultaneously in a quotation.

Each status fulfills two functions:

- It **informs** you that a specific status has been set (for example, the quotation was released).
- It **specifies** which step you can or cannot take next (for example, release is allowed).

If a business transaction is executed, this can set or delete one or several statuses for the relevant object. The switch between status and transaction can be portrayed as follows:



Statuses are displayed in the system in two different ways:

- As a 30 character text
- As a 4 character abbreviation (only in status profile)

Both methods of portrayal are language-dependent.

There are two types of status:

- **System status**: Status set by the system, which informs the user that the system has executed a specific business transaction on an object. You can only influence this status if you execute a business transaction that changes the system status.

Example

If you release a quotation, the system automatically sets the system status to **Released**.

- **User status**: Status that you set that you can create as additional information to the existing system status. You define a user status in a status profile that is created in Customizing for business transactions. You can define and activate as many user statuses as you wish. A status profile can then be assigned to the following in Customizing for business transactions:
    - Transaction types (for header status)
    - Item categories (for item status)

A status profile can be assigned to several transaction types and item categories.

System status and user status influence the business transactions in the same way.

Note

The header status is independent of the item status. One exception is the status **Completed**. If all items have the status **Completed**, the header status is also set to **Completed**.

Integration

A business transaction is not a static object, but it has its own life cycle, which starts when it is opened and ends when it is completed. During this time, other business transactions change this business transaction. It is, for example, forwarded for release or set to completed.

There are five system statuses that represent the life cycle of the business transaction:

- **Open**: Has been recently created and not yet processed

- In process: Administrator is clearing up questions
- Released: Transaction is complete and legal. Follow-up processes can be started (printing, billing, distribution, releases and so on)
- Completed: All actions directly linked to the transaction are complete
- Closed: Actions indirectly linked to the transaction are also complete

Prerequisites

If you want to display and maintain the user statuses, you need to have maintained a status profile and assigned it to the relevant transaction types and item categories. To do this, go to Customizing and choose Service > Transactions > Basic Settings > Status Management. You have made the assignments in the activities Define Transaction Types and Define Item Categories.

Features

The following functions are available to you at header level:

- Display of system and user status

  The system displays the user status that is assigned an ordinary status number, as well as the user status without a status number, on the Status assignment block. Only one user status with a number can be active at a time, while many user statuses without numbers can be active at a time.

  The system also displays the user status with a status number on the status dropdown list at header level. When you have not maintained a status profile, predefined system statuses appear in the status dropdown list.

  System statuses are shown in selected areas.

  Example

  The delivery status is shown on the Shipping assignment block.

The following functions are available to you at item level:

- System status / user status: the Status assignment block and status profile are also available at item level.
- Block / reason for rejection: you can reject an item or set a billing block. You must select a reason in both cases. You can also set a delivery block and credit block. The delivery block status for an item is set by the system when you maintain a delivery block reason.

The following functions are also available to you:

- You can lock or cancel all items at header level. You can set a billing block or cancellation for all items. This is a kind of “fast setting” of the flag at item level.

Authorization Check in Business Transactions

Use

Business transaction processing in Service is protected by an authorization check based on alternatives, so that only authorized users can create, change, display, or delete a business transaction.

Integration

The authorization check in the business transaction uses the functions available from Netweaver. This means that specific authorization fields and objects exist, and the user's corresponding authorizations are derived from the role assigned to him or her.

You can find general information about the authorization concept in the SAP Library under **SAP NetWeaver Library** > **SAP NetWeaver by Key Capability** > **Security** .

Process Flow of the Authorization Check in Business Transactions

Use

Business transaction processing in Service is protected by an authorization check based on alternatives, so that only authorized users can create, change, display, or delete a transaction.

The authorization check follows a specific sequence, that is, the check runs through several levels. This means that authorization can be granted at each level, and therefore no check is necessary in the subsequent levels. If for example, an employee is in a transaction as the employee responsible, he or she is allowed to process this transaction, regardless of whether further checks would lead to a positive or negative result.

The authorization concept in the business transaction in Service has the following characteristics:

- **Role-related protection**

  Users have access to existing business transactions, regardless of all other authorization checks, if they have been entered as a partner in these transactions. This can be an example of a user-defined partner function; the user does not need to have the **Employee Responsible** partner function.

  You can control the type and scope of the permitted activity, (for example creating or changing transactions), for each partner function and partner function category.

  See also: Partner Processing

- **Sales area/relationship in the organizational model**

  Users have access to new or existing business transactions in or below a certain level in the organizational model, regardless of all other authorization checks. If necessary, users can execute only specific activities.

- **Business transaction category**

  Users can create transactions only if they have authorization for the corresponding business transaction category (for example, activity - CRM_ACT).

- **Business transaction type**

  Users have access to activities only if they have authorization for the corresponding business transaction type. If necessary, users can execute only specific activities.

- **Sales area**

- **Payment card processing**

  Only authorized users are able to see the payment card number.

Prerequisites

Users for whom an authorization check is to be executed must be assigned in the organizational model.

→ Recommendation

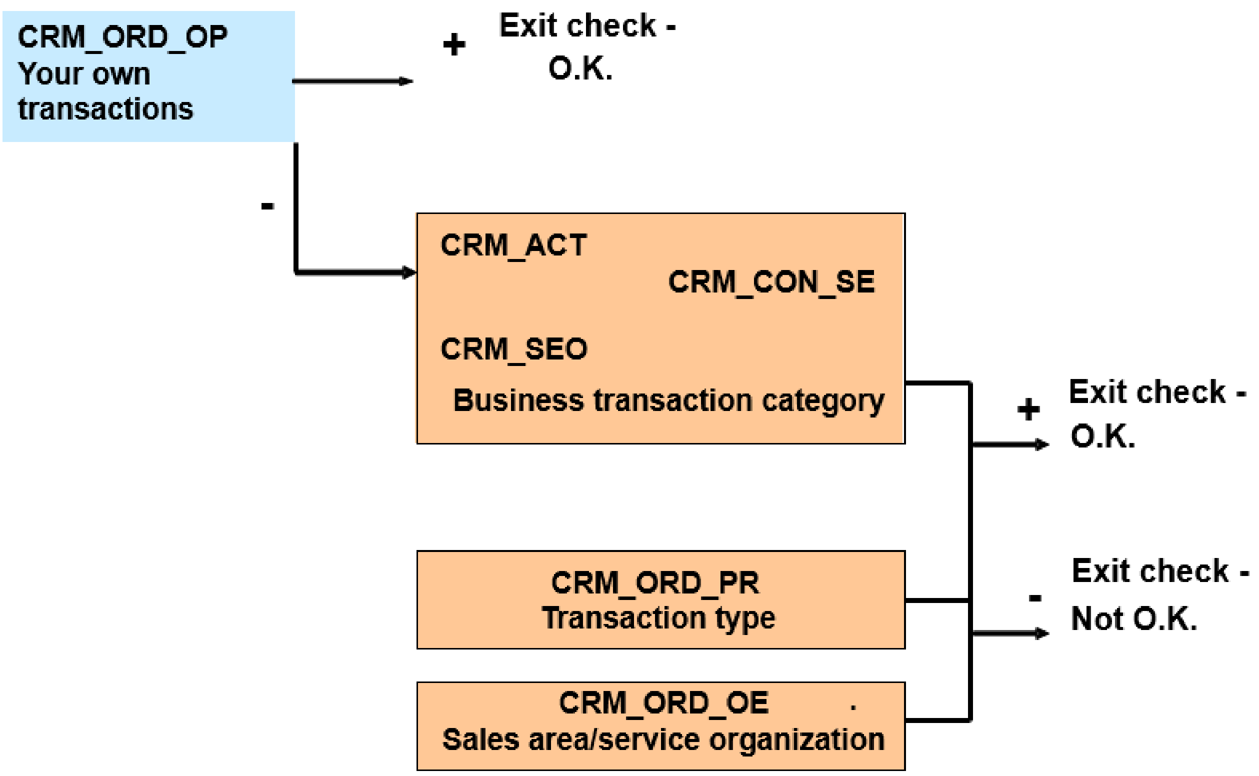
To execute other functions, for example partner determination, we recommend that you assign an employee to the position, to whom a user is assigned in the employee record.

See also:

Organizational Management in Service

Process

The authorization check is run according to the following procedure:



i Note

The authorization objects listed in the business transaction category box are a sampling of the available objects.

1. **Your own transactions (authorization object CRM_ORD_OP)**

   The system checks whether the user in the relevant transaction takes on a specific partner function for the activity executed, for example, whether he or she is the employee responsible. Furthermore, the system checks whether the user has the authorization to change, display, or delete a transaction. If the result of this check is positive, no further checks take place at transaction level.

2. **Combination of several authorization objects**

   If the preceding checks were not successful, this combination of different authorization objects is checked. All the checks must be successful before the user is authorized to process the required transaction. This means that the user receives the authorization to process only if he or she is authorized to:

   - Process the If the user is not authorized in this step of the check, he or she is not able to process the transaction in the required way. The user receives a system message that contains the corresponding authorization object, and refers to the missing authorization.If the preceding checks were not successful, this combination of different authorization objects is checked. All the checks must be successful before the user is authorized to process the required transaction. This means that the user receives the authorization to process only if he or she is authorized to:
   - Process the leading business transaction category in the corresponding transaction type
   - Process in the corresponding transaction type
   - Process in the corresponding sales area

     a. **Authorization objects CRM_ACT, CRM_SEO, CRM_CON_SE**

        Using these authorization objects, the system checks which business transactions the user is allowed to process, and whether he or she is allowed to carry out the functions create, display, or delete in these transactions. The system checks the relevant authorization object, depending on the activity executed:

        - Activities: CRM_ACT
        - Service transactions: CRM_SEO
        - Service confirmation: CRM_CON_SE

        i Note

        The authorization objects listed above are a sampling of the available objects.

     b. **Authorization object CRM_ORD_PR**

        Using this authorization object, the system defines which action the user is allowed to execute for each business transaction type.

     c. **Authorization object CRM_ORD_OE**

        Using this authorization object, the system defines in which sales area or in which service organization the user is allowed to process business transactions in Service, and which activities he or she can carry out there.

If the user is not authorized in this step of the check, he or she is not able to process the transaction in the required way. The user receives a system message that contains the corresponding authorization object and refers to the missing authorization.

Effects on the User Interface Layer

- When selecting the transaction to be created, only the transaction types of the business transaction categories for which the user is also authorized, are displayed. For example, if you only have authorization to create service transactions, only the transaction type for the **Service** business transaction categories is displayed.
- Buttons **Create/Change** and **Delete:** The system displays only the keys that the user is authorized to use. If, for example, you are only authorized to display, the key **Create/Change** is not active.

Incompleteness Check in Business Transactions

Use

The system uses the incompleteness check to determine whether important data, that you need to complete, is missing in a business transaction.

Prerequisites

You have defined the scope of the incompleteness check in Customizing under **Service** > **Transactions** > **Basic Settings** > **Incompleteness Check** > **Define Incompleteness Procedures**.

Features

- A message in the application log informs you of missing data in the transaction document. You can branch to the fields in which data is missing, and make entries directly from the incompleteness log. You can specify in Customizing whether a warning or an error message should be issued.
- The scope of the check can be different for each transaction type, for example, different fields can be checked in a service order than in a service request.

  Example

  In the incompleteness check procedure for quotations, you specify that the fields must be filled for the validity period. If you do not enter any values in these fields, the fields are checked as incomplete, and the system issues a message in the application log.

- Not only can you make the incompleteness check dependent on a transaction type, you can also specify restrictions for certain business partners.

See also:

Specifics for Interaction Center: Business Transactions

Authorization Objects and Authorization Fields

The following authorization objects and authorization fields are used for the check in the business transaction:

Authorization Object	Authorization Fields
CRM_ACT (authorization object CRM transaction – business transaction category: activity)	ACTVT (Activity)

Authorization Object	Authorization Fields
CRM_CON_SE (authorization object CRM transaction – business transaction category: service confirmation)	ACTVT
CRM_ORD_OE (authorization object CRM transaction – allowed organizational units)	SALES_ORG (sales organization) SERVICE_OR (service organization) DIS_CHANNE (distribution channel) SALES_OFFI (sales office) SALES_GROU (sales group) ACTVT
CRM_ORD_OP (authorization object CRM transaction – separate documents)	PARTN_FCT (partner function) PARTN_FCTT (partner function category) ACTVT
CRM_ORD_PR (authorization object CRM transaction – transaction type)	PR_TYPE (transaction type) ACTVT
CRM_SEO (authorization object CRM transaction – business transaction category: service transaction)	ACTVT

Examples for the Authorization Assignment

- A user should receive authorization to process service transactions with the transaction type SRVO in service organization 0001 and for distribution channel 01.

You assign the following authorizations:

CRM_ORD_PR: PR_TYPE 'SRVO', ACTVT '*'

CRM_SEO: ACTVT '*'

CRM_ORD_OE:SERVICE_ORG '0001', DIS_CHANNE '01', SALES_OFFI '*', SALES_GROU '*', ACTVT = '*'

- A user should receive authorization to process all types of service transactions (independent of the transaction type and the organizational assignment).

You assign the following authorizations:

CRM_ORD_PR: PR_TYPE '*', ACTVT '*'

CRM_SEO: ACTVT '*'

CRM_ORD_OE:SERVICE_ORG '*', DIS_CHANNE '*', SALES_OFFI '*', SALES_GROU '*', ACTVT = '*'

- Along with his or her other authorizations, the user should receive authorization to process a document in which he or she is entered as the employee responsible. This is automatically generated from the user role.

You assign the following authorizations:

CRM_ORD_OP: PARTN_FCT '*', PARTN_FCTT '0008 ', ACTVT ,*'

i Note

Service is delivered with business roles such as the service professional that already contain preconfigured PFCG roles.

Communication Management Software

Integration with Communication Management Software enables use of CMS functions in business roles in Service and allows you to achieve simplified interactions, faster response times, and improved end-to-end processes.

Use

You can integrate communication management software (CMS) into Service. You can then integrate CMS functions into business roles in Service. This allows you to establish communication-enabled business processes (CEBPs). This in turn allows your company to achieve the following:

- Simplified interactions with your customers
- Faster response times
- Improved ability to locate people with the right skills
- Better enabled end-to-end processes involving employees, customers, suppliers, and partners

CEBPs yield the most value in contact-intensive situations where service requests are complex. CEBPs can bring the greatest benefits when business requirements call for a high volume of interpersonal interactions, when customers demand service through multiple contact channels, and where there are globally distributed resources.

You can enable the use of an active browser component (ABAP Push Channel) for communicating real-time events from the server to a user's browser session. The client is notified in real-time when there is an update available, and only then does it retrieve information from the server, instead of continually polling. This feature is available in the Interaction Center and for all CEBPs. It can be used for all channels integrated through the Integrated Communication Interface (ICI), such as telephone, e-mail, or chat, and for all features that currently use polling, such as alerts, broadcast messaging, and third-party integration.

Prerequisites

If you want to enable an active browser component for messaging in the Interaction Center, you have done the following:

- Defined a new polling profile and assigned it to a new context area profile in Customizing for Service under Interaction Center WebClient > Basic Functions > Define Context Area Profile

  To avoid continuous polling, you have selected the APC Usage checkbox in the polling profile. If you select the checkbox, information is being retrieved from the server only in case of available updates.

- Assigned the new context area profile to your business role in Customizing for Service under UI Framework > Business Roles > Define Business Role
- Assigned the notification server to your new polling profile on the SAP Easy Access screen under Interaction Center > Interaction Center WebClient > Administration > Communication Management Software > Assign Notification Server to Polling Profile

Features

If CMS functions are integrated into business roles in Service, users who are assigned to these business roles can use the following features:

- Manual logon to and logoff of the CMS

  You can specify that users can manually log on, rather than being logged on automatically, to the CMS when they log on to the WebClient UI in Service. The **Sign In** pushbutton will be available in the toolbar. Once the user has signed in to the CMS, the **Sign Out** pushbutton becomes available in the toolbar.

- Assignment of users to queues

  The assignment of users to queues is maintained in the CMS based on user names. Each user in Service must be mapped to a user in the CMS and have the same user name. If users do not have fixed work centers, the user configuration can be based on the work center instead. For more information, see Free Seating.

  Users can choose queues for transferring, forwarding, consulting, or conferencing contacts instead of dialing other users' phone numbers. If queues are maintained in Service, users have to select which queue they want to contact. The CMS determines which users are available and working in that queue.

  If queues are not maintained, users have to manually enter the phone number that they want to contact.

  If more than one connection ID is maintained, you can decide which connection users use by assigning the connection in the CMS profile, which is then assigned to the business role.

- Toolbar

  The toolbar contains different sets of pushbuttons and icons for communication functions depending on the connection of the user to the CMS and the status of the user's interaction with a contact. For more information, see Toolbar.

- Inbound and outbound phone calls

  If a user is logged on to the CMS and is not in a call, he or she can receive phone calls from the CMS. When a phone call is sent to the user by the CMS, the **Accept** and **Reject** pushbuttons start flashing. In addition, if a user is logged on to the CMS and is not in a call, he or she can trigger phone calls by choosing the **Dial Pad** pushbutton. The dial pad opens and the user can dial a number and make a phone call.

- Communication hyperlinks

  You can specify that the telephone numbers of business partners are displayed as communication hyperlinks in assignment blocks related to business partners. If a user who is logged on to the CMS clicks a hyperlink, the dial pad opens, the number is prefilled, and the user can call it directly.

- Presence status of users

  The presence status of users can be displayed in the dial pad and in some assignment blocks on the WebClient UI in Service related to business partners. If a user is available, you have different options for collaboration with this user. For more information, see Presence Status.

- Contact attached data

  When phone calls are transferred between users who are logged on to the CMS, contact attached data can be transferred together with the communication event. Contact attached data can be used to transfer the business partner information of the caller. It can also be used to transfer the content of the current business process. The communication event and contact attached data can be integrated into a business process by the automatic creation of a business transaction, for example, an interaction log. For more information, see Integration of Communication Events into Business Processes.

- Soft phone of the CMS

  A link to the soft phone of the CMS is available in the **Communication Center** in the navigation bar on the WebClient UI in Service. The user can access the soft phone by clicking **Soft Phone Link**. The soft phone offers the same functions as a regular telephone and also allows the user to access various soft-phone-specific functions, such as audio and microphone volume controls.

For information about the CMS functions that are available in the interaction center, see Specifics for Interaction Center: Communication Management Software.

Free Seating

Use

This function allows the configuration of communication channel services from the communication management software (CMS) to be based on the user's current work center location, rather than on a fixed work center.

Prerequisites

- Your CMS supports and is configured for free seating.
- You have made the settings for users to automatically log on to the CMS when they sign in to Service. You do this in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Communication Management Software Profiles**.

Features

Users can work in a virtual environment where they do not have fixed seating locations, for example, users sit at the first available desk at the start of their shift. To support this virtual free-seating working environment, the CMS uses the work center ID to configure the communication channel services. Work centers are equivalent to a physical location, either a cubicle or a desk that includes electronic equipment, such as a computer and phone.

You must configure the work center ID as **Unspecified Identification (ID)** in the CMS.

Without free seating, the channel assignment is tied to users. This assignment only enables users to work with communication channel services from one fixed location.

Activities

The relationship between service solution and the CMS is as follows:

- The user signs in to service solution.
- Service solution determines whether the CMS is configured for free seating.
- If free seating is not supported, the user is automatically logged on to the CMS.
- If free seating is supported, the user is manually prompted to enter or select the relevant work center, and is logged on to the CMS based on the selected work center.

Example

ABC company uses a CMS that was recently configured for free seating. The following work centers exist:

- Work center A, phone extension X1234
- Work center B, phone extension X5678

- Work center C, phone extension X9012

Prior to free seating configuration, Anna, a user, was assigned to phone extension X5678 in the CMS, and could only sit at work center B. When free seating is enabled, Anna can sit at any work center. Once she logs on, the phone extension is determined by her current work center. Today, Anna sits at work center A, so her phone extension is X1234.

Toolbar

Use

You can integrate functions of a communication management software (CMS) into Service business roles. The toolbar provides an interface between the CMS and service solution. In service solution, the toolbar contains pushbuttons and icons for communication functions that apply to the telephony channel. In the interaction center (IC), the toolbar contains pushbuttons and icons for communication functions that apply to all communication channels. For more information, see Specifics for Interaction Center: Toolbar.

You can modify the content of the toolbar to better meet your business requirements. For example, you can specify that different sets of pushbuttons are available in the toolbar, depending on the user's connection to the CMS and the status of the user's interaction with a contact.

Prerequisites

- Your system administrator has set up the CMS and ensured that the user names there match those in Service. For more information, see Communication Management Software and the documentation for the CMS you are using.
- You have made the settings for the toolbar in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Toolbar Profiles**.

Features

You use the toolbar to work in the telephony channel. The system determines the pushbuttons that appear in the toolbar, depending on the Customizing settings.

Context Area

Use

The context area is part of the WebClient UI of Service and the Interaction Center (IC) WebClient UI and contains the following:

- Branding area, which includes the brand logo and message, as well as standard header area links of service solution such as **Personalize** and **Help Center**
- Context area components, such as the scratch pad, alerts area, and toolbar buttons

  Note

  The scratch pad is only available in the IC interface.

The context area resides in a fixed position at the top of the screen. However, you can customize the area for different business roles.

Prerequisites

- You have created context area profiles in Customizing for Service under Interaction Center WebClient > Basic Functions > Define Context Area Profile.
- You have assigned context area profiles to business roles in Customizing for Service under UI Framework > Business Roles > Define Business Role. If upgrading from a previous version of service solution, you have assigned the `CONTEXTAREA` function profile to your business roles.
- You have configured the context area to suit customer requirements. For example, you have configured the layout and content of individual components within the context area, or replaced the standard SAP logo and brand message. Configuration instructions are provided in SAP Solution Manager.

Features

- Context area customizing

  You can create different versions of the context area and assign them to separate context area profiles, which you then assign to business roles. The context area profile also allows you to set polling frequency.

- Context area configuration

  You can configure the context area layout and the components within the layout. You can also enhance individual context area components, or add new ones.

- Toolbar configuration

  You can customize button availability and organization in the toolbar.

More Information

Toolbar

Context Transfer

Use

You can use this function to transfer business context (for example, all recently accessed objects in a user's session in Service) between users while using communication-enabled business processes (CEBP) or the interaction center (IC).

Prerequisites

If you want to use multiple context transfer, you have enabled it in Customizing for Service under Basic Functions > Communication Management Software Integration > Define CMS Integration Profiles.

Features

With this function, you can do the following:

- Access the contact attached data (CAD) view and transfer CAD objects as context
- Import items from the activity clipboard or recent items to be transferred as context

  Note

The activity clipboard is available when using interaction center (IC) business roles, and the recent items is available when using communication-enabled business process (CEBP) business roles.

- Transfer context when interacting with customers using telephone, instant messaging (IM), or chat
- Transfer multiple objects at once

  You can add multiple objects (for example, all objects under **Recent Items**) to the contact attached data (CAD) view. The entire contents of the CAD view can then be transferred between users during phone call or instant message context transfers.

Presence Status

Use

If functions of a communication management software (CMS) are integrated into Service business roles, the following options are available:

- The presence status of users who are assigned to communication-enabled business roles can be displayed in the dial pad and in some assignment blocks on the WebClient UI of service solution related to business partners. If a user is available, you have different options for collaboration with this user.
- When a communication event is received and transferred, users can integrate the communication event in a business process by automatically creating a business transaction.

Prerequisites

- To display the presence status of users in the dial pad and in some assignment blocks on the WebClient UI of service solution related to business partners, you have made the settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Communication Management Software Profiles**.
- To define the UI objects that are displayed when a transfer is completed, you have made the settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define CMS Integration Profiles**.

Features

- Presence status and collaboration options

  The presence status and the collaboration options are retrieved from the CMS.

  Example

  The presence status can be one of the following:

  - **Available**

    The user is logged on to the WebClient UI of service solution and the CMS and is not in a call.

  - **Busy**

    The user is logged on to the WebClient UI of service solution and the CMS and is in a call or the user is only logged off of service solution.

  - **Logged Off**

The user is logged off of the WebClient UI of service solution and the CMS.

If a user is available, the collaboration options can be **Call**, **Blind Transfer**, or **Warm Transfer**.

- Integration of communication events into business processes

  For more information, see Integration of Communication Events into Business Processes.

Integration of Communication Events into Business Processes

Use

You can use this process to establish communication-enabled business processes. If functions of a communication management software (CMS) are integrated into business roles in Service, the following options are available:

- Users can trigger and receive communication events, such as phone calls.
- Users can send and receive contact attached data to and from the CMS.
- When incoming communication events are transferred between users who are logged on to the CMS, the communication event and contact attached data can be integrated into a business process by the automatic creation of a business transaction, for example, an interaction log.

i Note

For outbound communication events, business transactions are not automatically created.

Prerequisites

- To display the presence status of users in specific business partner assignment blocks in Service, you have made the settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define Communication Management Software Profiles**.

- To define the UI objects that are displayed when a transfer is completed, you have made the settings in Customizing for **Service** under **Interaction Center WebClient** > **Basic Functions** > **Communication Channels** > **Define CMS Integration Profiles**.

Process

1. An account makes a phone call to make an inquiry about a business transaction, for example, a service order.

   The CMS sends the phone call to a user whose business role is enabled for communication and who is logged on to the CMS and not in a call. The **Accept** and **Reject** pushbuttons in the toolbar start flashing. The account name and telephone number are identified from the contact attached data from the CMS and are displayed in the account information area on the screen. The account name is displayed only if it is identified from the incoming telephone number and if no other contact persons have the same telephone number. Otherwise, only the incoming telephone number is displayed.

2. The user accepts the phone call and navigates to the business transaction for which the account has an inquiry.

   The overview page of the business transaction is displayed. The employee responsible for the business transaction is displayed in the **Parties Involved** assignment block.

   i Note

A transfer to the employee responsible is only possible if the business role of the user is also enabled for communication, and the user is logged on to the CMS and is not in a call.

3. The user transfers the phone call to the employee responsible. This can be done in one of the following ways:
    - If the presence status of the employee responsible is displayed and is **Available**, the user clicks **Edit List** and selects a collaboration option, for example, **Warm Transfer**.

      The dial pad appears, prefilled with the telephone number of the employee responsible.
    - If the presence status of the employee responsible is not displayed, the user chooses the **Transfer**, **Warm Transfer**, or **Blind Transfer** pushbutton in the toolbar.

      The dial pad appears and the user has to enter the telephone number of the employee responsible.
4. The user chooses the **Dial** pushbutton.

   On the screen of the receiving user, the **Accept** and **Reject** pushbuttons in the toolbar start flashing. If the necessary Customizing settings have been made, when the receiving user accepts the phone call, the business transaction, which the sending user was handling, appears on the screen in display mode.
5. To complete the transfer, the sending user chooses the **Transfer** pushbutton in the dial pad.

   To view any changes to the business transaction the sending user has made and saved, the receiving user must click **Edit**.

   If the necessary Customizing settings have been made, the creation of a business transaction is automatically triggered. The **Interaction Log** page appears on the screen of the sending user.
6. To record important information from the call, the sending user creates the interaction log.

   Some of the information, such as the type of communication event and the account name, if identified, is automatically filled in the interaction log.

Instant Messaging for Internal Collaboration

Use

An instant messaging function is available for internal collaboration of employees in communication-enabled business processes (CEBP) and employees who use the interaction center (IC). You can use this communication channel independent from the other communication functions in the WebClient UI. In addition to the sending and receiving of messages, the instant messaging function also allows you to send e-mails to communication partners, initiate a telephone conversation, and send a business context to a communication partner.

Instant messaging on the WebClient UI is only available for internal communication between employees.

You can keep a list of personal contacts, into which you can copy all colleagues with whom you regularly use instant messaging.

Prerequisites

You have made the following settings and met the following requirements:

- The employee role is assigned to the users.
- Communication-enabled user roles are set.

  For more information, see Integration of Communication Events into Business Processes.

- You have checked to ensure that the Personal Contacts (CRMI01) business partner relationship was created.

  Note

  You can check business partner relationships in the Change Relationship Types view (transaction BUPA).

- Instant messaging is activated in the chat profile.

  Note

  You make this setting in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Chat Profiles.

- The polling profile is defined.

  Note

  You make this setting in Customizing for Service under Interaction Center WebClient > Basic Functions > Define Context Area Profile.

- The communication management software (CMS) that you are using supports a chat channel.

  For more information, see Communication Management Software.

- In the CMS profile of the user, display of the presence status is activated.

  Note

  You make this setting in Customizing for Service under Interaction Center WebClient > Basic Functions > Communication Channels > Define Communication Management Software Profiles.

Features

To start the instant messaging function, you can use the List of Personal Contacts, which is found in the broadcast messaging bar in the lower part of the work area. In communication-enabled business processes (CEBPs), you can also start instant messaging from a business object in Service if the colleague involved appears online.

The instant messaging function provides the following options:

- **List of personal contacts**

  You can add colleagues to the List of Personal Contacts, provided that these colleagues fulfill the prerequisites for the instant messaging function. You can also delete contacts from the list.

  If you copy colleagues' names into the list, your name also appears in their contact list. The opposite is also true.

  For each contact entry, the list shows whether the person is Online, Busy, or Offline.

- **Communication**

  If a person is Online and in status Ready, the following communication options are available:

  - Instant messaging
  - Phone
  - E-mail
  - Send business context

Depending on the communication method you choose, the appropriate dialog box appears, supporting the communication process.

You can switch among the different communication methods at any time. For example, you can switch from instant messaging to a telephone call or writing the person an e-mail.

You can have a maximum of five instant-messaging channels open at the same time.

- **Business context**

  In addition, the instant messaging function allows you to send a business context that you are processing, such as an opportunity, to a colleague. When you choose this function, you can choose the objects you want to send either from the context you are currently working on, or from a list of business objects that you accessed recently.

  For more information about transferring contexts, see Context Transfer.

Communication Channel Blending

Use

You can use multiple communication channels interchangeably in communication-enabled business processes in Service. For example, you can convert an inbound chat interaction with a customer into an outbound call. You can also transform an internal Instant Messaging (IM) conversation to a telephone call or an e-mail. As a result, you can establish multiple communication channels with a customer or colleague at any point in time.

For all blended communications, there is only a single interaction record in the system in service solution.

When you start an outbound call during a chat interaction, the channel information for both is displayed. You can toggle between the channels to see individual timers for the elapsed time.

Prerequisites

You have enabled the relevant business roles for communication. For more information, see Integration of Communication Events into Business Processes.

More Information

Communication Management Software

Chat

Partner Processing

Use

Partner processing controls how the system works with business partners in transactions. It ensures the accuracy of partner data in transactions by applying rules you specify in Customizing, and it makes your work easier by automatically entering certain partners and related information, like addresses.

One of the most important aspects of partner processing is partner determination, the process by which the system automatically finds and enters the partners involved in a transaction. In most transactions, you manually enter one or more partners, and the system enters the others through partner determination. Various sources of information make partner determination possible; two of the most important are business partner master data and organizational data.

Prerequisites

You have maintained business partner master data, organizational data, transaction types, and item categories.

You define transaction types and item categories in Customizing for **Service**, by choosing **Transactions** › **Basic Settings** › **Define Transaction Types/Define Item Categories**.

Features

- With Partner Functions, you define partners with the terminology used in your area of business and in your company. For example, if you have a real estate business, you might define the partner functions **Tenant** and **Mortgage Provider**.
- In Access Sequences, you specify the sources of data the system uses for automatic partner determination, and the order in which it checks those sources.
- In Partner Determination Procedures, you specify which partner functions are involved in a transaction, assign access sequences to the functions the system should determine automatically, and set other rules for working with partners in transactions.

More Information

Business Partners

Organizational Management in Service

Basic Elements of Partner Processing

Definition

The basic elements of partner processing are:

- Partner functions and partner function categories
- Access sequences
- Partner determination procedures

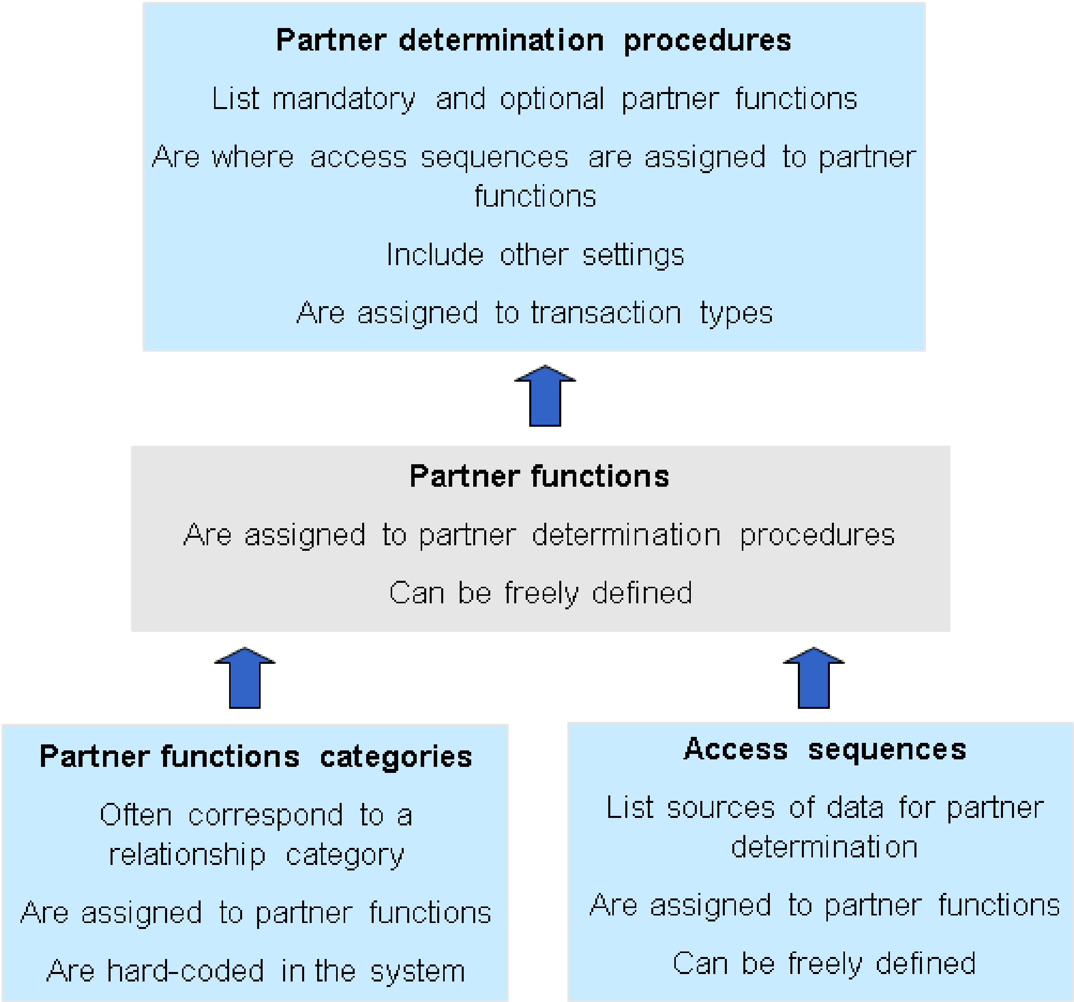
Use

The **Parties Involved** assignment block in business transactions gives you an overview of partners, determined by partner processing, involved in a business transaction:

- When you create a business transaction the system reads the partner determination procedure, finds the master data and automatically enters it in the document.
- You can add, delete, and redetermine any partners or partner functions manually in the **Parties Involved** assignment block.
- The address data shown for the parties involved is derived from the business partner master data. Any changes you make to it are specific to the transaction, and are not saved in, nor do they affect, the address details maintained in the business

partner master data. By changing address details here, instead of in the master data, you avoid the risk of deleting or altering essential master data records and you do not need special system authorizations for master data maintenance.

Structure



Partner Functions

Definition

Partner functions are terms that describe the people and organizations with whom you do business, and who are therefore involved in transactions. For example, when you create an activity, based on Customizing settings, it automatically includes the partner functions **Activity Partner, Contact Person,** and **Person Responsible**. Partner functions are always assigned to partner function categories, which are hard-coded in the system.

The system includes commonly used partner functions, but you can also define your own.

Use

You use partner functions in transactions, in all areas of partner processing, and in some areas of business partner master data. For example:

- You have a real estate business and define the function **Tenant** to refer to your customers.

- **Tenant** is displayed in the header and **Parties Involved** assignment block in transactions, and in the **Buying Center** assignment block, when relevant.
- It appears in the Customizing settings for access sequences, partner determination procedures, and partner teams.
- It appears in business partner master data when you assign partners to partner functions when creating relationships between business partners, and when you enter excluded partner functions.
- You define partner functions in Customizing for **Service**, by choosing **Basic Functions** > **Partner Processing** > **Define Partner Functions**.

Structure

When you define a partner function, you assign it to a partner function category. Therefore, the partner function consists of two basic elements: the new term itself and the category. The new term is your name for the function, and the category is the hard-coded key that allows the system to identify what the function means, and how to work with it. In some cases the functions have the same name as the categories to which they are assigned, but they need not.

In most cases, you also specify the business partner relationship category to which the partner function should correspond. Frequently, once you enter a partner function category, the system automatically enters the relationship category. It is able to do so because many partner function categories are hard-coded to correspond to certain relationship categories. If you use the partner function category **Undefined Partner** or relationship categories you have created yourself, you must enter the relationship category manually.

The following table shows several of the partner functions in the system, with the partner functions categories to which they are assigned, and the corresponding relationship categories:

Partner Function	Partner Function Category	Relationship Category
Sold-To Party	Sold-To Party	
Ship-To Party	Ship-To Party/Service Recipient	Is the Ship-To Party/Recipient Of
Sales Prospect	Activity Partner	Is Activity Partner For
Activity Partner	Activity Partner	Is Activity Partner For
Sales Manager	Employee	Is the Employee Responsible For
Executing Service Employee	Serv. Employee/Serv. Provider	

For a list of all partner functions and partner function categories in the system as well as relationship categories, see the Customizing activity **Define Partner Functions**.

To block a partner function, you can delete the partner function in the business partner's master data (sales-area-specific data).

“Undefined Partner”

Definition

The partner function category **Undefined Partner** is a free category that you can use if none of the other partner function categories suit your needs. The benefit is that you can create partner functions and include them in transactions, even if none of the predefined categories are appropriate.

Use

The system's processing of the category **Undefined Partner** is limited. Normally, there are certain system functions or rules associated with a function category. However, in the case of the **Undefined Partner** there are almost none.

- You can specify that a partner function assigned to this category must be included in a certain transaction type, just as you would with any other category. The system then searches for and enters a partner, or asks the user to do so manually.
- When you assign a partner function to this category, you can choose any relationship category, either an existing one, or one you create yourself to correspond to it. However, the relationship category **Is the Undefined Partner Of** corresponds only to the partner function category **Undefined Partner**.

Example

Using "Undefined Partner"

When you have determined that none of the pre-defined partner function categories suit your needs, undefined partner is the best choice:

1. You have an investment business, and to help keep track of your customers, you have created the new relationship category **Is the Stock Options Analyst.** You want the system to be able to handle partners of this category in transactions, and therefore you need a corresponding partner function.
2. You define the new partner function **Stock Options Analyst,** but there is no partner function category that fits your needs, and so you choose **Undefined Partner**. You enter **Is the Stock Options Analyst** as the corresponding relationship category.
3. You can now enter this partner function in transactions, and the system will recognize that the partners with the relationship **Is the Stock Options Analyst** can carry out this function.

Customizing activities:

- **Service** > **Basic Functions** > **Partner Processing** > **Define Partner Functions**

Avoiding "Undefined Partner"

Because the system's processing of this partner function category is limited, we recommend you avoid using it if one of the other categories works:

1. You have a real estate company, own several apartment buildings and rent the apartments to tenants. You want to create the new partner function **Tenant**, but the partner function category **Tenant** does not exist. However, you do not need to use **Undefined Partner** or create the new relationship category **Is the Tenant Of** because there is a simpler solution.
2. In terms of the business processes in which they are involved, your tenants are basically **Sold-To Parties**. Therefore, you maintain master data for them either with no specific role or the role **Sold-To Party,** and you create the new partner function **Tenant** and assign it to the partner function category **Sold-To Party**. There is no corresponding relationship category for **Sold-To Party,** so you leave this blank.
3. You can then include this new partner function in transaction documents, enter your tenants as the partners carrying out this function, and the system automatically enters the other required data.

Blocking Partner Functions in Partner Processing

Use

In the area of partner processing, “excluding” a partner function is known as “blocking”. If you do not create any relationships or partner function assignments for a partner, it can automatically perform all functions itself. In some cases this may not be appropriate and you may specifically want to exclude certain partner functions.

Features

- You set a block in Customizing for **Service**, by choosing **Basic Functions** > **Partner Processing** > **Define Partner Functions**, and set the **Block** indicator for the relevant partner function.
- The main partner in transactions, such as the activity partner in an activity, cannot perform the blocked functions in that transaction.

Example

Your transactions include the partner functions **Sales Prospect, Contact Person, Employee Responsible** and **Competitor**. To ensure that the sales prospect is never mistakenly entered as the competitor, you set the **Block** indicator for the partner function **Competitor**. This setting applies to the competitor in any transaction.

Access Sequences

Definition

Access sequences are search strategies that specify the sources of data the system uses for finding partners, the order in which it uses these sources, and related details.

Example

An access sequence occurs when a colleague asks to borrow your dictionary, and you tell her where to find it. You might say, “Look on my desk”, or “Look on my desk, and if it is not there, look in the shelves”. When you define access sequences in Customizing, you replace the desk and shelf concepts with sources of information such as business partner master data, organizational data, and preceding documents.

Use

Access sequences allow the system to carry out partner determination, the process by which the system automatically finds and enters partners in a transaction.

When you define a partner determination procedure, you can assign an access sequence to each partner function listed in the procedure. Then, when you create a transaction, the system knows how to search for partners to carry out these functions. If you do not assign an access sequence, or the system cannot find partners in the sources listed, you enter the partner manually.

The system includes a number of commonly used access sequences. We recommend that you check to see if you can use the existing ones before defining your own.

You can view and define access sequences in Customizing for **Service**, by choosing **Basic Functions** > **Partner Processing** > **Define Access Sequences**.

Structure

An access sequence is a list of one or more steps called **single accesses**. Each single access names a source, such as master data or the preceding document, and specifies other details. The description of an access sequence lists the sources named in the single accesses, separated by arrows.

Example

Preceding document -> BP relationships by sales organization: sold-to party -> Business partner relationships: sold-to party

When the system determines a partner, it checks the first source, and, if it does not find a partner there, it checks the next source. It does this for each partner function in a transaction, aside from those entered manually. Understanding the sources is crucial to understanding the access sequences.

More Information

For more information, search for *Business Partner Relationship* in the product assistance on SAP Help Portal.

See also Organizational Management in Service

Partner Determination Procedures

Definition

Partner determination procedures are sets of rules for how the system works with business partners during transaction processing. They bring together partner functions and access sequences, and include additional information.

Use

You define a procedure and assign it to a transaction type or item category. Then, when you create transactions of that type or containing items of that category, the system works with the partners involved according to the rules you specified in the procedure. Therefore, the system can automatically handle partners for an Internet service transaction differently than it does for a business activity, for example, without end users concerning themselves with these differences. The settings in the procedure apply to both partners determined automatically and to those entered manually.

Structure

You define partner determination procedures in Customizing for **Service**, by choosing **Basic Functions** > **Partner Processing** > **Define Partner Determination Procedure**.

The settings include:

- Which partner functions are mandatory and which suggested, which are entered automatically and which manually, and how many partners can be entered for each function.
- Which access sequence applies to each partner function that is determined automatically by the system.
- At what point the system starts partner determination for each function. This could be as soon as you enter data, when you enter a product, or when you save the transaction.
- Whether the system enters an activity in the calendars of the partners involved (relevant to activities only).
- Whether you can change a partner's address in transactions, and, when a partner has more than one address, which address is used.
- Which partner functions are available in a business transaction and which are not.

The procedure consists of three levels:

- Procedure User identifies the transaction categories and link item object types to which the procedure applies.
- Partner Functions in Procedure lists the partner functions involved and specific settings for each function.
- User Interface Settings allows you to influence which partner function appears in individual partner fields (such as contact).

Integration

For a procedure to be available for a particular transaction type, both it and the transaction type must be assigned to the same transaction category.

For a procedure to be available for a particular item category, both it and the item category must be assigned to the same item object type.

You define transaction types and item categories in Customizing for Service, by choosing Transactions > Basic Settings > Define Transaction Types/Define Item Categories.

Partner Redetermination

Use

Using this function, you can manually trigger the partner determination procedure from within a business transaction, and redetermine all parties, such as contacts, employees, and other partners, involved in the transaction. This enables you to ensure that the correct parties are identified and associated with a transaction

This could be necessary because partners determined originally, based on a partner determination procedure, are no longer up-to-date. You could have switched to new partners, for example, which in turn might require changes to all dependent partner functions.

Prerequisites

- You have set up the partner determination procedure appropriately. You do this in Customizing for Service, by choosing Basic Functions > Partner Processing > Define Partner Determination Procedure.
- You have activated the enhancement implementation ES_CRM_SET_ACTIVE of the enhancement spot ES_CRM_PARTNER_REDETERMIN.
- You have to set the parameter value X for the user parameter CRM_REDETERMINATION (System > User Profile > Own Data > Parameters).

Note

Partner redetermination is only possible for those business objects listed in the table COMS_PARTNER_DET.

Features

You can start partner redetermination manually from the overview page of a lead or opportunity. Depending on your Customizing settings, partners are redetermined as follows:

- **No Redetermination**

The partner functions in the partner determination procedure that have this value are not changed. All existing partners remain, with no new partners of the given partner function being determined or added to the partner set. Similarly, any existing partners are not replaced.

- **Add New Partners**

  The partner functions in the partner determination procedure that have this value are redetermined. All existing partners remain and any new partners determined based on the partner function are added to the partner set. Before a new partner is added to the set, the system checks whether the same partner with the same partner function already exists. If this is the case, the newly determined partner is not added to prevent a duplicate entry. The system indicates when the maximum number of partners has been reached.

- **Replace Existing Partners**

  The partner functions in the partner determination procedure that have this value are redetermined. All existing partners of the given partner function, irrespective of whether they were entered manually or determined automatically, are discarded and the newly determined partners with the given partner function are added to the partner set. The system indicates when the maximum number of partners has been reached.

  A partner might be replaced by the same partner. However, if the partner previously had any notes, these will be lost.

i Note

- If partner determination has been blocked in Customizing for the partner determination procedure, redetermination does not take place for this partner function, irrespective of the settings made for partner redetermination.
- Partner functions that are not part of the partner determination procedure are unaffected by redetermination.

Alternative Partner Proposal

Use

This function allows you to trigger partner determination within business transactions to change previously automatically determined partners, quickly and easily within the application itself. This means you no longer have to search manually for partners you want to replace.

Prerequisites

You have maintained the appropriate business partner master data to enable partner determination.

Features

You use the alternative partner proposal to manually trigger partner determination for partner functions that have already been automatically determined. You do this for one partner function at a time:

- The system searches for and displays the current and alternative partner options for the partner function, based on current document data and the Customizing settings for the partner function. You make your selection and the system automatically replaces all partners in the partner function in the transaction.
- Changes made using the function are limited to individual partners and partner functions in the business transaction and do not affect business partner master data and other data dependent on this document.
- The function can be used to trigger partner determination for empty partner functions stored in the transaction. You can use it to determine one partner per empty partner function at a later date, even if no partners were originally saved for the

partner functions in the document.

- The function can also be used for the same purpose in saved business transactions to change partners previously saved in the document.

Partner Processing and Business Partner Master Data

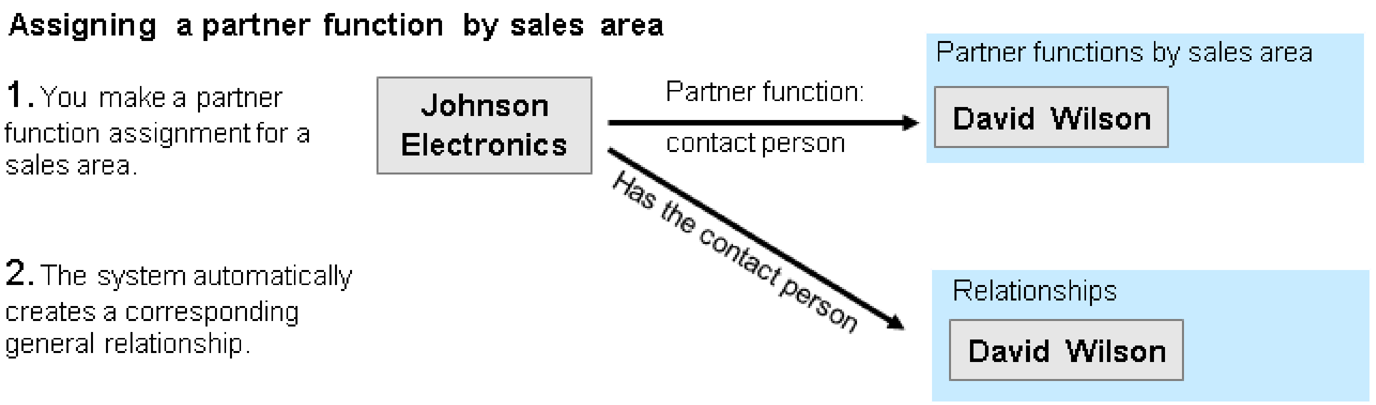
Use

Business partner master data is a crucial source of information for partner processing, and relationships are the most important aspect of this information. Relationships exist in two forms: general relationships and sales area-specific relationships. A general relationship is a relationship that does not include any sales area data, while a sales area-specific relationship includes data on one or more sales areas, and is only valid for those areas:

- A general relationship is valid for situations in which the system does not require sales area-specific information. You determine that it is general by not assigning it to any sales areas when you create it.
- A sales area-specific relationship is valid only for sales areas that you specify when you create the relationship. You can create sales area-specific relationships only for partners maintained in roles where sales area data is relevant, such as the sold-to party. Sales area-specific relationships are known as sales area-specific partner function assignments.

When you create a sales area-specific partner function assignment, the system automatically creates a corresponding general relationship at the same time. For example, if you define David Wilson as the contact person for Johnson Electronics for sales area 1000, the system also enters him as a general contact person for Johnson Electronics. The system sees the sales area-specific partner function assignment as one use of a general relationship.

It is possible to assign a relationship category in one case and a partner function in the other based on mapping defined in the system. For more information on how this is defined, see Partner Functions.



Example

Creating a general relationship:

1. You create your company's customer, Johnson Electronics, as a corporate account.
2. You create David Wilson, your contact at Johnson Electronics, as a contact.
3. In Johnson Electronics' master data, you enter David Wilson as a contact in the **Contacts** assignment block.
4. In David Wilson's master data, the system automatically creates the relationship in the **Work Addresses** assignment block.

Although your company has three sales areas, you do not assign the relationship to any of them. This makes it a general relationship.

Creating a sales area-specific partner function assignment:

1. You create your company's customer, Smith Plastics, as a corporate account and assign it the role **Sold-To Party**.
2. You create Ken Lee, your contact at Smith Plastics, as a contact.
3. In Smith Plastics' master data, you assign Ken Lee to the partner function **Contact Person** in **Sales Area 1000/Retail/10**.

   In Ken Lee's master data, the system automatically creates the relationship in the **Work Addresses** assignment block.

Although your company has three sales areas, this relationship is only valid for the one you specified.

Partner Determination Using Business Partner Relationships

Use

When the system uses business partner relationships for partner determination, it searches through the business-relevant connections between various companies and individuals that you have stored in master data.

Prerequisites

You have created relationships and sales-area-dependent partner function assignments in business partner master data.

Activities

1. You define an access sequence that specifies business partner relationships or sales-area-dependent partners as the source(s).
2. You assign this access sequence to the partner functions that should be determined in this way.

   You define access sequences and assign them to partner functions in Customizing for **Service**, by choosing **Basic Functions** > **Partner Processing** > **Define Access Sequences/Define Partner Determination Procedure**

Contact Relationship

Use

In a work step, you can assign multiple contacts to an account, for example a sold-to party. This assignment function is especially important for internal service scenarios.

Features

You can search for the **Contact Relationships** of one or more contacts, and create or delete contact relationships:

- If you want to search for contact relationships, specify the appropriate search criteria for one or more contacts.

  A result list displays the contact relationships that a contact has with one or more sold-to parties. For each relationship, the contact ID and name of the account is displayed, as well as its ID and the validity period of the relationship.

- If you want to create a new contact relationship from the result list, select one or more contacts and choose **Create Relationship**. Specify the ID of the account and the validity period for the new contact relationship, and choose **Create Relationship**.
- If you want to delete contact relationships from the result list, select one or more contacts and choose **Delete Relationship**.

Partner Processing and Organizational Data

Use

Organizational units, employees, and users from your company's organizational model can act as partners in transactions. In other words, they can perform partner functions in the same way that other companies or outside people do. In these cases, they are entered in the **Parties Involved** assignment block rather than in the **Organizational Data** assignment block. For example, a sales representative could perform the partner function **Employee Responsible**, or, in inter-company billing, an organizational unit within your own company could be the **Bill-To Party**. Therefore, organizational data is a crucial source of information for partner determination.

To use organizational data in partner determination, the system uses the determination rules defined in organizational management. Therefore, when you specify organizational data as the source in access sequences in Customizing for partner processing, you must enter a determination rule.

Partner processing is able to recognize organizational objects because, when you create an organizational object, the system automatically creates a corresponding business partner with the role **Organizational Unit**. Therefore, an object, such as a sales group, has two identities: one as an organizational unit and one as a business partner.

More Information

Organizational Management in Service

Determining Organizational Data

Rule Resolution Using Organizational Model

Rule Resolution Using Responsibilities

Attributes and Rules Used in Partner Determination

Use

Organizational attributes are characteristics that help define organizational units. You assign them to units in the organizational model. They define, for example, whether a unit is active in sales or service, or for which distribution channel or product group it is responsible.

Organizational determination rules find organizational data in transaction processing by evaluating attributes in the organizational model and information in the transaction.

You can use these rules to determine partners from the organizational model. You must specify a determination rule whenever you enter the source **Organizational Data** in an access sequence. The following rules are used frequently in partner determination.

Rule	Name	What It Does	What It Evaluates
Partner_1	Employees for responsible organizational unit	Finds the employees assigned to the responsible organizational unit	Responsible organizational unit in transaction, and attribute SA_RESPORG in organizational model
Partner_2	Responsibility according to sales office and district	Finds the responsible organizational unit	Sales office and district of main partner in transaction, SA_ORG and DISTRICT in model
Partner_3	Responsibility according to product group and district	Finds the responsible organizational unit	Sales office and district of main partner in transaction, PRODGRP and DISTRICT in model
Partner_4	Organizational unit determined from partner attribute (postal	Finds the responsible organizational unit	Postal code of main partner in transaction, POSTCODE_1 in

	code)		model
Partner_5	Organizational unit employee	Finds the employees assigned to the responsible unit	Responsible organizational unit in the transaction, PARTNER in the model

More Information

General Attribute Maintenence

Determining Organizational Data

Date Management

Use

Dates are important for the correct processing of all business transactions. In global business processes that cover several countries and time zones, it is particularly necessary to have a date management system that can convert time zones and the factory calendar in documents of business partners involved in a transaction.

The component **Date Management** enables you to process as many dates as you require in a document. You can either enter dates yourself, or, by using date rules, have the system calculate these. In Customizing, you can define date types, durations, and date rules to meet your requirements.

Integration

Date Management is used in business transactions in Service.

Furthermore, you need Date Management to define time-dependent conditions for actions, as well as date rules and validity periods of cancellations.

To use Date Management, you have to have set the **time zones** in Customizing.

Features

The component **Date Management** offers a flexible tool for defining dates and processing these in documents. It also provides the same interface for all documents in service solution, as well as the same processing and controlling (Customizing) of dates. Date Management covers the following functions:

- You can define dates (date types) and durations to meet your requirements. That means you can name the date types and durations according to your company terminology.
- For every transaction at the header and/ or item level, you can define which date type you need for the transaction, that is, which item type you need.

  By doing this, you avoid saving unnecessary date types on the database. The same applies for durations (for example, contract run time).

  To assign dates and durations to an item category, you group the date types and durations you want together in a date profile and assign this to a transaction type or an item type.
- You can calculate dates using predefined rules.

In date rules, you can link whichever date types and durations you require, as well as other date rules, with each other so that you get calculation chains.

By using date rules in the cancellation procedure, you can also determine the cancellation dates (deadlines, cancellation date) of a cancellation.

Note

You can create date rules in Customizing according to your requirements. You make the required settings in Customizing for **Service** by choosing **Basic Functions** > **Date Management** > **Define Date Types, Duration Types, and Date Rules**.

A rule editor is available for this, in which you can define several rules using XML.

You can use Reference Objects to control the time zones and factory calendar to which the dates in the document refer. Instead of automatically using the time zone of the user or system, you can make your settings so that the system uses a different time zone when calculating dates, for example, the time zone of a customer.

Date

Definition

A defined time period that consists of a start time and an end time.

Use

You use dates in calendar functions (for example, in activities), as well as for all dates that play a role in business transactions (for example, valid-to dates, cancellation dates) and in actions (date-dependent conditions).

Structure

A date is made up of a start time and an end time. The start time is within the time period, the end time is not. Dates that have the same start and end time are also called **milestones**.

Dates are further described by the **date type** and distinguished in the system by a unique indicator (GUID).

Dates need a **reference object** so that the system can interpret them uniquely.

Example

The end date of a contract with the reference object **Main Partner** refers to the time zone and the factory calendar of the sold-to party, and not to the time zone of the user who created the contract.

Note

The reference object for a date type can vary according to the date profile.

Example

Examples of dates:

- Valid from

- Planned date of a sales visit
- Signature date
- Cancellation date

Duration

Definition

A timeframe between two points in time consisting of a number value and a time unit, for example, 1 month.

Use

The system uses durations when calculating dates.

Durations are always given in entire numbers. You cannot combine two different units, for example the duration 1 hour 20 minutes must be expressed as 80 minutes.

Structure

A duration always consists of a number value and a time unit, for example, five days. The number value has no defined meaning without the time unit.

The duration always has a local reference, that is, a reference object from which the time zone and the calendar can be determined.

Date Rules

Definition

Rules for calculating times. The calculation can depend on other times, durations, and reference objects.

Use

The system can calculate times (dates) in transaction documents by using date rules, for example, the cancellation date of a contract. You can use predefined date rules or define your own date rules.

You define date rules in Customizing for **Service** by choosing **Basic Functions** > **Date Management** > **Define: Date Types, Duration Types, and Date Rules**.

Date rules are defined in XML.

> **Note**
> You cannot delete date rules.

Structure

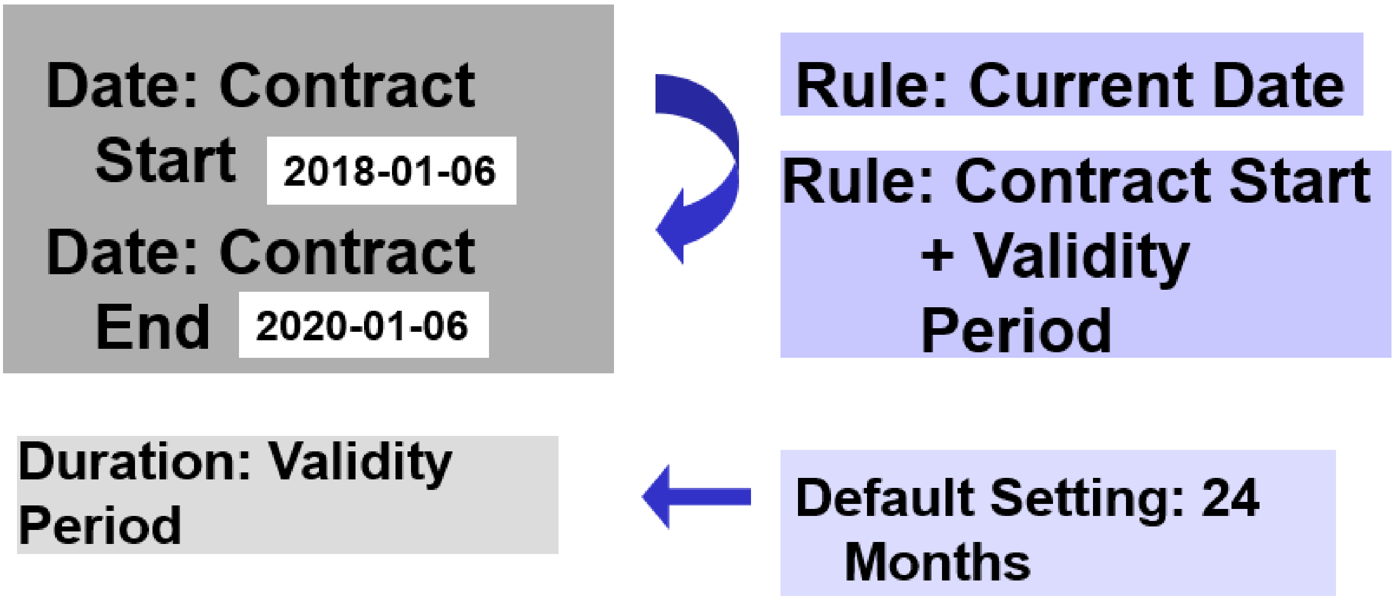
Date rules can use times, date types, durations.

Date rules have different versions. When date rules are changed, a new version of this date rule is created automatically. This ensures that date rules used in an existing document cannot be changed.

The new version of the date rule is valid from the time it is created, and is used automatically when a new document is created.

Example

The graphic shows date rules for calculating the start date and the end date of a value contract:



For the date **Contract Start**, the current date is taken as the date rule. The system automatically enters the date of the current day.

For the date **Contract End**, the date rule is defined as follows: **Contract Start + Validity Period**. The validity period has been preset as a default value in Customizing to 24 months.

The system calculates the contract end as `2020-01-06` from the contract start date of`2018-01-06` and the validity period of `24 months`.

Reference Objects

Use

To calculate dates, the system needs reference objects.

By referencing to a reference object, each time becomes a local date and a local time. Because the system works internally with time stamps (date, time) in global time Universal Time Coordinated (UTC), it needs the reference object to translate the time into UTC.

For durations, the system needs a reference object to determine the correct factory calendar (workdays and holidays).

You can use the reference object to control the time zone and factory calendar to which a date type or duration refers. For example, you can determine that the date type **valid to** in quotations is calculated in the time zone of the user who creates the quotation, regardless of the main location of the organization.

The reference object for a date or duration is always defined independently of the date profile. Therefore, a date type or duration can have different reference objects, depending on the date profile.

Different date types can have different reference objects within a date profile.

Prerequisites

You have entered the time zone in the appropriate master record to use the user or a business partner (for example, sold-to party) as a reference object.

Note

The system determines time zones in the standard address in the business partner master record.

Activities

Reference objects are delivered as default settings.

You have selected the valid reference objects for each date profile in Customizing for **Service** by choosing **Basic Functions** > **Date Management** > **Define Date Profile**. Then, in the date profile, define a reference object for every date type and duration type.

Example

The Company Brown Electronics in Vienna, Austria draws up a maintenance contract on April 20, 2017 with Medsys Inc., USA. The contract start date is to be on the first working day of the following month.

- If the reference object is **User** or **System**, the contract begins on May 2 (in Austria, May 1 is a public holiday and May 2 a working day).
- If the reference object is **Customer**, the contract begins on May 1 (in the USA, May 1 is not a public holiday).

Dates in Business Transactions

Use

You can display and change dates in business transactions.

In business transactions, on a special tab page, you can edit dates and durations according to transaction types, or you can use other tab pages more suitable for such dates (for example, the **Schedule Lines** tab page).

You can edit dates and durations at header or item level.

Prerequisites

To be able to use dates in the transaction, you must have assigned a date profile to the appropriate transaction or item type in Customizing.

You define the date profile in the Customizing for **Service** under **Basic Functions** > **Date Management** > **Define Date Profile**.

Ensure that you have made the following settings:

- You have grouped together all dates, durations and date rules as well as [Reference Objects] that are valid in this transaction type (or in this item category).

- In the views **Screen Areas: Dates** and **Screen Areas: Duration**, all date types and duration periods have been assigned to the tab pages on which they are to appear.
- If you want to change dates and durations in a transaction, do not mark the **Display Only** field in the views **Screen Areas: Dates** and **Screen Areas: Durations**.
- The display format of dates is preset. You can choose whether you want to see a date, for example, with or without a time, in a transaction.

For more information about controlling dates, see the Customizing for **Service** under **Basic Functions** > **Date Management**.

Activities

When processing business transactions you can:

- Display dates

  The system displays all date types and durations that are assigned to the tab page in a table. If dates or durations have already been entered or calculated, the times are shown according to the display format set in Customizing.

  If a date rule is assigned to a date type, this too is displayed in the table.

- Enter dates

  You can enter dates and durations in empty fields.

- Change dates and durations

  You can change dates and durations by overwriting existing ones.

  If a date was calculated using a date rule, and you change the date in the transaction manually, the corresponding date rule is automatically faded out.

  If a date or duration has been marked as a display field in Customizing, you cannot change it in the transaction.

- Recalculate dates

  If you have changed a date or duration, the system recalculates all dependent dates. To do this, the system uses the calculation sequence assigned to the date type in Customizing.

  In Customizing, you can turn off the function responsible for recalculating date types (for example, the date type **Valid from** should not display the current date). In this case, the time for this date type is only calculated once and remains the same even if other dates are changed.

  You can select a new date rule for a date if this is permitted in the date profile. Permitted rules are displayed in the input help.

Related Information

[Dates in Service Orders and Service Quotations](#)
[Dates in Service Contracts and Service Contract Templates](#)

Availability Check

During the availability check (ATP check), the system checks whether a product contained in a service order can be confirmed as available (sufficient stock is available or can be produced or purchased on time). The required quantity of the product is reserved, and the ATP requirements are transferred to production or procurement.

Use

You can perform an ATP check for sales items and service part items. For service part items, the system performs a simulative ATP check only. During the check, the system doesn't lock the stock service parts available. That means that, for example, two users can in parallel access the same stock during the check. For sales items, also a non-simulative ATP check is possible: If the ATP type is set to C, the system temporarily locks the items during the check.

You set up the ATP check in Customizing under Service > Basic Functions > Availability Check. You can activate the availability check at item category level. To activate it, you first define an ATP profile and then assign this ATP profile to an item category.

Features

When you create or change a service order and enter a product with a requested delivery date, the data is transferred to the system where the ATP check is performed. The results of the availability check are returned to the service order.

When you create a service order, it consists of header data and one or more items. Each item has a requested delivery date and quantity (figure 1). After you've executed the availability check, the system displays the results in the fields Confirmed Quantity and Confirmed Date (figure 2).

Fig. 1: Service Order with Requested Quantity/Date (Before Availability Check)

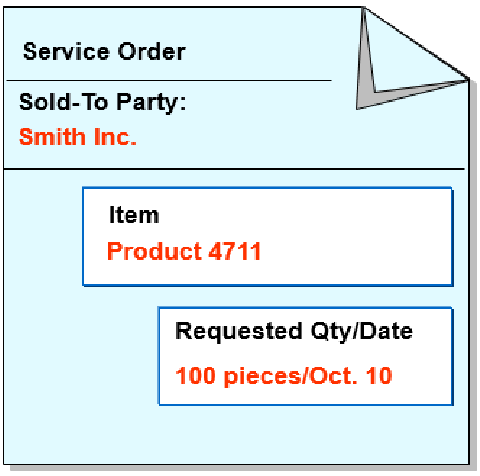
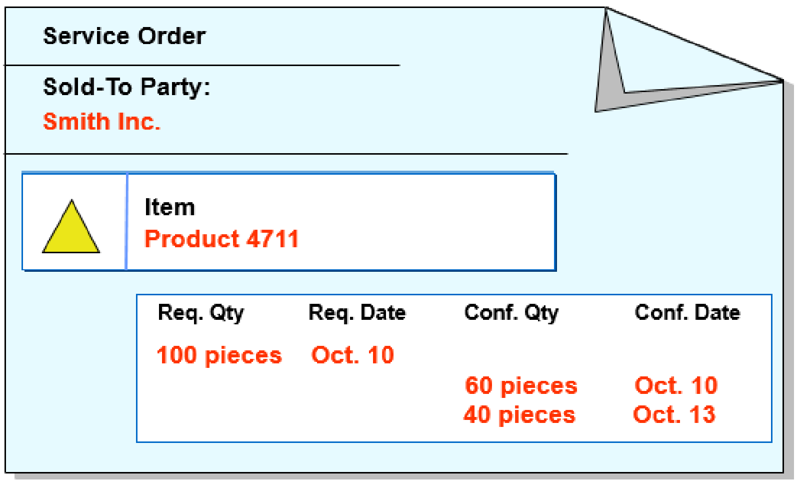


Fig. 2: Service Order with Confirmed Quantity/Date (After Availability Check)



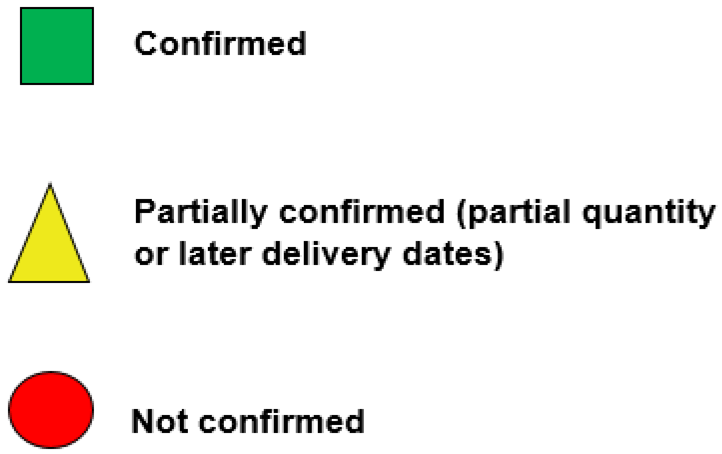
You can see the availability of an item in the Items assignment block in the Availability column (figure 3).

→ Tip

Before checking the current availability of the stock service parts, you need to restrict the item list to service part items.

The system indicates the availability of items using traffic lights. You can find detailed information about requested quantities/-dates and confirmed quantities/-dates in the item in the Schedule Lines assignment block.

**Figure 3:
Results of the Availability Check at Item Level
in the *Availability* Column**



More Information

For more information, see Schedule Lines.

Availability Check in Service Transactions

Use

If you perform material requirements planning, you can carry out an availability check (ATP check) in service transactions in Service.

Features

When you carry out the availability check, the system performs the following functions:

- Transfer of requirements to production or purchasing
- Reservation of stock
- Shipping and transportation scheduling

The system automatically triggers a new availability check when you perform one of the following actions:

- Change the requested quantity
- Change the requested delivery date
- Change the schedule line
- Remove the cancel reason code (sales items) or rejection reason code (service parts)

During an availability check, the system takes the following data into account:

- Order type

- Sold-to party
- Product
- Quantity
- Account
- Unit of measure type
- Requested delivery date
- Sales organization (at header level)
- Plant or vendor
- Ship-to party
- Batch
- Item category
- Requirement rule (fixed date or quantity)
- Shipping conditions
- Weighting

Constraints

The ATP check doesn't support or only partially supports the following functions and processes:

- All types of product substitution
- Configurable materials
- Checking of subitems

  Subitems created by an availability check in Service (for example, by a bill of material explosion) can't be checked. However, you can check manually created subitems individually for availability.
- Availability check against planned individual requirements
- Total requirements

  Individual requirements are supported.
- Availability check against product allocation

  Checking against product allocation is possible. However, it isn't possible against custom product allocation characteristics.
- Third-party order processing
- Start production from availability check
- Assembly from catalog
- Production
- Special stock (consignment and returnable packaging)
- Rush orders and cash sales

Specifics of Availability Check for Sales Items

To perform an ATP check for sales items, you must activate the business transaction events (BTEs) for the ATP check. To activate the ATP check, do the following:

1. Enter transaction `FIBF`.
2. Choose Settings > Identification > SAP Applications.
3. Enable the `CRMATP` application.

The material must be relevant for the availability check.

For information about the settings for the availability check, see the Customizing for Sales and Distribution > Basic Functions > Availability Check and Transfer of Requirements.

For sales items, you can use the following processes in combination with the availability check:

- Transportation and shipment scheduling to determine the material availability date
- Follow-up activities such as outbound delivery and goods issue
- Product allocation
- Backorder processing

> **Note**
> The system doesn't set the order status to **Newly Scheduled**.

Schedule Lines

Use

Dates and quantities for a service order item are stored in schedule lines.

Features

The following schedule line types are available:

- Requested Quantity/Date

  The request schedule line contains the quantity for the first requested delivery date for the customer.

- Confirmed Quantity/Date

  The confirmation schedule line contains the quantity that is deliverable for the date entered, according to the availability check.

The schedule line dates are assigned to specific time zones. In this way, you can always guarantee that you deliver the products to your customer at the requested time in the relevant time zone. The system copies the time zone from the central address data for the customer.

Activities

You can view the schedule lines for a service order item on the item overview page in the assignment block **Schedule Lines**.

Location Determination in the Service Order

Use

When you enter an item relevant for the availability check in a service order, location determination is performed automatically. Once the ATP system has determined the location (delivering plant), it is transferred to the service order in service solution. The ATP system executes the availability check on the basis of this location.

Process

The system determines the location for each confirmed item in the service order as follows:

1. The ATP system determines a default location based on the item category.
2. The item category has an ATP profile assigned to it.
3. The ATP profile is used to specify that the item category is relevant for the availability check.

For information on how to set this up, see Customizing under **Service** > **Basic Functions** > **Availability Check**.

If you make changes to an item in a service order, such as changing the product, the quantity, or the item category, the system runs a new availability check at this location.

Product Substitution, Listing, and Exclusion in Service Transactions

Material determination is the process of identifying alternative product IDs based on the entered product in a service transaction, and is a key functionality for implementing product substitution, listing, and exclusion. Below are the supported service transactions:

- Service orders
- Service quotations
- Service confirmations
- Service order templates
- Service contracts

For more information, see Product Substitution, Listing, and Exclusion

Text Management

Use

The definition of text objects and their text types are the basic building blocks of text management within applications in Service. All of the other text management Customizing activities are based on these definitions. There are various settings that you must make for each text object and text type combination. All applications in Service fully support SAPscript text, which can have basic formatting options. You can use the Text Format Customizing activity to control text formatting.

You define the text type to be used for a text object and the grouping of text types in Customizing for **Service** under **Basic Functions** > **Text Management**.

Note

Customizing text management in Service is application-independent. You have to perform the assignment to a text determination procedure for each application.

More Information

- Text Object
- Text Type
- Text Format
- Text Determination Procedure

Standard Texts in Service

If you want to use the same text repeatedly, you can create it as a standard text and then include it within other texts. The standard texts that you create are SAPscript-based. Unlike other texts, such as the text for a master record, these are not object-specific. When you include a standard text, it remain SAPscript-based if that corresponds to the respective text's format.

Creating a Standard Text

To create a standard text, proceed as follows:

1. Enter transaction `SO10`.
2. In the **Text Name** field, enter the required name for the text to be created.
3. For standard texts in the **Text ID** field, enter the identifier **ST**.
4. In the **Language** field, enter the identification for the language in which you want to create the standard text.
5. Choose **Create/Change**.

Inserting Standard Texts

To insert an existing standard text, you select the **Insert Standard Text** button in the text editor in service solution. A pop-up will open where you can search and choose a standard text. The list is filtered depending on the IDs that you have set in the **Define Text Determination Procedure** Customizing activity.

The text editor used is dependent on the settings of the text object and ID combination where you are inserting the standard text. For example:

- You are using a text object and ID combination with unformatted SAPscript-based text (i.e. plain text). the standard text is inserted without formatting. No formatting options are available for maintaining the text.
- You are using a text object and ID combination with formatted SAPscript-based text. The standard text is inserted and formatted the way it was when it was created. You can format the text further using the SAPscript formatting options

available for this text.

To display the **Insert Standard Text** button in the text editor in Service, go to the Customizing for **Service** under **Basic Functions** > **Text Management** > **Define Text Determination Procedure** and refer to the associated system help.

Text Object

A text object is used to define the connection between a business object, for example a transaction, and associated texts. SAP provides text objects which are preset to application programs. Do not delete any of the text objects contained in the standard system.

You can maintain objects in Customizing for **Service** under **Basic Functions** > **Text Management** > **Define Text Objects and text Types** (transaction SE75).

i Note

This Customizing is client-independent and thus is effective immediately in all system clients.

Text Type

Text types, also known as text IDs, represent different types of texts in a transaction. They are used to give the texts a business context. For example, you can customize the system so that you have customer hints and internal notes as available text types in a Service Incident.

SAP provides numerous text types, but you can also create new ones in the system. You can maintain text objects in Customizing for **Service** under **Basic Functions** > **Text Management** > **Define Text Objects and text Types** (transaction SE75).

i Note

This Customizing is client-independent and thus is effective immediately in all system clients.

More Information

For more information about formatting text, see [Text Format].

Text Format

Use

You can define text formatting in Customizing for **Service** under **Basic Functions** > **Text Management** > **Define Text Format** (transaction `CRMC_TEXT`). For each text object and ID combination, you can select or deselect the **Formatted** option to result in the following types of formatting.

1. **Formatted** deselected.

   No formatting (plain text): the text content is SAPscript-based without formatting. In Service, a text editor that does not provide any formatting options is used for such texts.

2. **Formatted** selected.

SAPscript-based and formatted: The text content is SAPscript-based and can be formatted. In Service, the text editor provides the formatting options for bold, underlined, italic, alignment and block text.

Features

Style and Form

In addition to the **Formatted** option, you can also use the **Style** and **Form** options. These are only applicable when you have selected the **Formatted** option.

If both **Style** and **Form** are assigned, the options defined in **Style** are used.

If neither are defined, the formatting options available are the ones described in the `SYSTEM` form.

Text Determination Procedure

The text determination procedure is used to represent a group of text types that you want to maintain or display in a business context.

Text Customizing in Service is application-independent, that is, you have to perform the assignment to a text determination procedure in each application in Service.

You perform the assignment for the transaction in Customizing by choosing **Service** > **Basic Functions** > **Text Management** > **Define Text Determination Procedure** (transaction `COMC_TEXT`).

Definition of Text Determination Procedure

Use

You use the **Define Text Determination Procedure** Customizing activity to define which text types are assigned to a business context.

This is available in Customizing for **Service** under **Basic Functions** > **Text Management** > **Define Text Determination Procedure** (transaction `COMC_TEXT`)

Here, you have to make additional settings for each text type, such as controlling text changes and the access sequence for texts.

More Information

For more information about **Define Text Determination Procedure**, see the system help associated with the Customizing activity.

Text Determination

Use

You can use text determination to define from where the system is to take a text.

Prerequisites

The communication structure is a prerequisite for text determination. For example, you enter a product in a transaction. The system passes this information onto text determination in the communication structure. Text determination can then be used, for example, to copy a text from the product master, providing that the product master is set in text determination.

Example

You have an internal dispatcher and cannot guarantee a delivery within 24 hours and want to describe this in an additional standard text. In this case, you can create your own function module, which checks the data from the communication structure for special cases and determines whether a corresponding standard text is to be used or a new text created.

The function module must correspond to certain rules, and function module CRM_TEXT_DETERMINE_TEXT serves as an example.

The new function module must have the same import and export data.

Type of Text Transfer

The following types of text transfer exist between the target text and the source text:

- ' ' = Not Yet Defined

  This is an initial value. It is only to be set in cases where no access sequence, and therefore no text determination, has been entered for the text type in the definition of text determination procedure.

- 'A' = Copy

  The system saves the current source text by using a new key for the target text. In this way, the system creates a copy of the current source text. The copy has no connection to the original source text. Therefore, changes to the source text do not affect the target text.

- 'B' = Save Reference Only

  The system saves a reference to the source text. The system reads the current version of each source text. As soon as the user changes the text for the target object, the reference is deleted.

- 'C' = Read Dynamically from the Template

  A text template is created by using text determination and is made available to the application. The text or reference is not saved.

Definition of Access Sequence

Use

In definition of the access sequence, you define the name for the access sequence in Customizing for Text Management. In addition, you define which text is to be used as the source text and the sequence in which the system searches.

Example

You have a ship-to party and cannot guarantee a delivery within 24 hours. You want to describe this in an additional standard text. In this case, you can create your own function module that checks data in the communication structure for special cases and determines whether a corresponding standard text is to be used or a new one created. The function module must correspond to certain rules and the function module CRM_TEXT_DETERMINE_TEXT serves as an example.

The new function module must have the same import and export data.

Integration

You assign an access sequence to a text type in the definition of text determination procedure.

Assignment of Fields

Use

In Customizing for text management, you use this function to define how the system finds the source text defined in the access sequence.

Integration

Since the field names can differ from one another, the fields in the communication structure must be assigned exactly to the fields in the key structure for the source text.

The **Sold-to Party** field in the transaction is called SOLD_TO_PARTY in the communication structure and PARTNER in the key structure.

Check Text Customizing for Consistency

Use

You can use this function in Customizing for text management to check whether your settings are consistent. You can limit the check specifically to text objects and text determination procedures.

Features

The following checking options are available:

- Text-specific
- With or without messages

  The system displays a list of messages as a result of the check.

  If the indicator **Also Positive Messages** is set, the system also displays those messages that contained no errors.

The following statuses exist:

- Red – Contains errors
- Yellow – Warning
- Green – No errors

Setting New Text Types

Use

You require this function to set new text types in **Text Management**.

Integration

For this function, you require the following procedures:

- Translating Text Types
- Translating Procedure Descriptions and Access Sequence Descriptions

Translating Text Types

Use

You want texts to be available in different languages.

Prerequisites

You are logged onto the system in each language required

Process

1. Select a text object and choose **Text IDs**.

   The screen **Change Text IDs for Object** appears.

2. Enter your translation.

3. Save your changes.

Translating Procedure Descriptions and Access Sequence Descriptions

Use

You want to translate the procedure and access sequence description. You can do this directly for each Customizing activity in Customizing for **Text Management**.

Process

1. Choose **Tools** > **Choose Language on the detail screen for procedure or access sequence.**

   Select the required languages

2. Select a procedure or an access sequence and choose **Goto** > **Translation**

3. Enter the text in the language selected previously

4. Save it

Attachments

Attachments in service transactions may include documents, images, or other files, such as inspection reports, photographs, or product manuals, which provide additional context or detailed instructions throughout the service delivery process. The attachments are important for record-keeping, ensuring the accuracy of data, and enhancing the overall service process by providing comprehensive information.

Service uses different attachment technologies to manage attachments of service transactions.

[Harmonized Document Management](#)

You can use Harmonized Document Management (HDM) to manage attachments from different document frameworks, including Document Management (DMS), Generic Object Services (GOS), and ArchiveLink, in a single user interface.

[CRM Content Management (Legacy)](#)

Only use CRM content management if you've started to use this feature in an earlier release. CRM content management enables you to add documents, graphics, or multimedia objects as attachments to business objects such as service transactions, service requests, or business partners.

Harmonized Document Management

You can use Harmonized Document Management (HDM) to manage attachments from different document frameworks, including Document Management (DMS), Generic Object Services (GOS), and ArchiveLink, in a single user interface.

HDM harmonizes the process of handling attachments across applications that use these frameworks. By default, the DMS document framework is the recommended and primary solution in HDM, as it offers a more robust feature set for document management compared to other technologies.

Service transactions support the following document frameworks:

Service Transaction	Supported Document Framework
Service order	DMS, GOS, ArchiveLink
Service confirmation	GOS
Service contract	GOS
Service request	GOS
Service quotation	GOS
In-house service	DMS

Processing Attachments

In Service apps, you can manage attachments for service transactions and items under **Attachments**. This assignment block allows you to do the following:

- Upload attachments
- Add URLs as attachments

i Note

Only the URL is saved, not the document itself.

- Download and delete attachments
- Rename attachments
- Define the sequence and filters for displayed attachments

You can also use the OData API **Attachments** (API_CV_ATTACHMENT_SRV) to create, read, rename, and delete attachments.

For more information, see Attachments.

Storing Attachments

All documents in Service are saved to an SAP HANA database by default. If you're planning to store larger volumes of data, SAP strongly recommends using an SAP-certified external content server.

To store attachments outside the database to free up database storage, you can use transaction code `OAC0` (**CMS Customizing Content Repositories**).

If you use GOS, the system stores attachments in table `SOFFCONT1`.

For more information about the attachment storage, see SAP Note 3051373.

For more information on content repositories, see Attachment Service.

Archiving Attachments

Attachments are archived together with their related service transactions.

For more information, see Data Archiving in Service.

Configuration

To use HDM in your business apps, configure the relevant settings in Customizing for **ABAP Platform** under **Application Server** > **Basis Services** > **Harmonized Document Management** > **Activate HDM for Attachment Object Types**.

In this activity, set up the following:

- **Activate HDM for Attachment Object Types: Overview**

  Activate HDM for your attachment object types.

  The attachment object type uniquely identifies the SAP object type or SAP object node type. In Service, the following attachment object types are available:

SAP Object Type / SAP Object Node Type	Attachment Object Type
In-House Service	`INHOUSEREPAIR`
Service Order	`SERVICEORDER`
Service Item	`SERVICEORDERSERVICEITEM`
Expense Item	`SERVICEORDEREXPENSEITEM`

SAP Object Type / SAP Object Node Type	Attachment Object Type
Service Part Item	SERVICEORDERSERVICEPARTITEM
Sales Item	SERVICEORDERSALESITEM
Execution Order Item	SERVICEORDEREXECUTIONORDERITEM
Tool Item	SERVICEORDERTOOLITEM

- **Activate Document Frameworks**

  Activate the document frameworks GOS and DMS for your attachment object types.

- **Maintain Default and Restricted Document Types**

  Set a default framework (for example, DMS) and maintain the document types, such as SRV for DMS and GOS for GOS. To define the DMS document types, use the activity **Define DMS Document Types** in Customizing for **Cross-Application Components** under **Document Management** > **Control Data**.

For more information, refer to the documentation of the corresponding Customizing activities.

Related Information

Attachment Service
Harmonized Document Management (HDM)

CRM Content Management (Legacy)

Only use CRM content management if you've started to use this feature in an earlier release. CRM content management enables you to add documents, graphics, or multimedia objects as attachments to business objects such as service transactions, service requests, or business partners.

You can use CRM content management to do the following:

- Link content and business objects

  The system doesn't store content objects (documents and folders) as standalone objects. It automatically assigns content objects to business objects. You can then use these business objects to access and maintain content objects.

- Link internet content and business objects

- Use third-party content servers

In addition, CRM content management provides the following functions under **Advanced**:

- Metadata management under **Properties**

- Content versioning under **Versions**

- Managing folders and grouping related content in folders

- Where-used list

Configuration

Implementing CRM content management

You set up CRM content management in Customizing for Service under Basic Functions > CRM Content Management.

Configuring the Attachments assignment block

Attachments that are processed with CRM content management, are displayed in a separate assignment block. If this Attachments assignment block isn't available on the UI, you can add it from the UI Component Workbench. To do so, call transaction `bsp_wd_cmpwb`, enter the component you require, for example `SRQM_INCIDENT_H` for service requests, and from the list of available assignment blocks, add `GS_CM` to the header or overview as required.

> **i Note**
>
> The Activity and Opportunity sales document types only support CRM content management.

Managing Attachments

Get to know how to add and process attachments using CRM content management.

Configuration

By default, you can only add attachments that have a size less than 100 megabytes. However, you can create your own size restrictions for attachments in Customizing for Service under Basic Functions > CRM Content Management > Business Add-Ins (BAdIs):

1. BAdI: Confirmation of Changed Documents (`CRM_DOCUMENTS`):
    - Use the `CARRY_OUT_ACTION` method to restrict the number of allowed attachments and/or the size of these documents.
    - Use the `GET_FILE_INFO` method to find the attributes of the file
2. In the BAdI: Authorization Check on Document Level (`CRM_DOC_AUTHORITY`), implement the following for `PREPARE_DOC`:

    Get the file size in bytes: `lr_upload TYPE REF TO cl_thtmlb_fileupload.`

    Restrict the upload: `lr_upload->file_length.`

    For more information about using this BAdI, see SAP Note 720434

If you do want to attach documents without size limitations, use the `RZ11` parameter `icm/HTTP/max_request_size_KB`. If you want to remove the file size check, set this parameter to `-1`.

Adding Attachments

You can add attachments to business objects, for example service orders, by performing one of the following actions:

- Select the attachment from CRM content management or from your local hard disk.
- Enter a corresponding URL.

Adding attachments

1. In the **Attachments** assignment block, click **Attachment**.
2. Use the document search to select a document from CRM content management, or browse your local hard disk to find the document that you want to attach.

   Under **Search: Document from Content Management**, you can search for documents using attributes or text. After you have found the required document, you can display it in the corresponding editor and, if necessary, assign it to the business object (for example, a service transaction).

   You can enter a name and description for the document. Otherwise, the attachment uses the document file name.
3. To attach the document, click **Attach**.

i Note

Alternatively, you can use Drag and Drop to upload attachments.

The system creates references to documents that are stored in CRM content management.

Adding URLs that refer to documents

1. In the **Attachments** assignment block, click **URL**.
2. Enter the URL of the document that you want to attach, and define a name for it. Optionally, you can enter a description of the document.
3. To attach the document, click **Attach**.

The system only saves the URLs and not the documents.

Processing attachments

You can process attachments as follows:

- **Change properties of attachments**: Change the properties of attachments, for example, name, description, keyword, or language.
- **Create folders**: Create new folders. If you highlight an existing folder and then choose to create a new folder, the system creates the new folder as a subnode in the folder hierarchy.
- **Create hierarchies**: Create hierarchies of folders and attachments. For example, you can highlight a folder and attach a document or URL to the folder. You can also highlight an existing attachment and attach a new document or URL to the hierarchy.

  #### i Note

  Hierarchies are business object-specific. If you organize attachments into a hierarchical structure within one business object (for example, a service order), this structure is not inherited by any other business object (for example, a service contract).
- **Delete attachments**: If you delete attachments, the links to the documents are deleted, but not the documents themselves.
- **Delete folders**: If you delete folders, the system deletes all subfolders and attachments in these folders.
- **Copy, cut, and paste folders and attachments**: Use copy, cut, and paste functions to restructure the hierarchy of the attachments.

  You can also copy an existing attachment and save it under a different name and copy cut content to another attachment.

- **Check documents in and out**: Check out a document that is locked by other users and edit it. After you have saved your document locally, you can check in the document that you have edited.

Version Management

The system distinguishes between **logical** and **physical** documents. A logical document comprises one physical document or several physical documents with different versions. This could be different language versions, content versions, or format versions.

When you check out a document and subsequently check the same document in again, the system creates a new version of that document.

Related Information

[Adding Attachments]

Setting Up Content Servers

All documents in CRM content management are saved by default to the HANA database. If you are planning on storing larger volumes of data, SAP strongly recommends using an SAP-certified external content server.

To use an external content server, you must define a minimum of one new content repository and storage category in the Knowledge Provider (KPro). Use document area `CRM`.

> **Note**
> You can use report `RSIRPIRL` to migrate existing documents to your content server.

For information, see [SAP Content Server].

Configuration

You set up data storage in Customizing for **ABAP Platform** under **Knowledge Management > Settings in the Knowledge Warehouse System > Document Management Service > Define Standard Category**. There, assign the classes `BDS_POC1` (business partner) and `CRM_P_ORD` (business transactions) to the storage category you have created.

Under **HTTP Service Hierarchy Maintenance** (transaction code `SICF`), activate the SICF service `/sap/bc/contenserver` by creating an external alias `//default_host/sap/bc/contentserver` which contains anonymous logon data and refers to the target element `//default_host/sap/bc/contentserver`.

> **Note**
> To display documents, the system requires anonymous logon data for the service `//default_host/sap/bc/contentserver` for HTTP access to the content server.

You activate the service `//default_host/sap/bc/contentserver` as follows:

1. Under **External Alias > Virtual Hosts/Services**, select **default_host > /sap/bc/contentserver**.
2. On the **Logon Data** tab, select the **Required with Logon Data** procedure. Then, enter your logon data.

   Ensure that you have a service user or a dialog for content servers.

Data Storage and Archiving of Attachments

Data Storage and Archiving of Attachments explains how attachments processed with CRM content management are stored and archived. It also provides information about how to free up database storage and display attachments of archived business objects. It's useful for managing and organizing large volumes of data.

Saving Documents in a Data Storage

Attachments that are processed with CRM content management are stored in the following tables:

- `CRMORDERLOIO`
- `CRMORDERPHIO`
- `CRMORDERCONT`

When an attachment is transferred to CRM content management, it's actually stored in a content repository, which is a logical unit within a storage system. A content repository can physically reside in a database, a file system or the ArchiveLink.

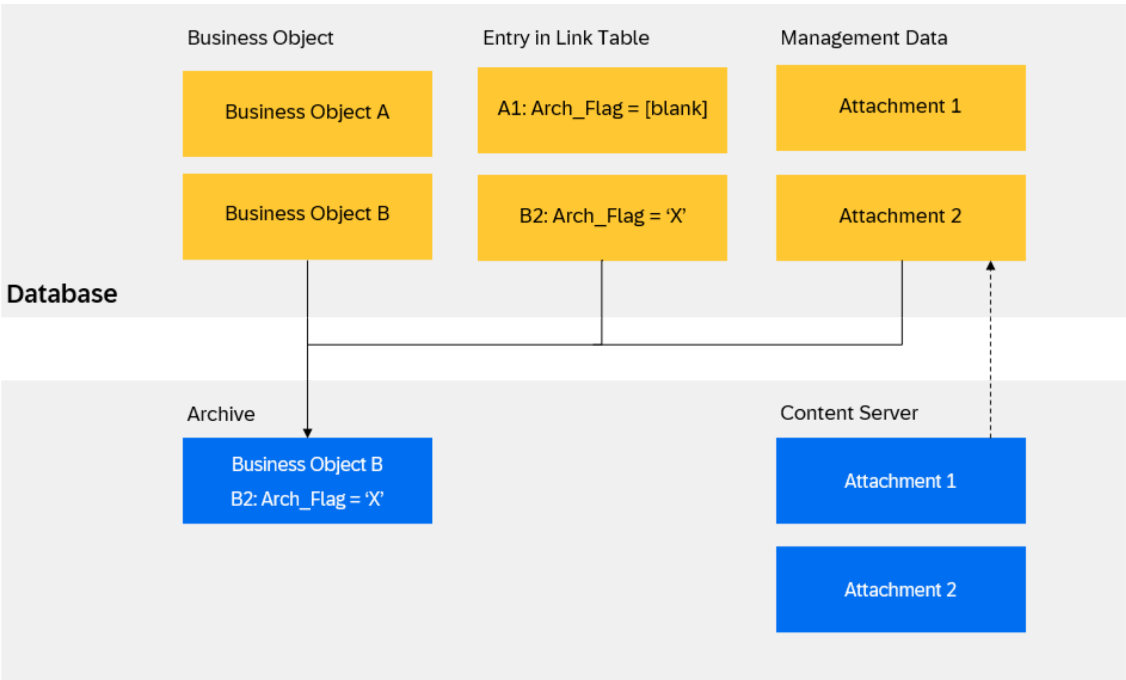
If you want to store attachments elsewhere to free up database storage, you can use transaction code `OAC0` (**CMS Customizing Content Repositories**).

Archiving Attachments

Attachments that are processed with CRM content management are archived along with the business objects to which they're attached. The attachment itself is never archived, because it has already been saved securely in a storage system. During archiving the corresponding link entry is marked automatically with the archiving indicator `ARCH_FLAG`. The management data is archived and then deleted.

You can display attachments of archived business objects at any time, as long as the management data and the attachment still exist.

The following graphics shows the archiving of a business object and its attachment:



For more information about archiving in Service, see Data Management in Service.

Integrating Product Lifecycle Management (PLM)

Integrating SAP Product Lifecycle Management (SAP PLM) lets you link documents from the PLM Document Management (DMS) with business objects, for example, service transactions in Service.

Configuration

You activate SAP PLM for use with Service in Customizing for **Service** under **Basic Functions** > **Content Management** > **Activate PLM Integration**.

Features

You can use the PLM DMS to link PLM documents directly to business objects or to delete the links:

- Business objects

  If you assign a PLM document to a business object, it appears in the **PLM Documents** folder.

- Finding PLM documents

  You can use the following search attributes for PLM documents:

  - User-defined text
  - Document number
  - Document type
  - Version number
  - Status
  - Description
  - User
  - Part

Note

For PLM documents, the **Creator** field in Content Management specifies the person responsible from PLM. The full-text search only searches the document long text and not the content of the document itself.

Advanced Variant Configuration

You can use this function to configure products in service orders. This means that you can offer products, with various characteristics that are managed in their own master records (product model).

Service uses Advanced Variant Configuration (AVC). The variant configurator offers you a toolset for managing customizable products with process and data integration.

For more information, search for *Advanced Variant Configuration* in the product assistance on SAP Help Portal.

Prerequisites

You must be assigned to the following roles to use the variant configurator:

- SAP_BR_PRODUCT_CONFIG_MODELER
- SAP_BR_INTERNAL_SALES_REP

The product is marked as configurable in the material master data.

Related Information

Configurable Products and Product Variants

Output Management

Use

SAP S/4HANA and SAP S/4HANA Cloud Private Edition support various output management frameworks. The following output management frameworks are used for Service:

- Conditions for Output
- SAP S/4HANA Output Control

Further Information

For more information about output management for service orders, service quotations, and service contracts, see Output Management for Service Orders, Service Quotations, and Service Contracts

Output Management for Service Orders, Service Quotations, and Service Contracts

Business context information about output management for service orders, service quotations, and service contracts.

Use

SAP S/4HANA output management enables you to use output functions such as printing, sending e-mail, or editing form templates for service orders, service quotations, and service contracts.

For more information, see SAP S/4HANA Output Control.

Output Management Application Object Types

Business Transaction Object Type	Output Management Application Object Type
Service Order	`SERVICE_ORDER`
Service Quotation	`SERVICE_QUOTATION`

Business Transaction Object Type	Output Management Application Object Type
Service Contract	SERVICE_CONTRACT

Output Types and Corresponding Form Templates

Business Transaction Object Type	Output Type	Form Template
Service Order	SERVICE_ORDER_SUMMARY	CRMS4_SERVICE_ORDER_SUMMARY
Service Quotation	QUOTATION_SUMMARY	CRMS4_SERVICE_QTAN_SUMMARY
Service Contract	SERVICE_CONTRACT	CRMS4_SERVICE_CONTRACT_DEFAULT

Activate Output Management.

You can activate output management for each service object and the corresponding application object type under Cross Application Components > Output Control > Manage Application Object Type Activation.

→ Remember

You can't use output management and PPF (Post Processing Framework) for output control at the same time. If you activate output management, the following PPF actions can't be scheduled:

- External Communication
- Smart Forms Fax
- Smart Forms Mail
- Smart Forms Print

i Note

- When you switch from output management using PPF to output management using output control, existing documents use PPF actions for output control. Conversely, if you switch from output management using output control to output management using PPF, existing documents use output control actions.
- If output management is activated for an object, the buttons **Print** and **Print Preview** aren't available anymore (these buttons are based on PPF actions). Instead, a new assignment block **Output Control** becomes available where you can trigger output per output request line.

For Service Quotations

A button **Send to Customer** becomes available on the service quotation header if you activate output management. This button can set the status of the service quotation to **Released** (only possible if the status of the service quotation is error free) and trigger all available output request lines. Using this button is optional. You can also set the status of the service quotation header directly and then use the assignment block **Output Control** to trigger the output. With the rules for output relevance, you can ensure that an output to the customer is only triggered after a service quotation is released.

Schedule Output for Service Contracts

To trigger the print output function, you can schedule output for the service contract after saving it. You can use the report **Service Documents Output Run** (`CRMS4_SERVICE_OUTPUT_RUN`) to select the relevant output items scheduled for print and send them to the print queue.

Output Management tracks the status of each output to be issued for a service contract. The following statuses can be generated by Output Management during the process of creating and issuing the output:

Value	Status	Description
1	In Preparation	Initial status assigned to the output when created
2	To Be Output	Output is being prepared for processing
3	Completed	Output has been processed

Related Information

Output Management

Conditions for Output

Use

You can define transportable conditions for output issue in Customizing for output determination.

You use the attribute of the business object type or the date profile assigned to the transaction type when determining transportable conditions. For the definition of conditions, use the condition editor from **SAP Business Workflow**.

Prerequisites

You have defined an action profile in which you have defined the required output types and assigned these to output mediums (processing types).

You can define conditions in the screen area **Schedule Condition** for each condition in this overview.

An output medium and detailed information are given for each task in the screen area **Actions and Processing**. The details for the processing method (time of processing) and output medium are copied from the action profile. You must enter detailed information here (recipient country, spool settings, or Smart Forms).

The system checks the conditions for each form of output (for example, is e-mail the standard communication type in the business partner master?). If the conditions are fulfilled, it issues output using the given output medium. It then checks the next condition.

If a condition is marked with a **Stop** symbol, the system does not continue to the next condition after issuing this output. It only checks the next condition if the conditions in the first line are not fulfilled. This stops you issuing output, for example, by fax after an e-mail has been sent, because the conditions for sending e-mail and fax were applicable to both.

More Information

Conditions for Scheduling and Starting Actions

Output Determination

Use

In applications in Service, the **Output Determination** component enables you to:

- Print documents
- Fax documents
- E-mail documents

Output determination uses conditions that you define in Customizing. You can define conditions for each output type and processing medium.

Integration

The **Output Determination** component uses the Post Processing Framework (PPF). You can use this component to automatically trigger output, follow-on documents, and workflows. The PPF calls the component **Output Control**.

You design the forms for your output under **SAP Smart Forms** (transaction `smartforms`).

Features

You can control output determination by using:

- Simple determination

  You do not need to define your own requirements in Customizing for simple determination.

- Conditions (for Output)

  When determining conditions you can select the transmission medium. You can also define the access sequence for several transmission media.

- Transportable conditions

  You use the conditions for action determination when determining transportable conditions.

If the conditions are met, the system creates the output:

- Immediately
- After the document has been saved
- After running a selection report (`RSPPFPROCESS`)

The action monitor gives you an overview of output that has been issued successfully. You can trigger repeat prints from the monitor (for example, for errors).

Constraints

You cannot issue output for items. When an item is changed, the complete document is reissued.

More Information

Conditions for Scheduling and Starting Actions

Issuing Output from Transactions

Prerequisites

In Customizing for **Service**, you have created an action profile with an action definition for issuing output in the Customizing activity **Define Action Profile**.

- You have selected the following fields in the action definition:
  - **Schedule Automatically**
  - **Execute in Dialog**
- You have selected the following fields in the **Processing types** view:
  - **Smart Forms Print**
  - **Smart Forms Fax** or **Smart Forms Mail**

Context

You can manually issue output for transactions, for example, service quotations, directly from transaction processing.

To do this, you can use the output media in **Service**, print, fax, and e-mail.

Procedure

1. Create a transaction.
2. Click **Output**.

Results

The action is executed according to the settings in Customizing.

The status of the action changes to **Output was triggered successfully**

Next Steps

Conditions for Output

Printing Output Using the Action Monitor

Use

You can use the action monitor to display and print several types of output together. If errors occur during printing, you can also repeat the printing of output.

Prerequisites

You have made the following settings in Customizing for output determination:

- Under **Define Action Profile**, you have selected the processing type **Smart Forms Print** for the action definition and entered a form.

For more information about output control, see the Customizing for output determination.

Procedure

To print documents that have errors and that have not been printed to date, proceed as follows:

1. Call up transaction SPPFP.

   The screen **PPF: Selection and Processing of Actions** appears.

2. Enter selection criteria, for example:

   - In the **Action Status** field: **0** - unprocessed and **2** – with errors
   - In the **Processing Type** field: **PRN** - Print
   - In the **Processing At** field: Choose **Processing using selection report**

3. Enter additional selection criteria as required to refine the selection.

   - If you only wish to print output of a particular action definition (for example, only send by e-mail), enter this action definition in the **Action Definition** field.
   - If you only want to print output for a particular action profile (for example, output only for opportunities), enter the appropriate action profile in the **Action Profile** field.
   - If you want to print output for a particular transaction document, enter the document number in the **Application Key** field.

4. Choose **Execute.**

   You get a list of the output created in the system up to the date you entered and that has either not yet been processed or has not been processed successfully.

   To display output of a selected transaction on the screen, choose **Preview**.

5. Select the required output and select **Execute**.

Result

The output is printed out by the printer defined in Customizing for output determination.

Related Information

[Action Monitor]

Actions

Use

Actions are important for maintaining and improving business relationships. You can schedule and start predefined processes with the **Actions** component by means of user-definable conditions from transaction and marketing objects.

You can tailor the type and time of actions to the requirements of your customers and the processes in your company. This component enables you to match your service management closely to the needs of your customers, and, simultaneously, to automate it.

The **Actions** component is also used for output determination.

Integration

Actions in Service use the **Post Processing Framework** (PPF) component as the uniform interface for different processing methods.

Date types and date rules for Date Management in service solution are used for time-dependent conditions.

The following components use the **Actions** component in service solution:

- Service Request

Features

You can define actions dependent on conditions so that the system automatically schedules and starts them when the conditions are met.

Using actions you can:

- Create follow-up transactions automatically
- Execute changes in the transaction currently being processed
- Trigger output

The system processes the actions automatically. You can, however, also schedule and start actions manually. You can use the attributes of the transaction type used in the business object as conditions. You can create time-dependent conditions using the dates and date rules from the appropriate date profile. You can also define partner-dependent actions, for example, to send a reminder e-mail to the employee responsible.

The system uses different processing types of the PPF during processing. There are various processing types for actions:

- Methods (Business Add-Ins)

  This is suitable for simpler processes, such as follow-up transactions or creating positions, for example, calling a customer when you have received a cancellation.

  Templates are delivered for defining methods. You can use these to create as many follow-up transactions as you require, and to define your own methods.

  The copy control is used when processing methods. You can use the BAdIs in the copy control to influence follow-up transactions freely.

- SAP Business Workflow

  This is suitable for more complex processes, for example, a follow-up transaction that includes an approval process.

- Smart Forms

Use this for printing, faxing or e-mailing documents.

More Information

[Processing Actions in Transactions]

[Date Management (CRM-BF-DAT)]

[Follow-Up Transactions]

[Output Determination]

The Post Processing Framework in Service

Use

The Post Processing Framework (PPF) is used in Service to schedule and process actions and outputs with an application document (such as a contract or an activity).

Integration

The PPF provides basic functions for the settings that are supplied with default settings by the relevant application. You can adjust the settings to meet your requirements. In service solution, by transaction type, you can create separate actions with schedule and start conditions.

Features

The PPF provides the following functions:

- Determination of actions and messages
- Merging of actions found
- Media for processing actions
- User interface for the Customizing of action templates, including determination and merge technologies and also processing media
- User interface for the Customizing of conditions, when determination using conditions is used

Controlling Actions

Use

You can set up actions in Customizing according to the demands of your company's processes.

Prerequisites

- You know the business object type of the transaction type or item category for which you want to define actions. You find the business object type in Customizing of transaction types.

- To define date-dependent actions, you must have defined date profiles in Customizing for **Service** under **Basic Functions** > **Date Management** > **Define Date Profile**. For more information, see Date Management.
- You have made the necessary settings for actions in Customizing for **Service** under **Basic Functions** > **Actions**.

Features

You use general attributes to control actions. For each action, you can define the following:

- When processing is to take place (processing time)
- Whether scheduling is to be automatic as soon as the schedule condition has been fulfilled
- Whether changes can be made in the document to condition and processing parameters
- Whether it can be started from the action list in the document
- Whether it depends on certain parameters, that is, it can only be executed for certain business partners
- Processing Types

  Several processing types are available in Service :
- Methods (Business Add-Ins)
- Workflow
- Smart Forms
- Conditions

  You can define one or more schedule and start conditions in the action definitions within a profile.
- The assignment of action profiles to transaction types or item categories

  You always define actions in an action profile, and assign the action profile to a transaction type or to an item category. As a result, these actions are always available in a document of this transaction type or in an item of this category.

Processing Time

Use

The processing of an action can take place at different times.

You can define when the started action is to be processed, that is, when the system actually executes the action, such as when the system is to create a follow-on transaction or print output.

In Service, the following possibilities are available for the time an action is processed:

- Immediate processing

  The action is started as soon as the start condition is fulfilled.
- Processing when saving the transaction.

  The action is started when the transaction is saved.

Integration

The processing time Process Immediately has consequences for the Display in Toolbar function.

If you choose an action with processing time Process Immediately from the application toolbar, the action is started immediately if the start condition has been met.

i Note

Use this setting to start actions directly from transaction processing. Define an action definition by using the following attributes:

- Select Display in Toolbar.
- Select the processing time Process Immediately.
- You can start this action directly from transaction processing because the start condition is always regarded as fulfilled.

Activities

Set the processing time in Customizing for Service, by choosing Basic Functions > Actions > Actions in User Management > Change Actions and Conditions > Define Action Profiles and Actions.

Scheduling Actions

Use

You can schedule actions during transaction processing. Click Schedule Actions to display available actions.

Prerequisites

To activate Schedule Actions, the following criteria must be met:

- The actions must be inactive, that is, set to be executed manually
- The actions cannot be print actions. Print actions are displayed either on the Print or Scheduled Actions pushbuttons in the toolbar
- At least one action must be set to Display in Toolbar

Scheduled Actions

Use

The Scheduled Actions assignment block displays all actions that have been scheduled, either manually or automatically.

Features

The following information is displayed in the Scheduled Actions assignment block:

- Status

  A colored icon indicates the status of the action

- Action Definition

  A hyperlink that leads to a detailed list of action definition information

- Processing Type
- Created By
- Status

  A hyperlink that leads to a detailed list of status information

- Processing
- Created On
- Created At
- Executable

  Indicates whether the action is yet to run or has been completed

Activities

In the header bar of the **Scheduled Actions** assignment block, you can do the following:

- Schedule New Actions

  Click this to open a dialog box for scheduling a new action

- Repeat

  Click this to copy a selected action

- Action Details

  Click this to see details of the selected action

- Determination Log

  Click this to see details of the determination log for the selected action

- Storage System

  Click this to open a dialog box for specifying the location for archiving documents

More Information

Actions

Conditions for Scheduling and Starting Actions

Use

In Service, you can make the schedule and start of actions dependent on conditions. Actions can only be scheduled when the schedule condition has been met and started, and when the start condition has been met.

You can adjust the conditions for scheduling and starting actions to meet the requirements of your company. When doing this, you can use the attributes of the business object you used from the Business Object Repository (BOR), as well as dates and rules of a date profile as condition expressions.

> **Note**
>
> If you do not want an action to be scheduled or started depending on conditions, do not define schedule or start conditions for it. The condition for processing a document is then regarded as fulfilled.

Prerequisites

You always define the conditions for an action definition of a particular action profile. In the definition of action definitions and action profiles, you have specified the business object type and the date profile whose attributes you wish to use for the definition of conditions.

Activities

The determination of conditions is described in Determining Conditions.

Changing Processing Parameters

You can change the processing and condition parameters during transaction processing. An integrated search help is also available, for example, allowing you to search for available printers. You can change the following processing parameters:

- Target document

  This allows you to specify what type of follow-up transaction you want to create using an action, that is, what transaction type

- Printer (using processing type smart forms print)

- Fax number (using processing type smart forms fax)

- E-mail address (using processing type smart forms mail)

  For example, you can change the print that the system proposes, or the spool parameters, or the fax number or e-mail address, for example, if a customer calls and asks the call center employee to fax a quotation to a different fax number than normal because the normal fax machine is out of order, or the type of subsequent document to be created.

  > **Note**
  >
  > You can only change action parameters as long as the action is unprocessed, and only if the Customizing settings for the action allow manual changes. Otherwise, you can only display the action parameters. The changes you make during transaction processing do not overwrite settings made in Customizing.

Determining Conditions

Use

You can define conditions for the schedule and start of actions according to your requirements.

Prerequisites

You have created action definitions in Customizing for Service. You have four options when creating action definitions:

- You use the wizard under Basic Functions > Actions > Actions in Transaction > Create Actions with Wizard
- You use the wizard under Basic Functions > Actions > Actions in User Management > Create Actions with Wizard
- You use the basic transaction under Basic Functions > Actions > Actions in Transaction > Change Actions and Conditions > Define Action Profile and Actions > .
- You use the basic transaction under Basic Functions > Actions > Actions in User Management > Change Actions and Conditions > Define Action Profile and Actions > .

Process

Calling Condition Definitions

1. Select Basic Functions > Actions > Actions in User Management > Change Actions and Conditions > Define Action Profile and Actions > . .
2. To select the action profile for which you would like to define conditions in the selection area, double-click it.

   The actions appear with the conditions in the list assigned to the action profile.

   The first time you call up the action profile you have to import the action definitions to the list via Create.

   The following information is displayed in the list:

   - Test status indicator

     Green means that the condition has been defined without errors. Red indicates mistakes.

   - Name of the action definition
   - Number

     This tells you the sequence in which the actions will be executed

   - Processing type

     This tells you which technique is used to process actions (for example, workflow or method call)

   - Name of processing (for example, send quotation, create telephone call or maintenance contract or collect purchasing data for campaign)
   - Name of schedule condition
   - Name of start condition
   - Stop indicator

     The stop indicator ensures that the conditions of following actions are not evaluated if the condition is fulfilled.

   You can execute the following functions in addition to the functions in the standard list:

   - Create an action template according to an action definition.

     You can select the action definition from the symbol list box.

   - The following functions appear when you have selected an action template:
   - Change selected action template

- Delete selected action template
- Copy selected action template
- Stop indicator to place or remove the condition for the selected action template
- Call up error log for the selected action template
- Refresh list
- Transport action template

  i Note

  If you call the delete and transport functions using the **Action Profiles** screen area, you obtain all action templates that are assigned to the selected action profile.

Conditions for Defining Action Templates

1. On the **Schedule Condition** tab page, enter a name for the schedule condition.
2. Go to change mode.
3. Enter the conditions.

   For more information about this procedure, search for *Condition Editor* in the relevant version of SAP NetWeaver on SAP Help Portal.

   i Note

   If you use system fields, note that the date and time fields in the system are of a different format to the dates in the date profile. For this reason, use the current date of the date profile as comparison date, instead of the system date.

4. On the **Start Condition** tab page, repeat steps 1 to 3.
5. Save your changes.

Changing Action Templates

1. If you do not want to adopt the settings from the action definition for the action template, deselect the option **Default Setting from Action Definition** on the **Overview** tab page. You can then change settings.
2. If actions are to be scheduled automatically when the schedule condition is fulfilled, select **Schedule Automatically**.
3. Save your changes.

   i Note

   We recommend that you retain the **Default Setting from Action Definition** and that you make changes to the action definition as required.

Processing Actions in Transactions

Use

You can use **Actions** to schedule and start follow-on processing steps, depending on the transaction type and different schedule and start conditions.

Prerequisites

You have done the following in Customizing for Service, under: Basic Functions > Actions :

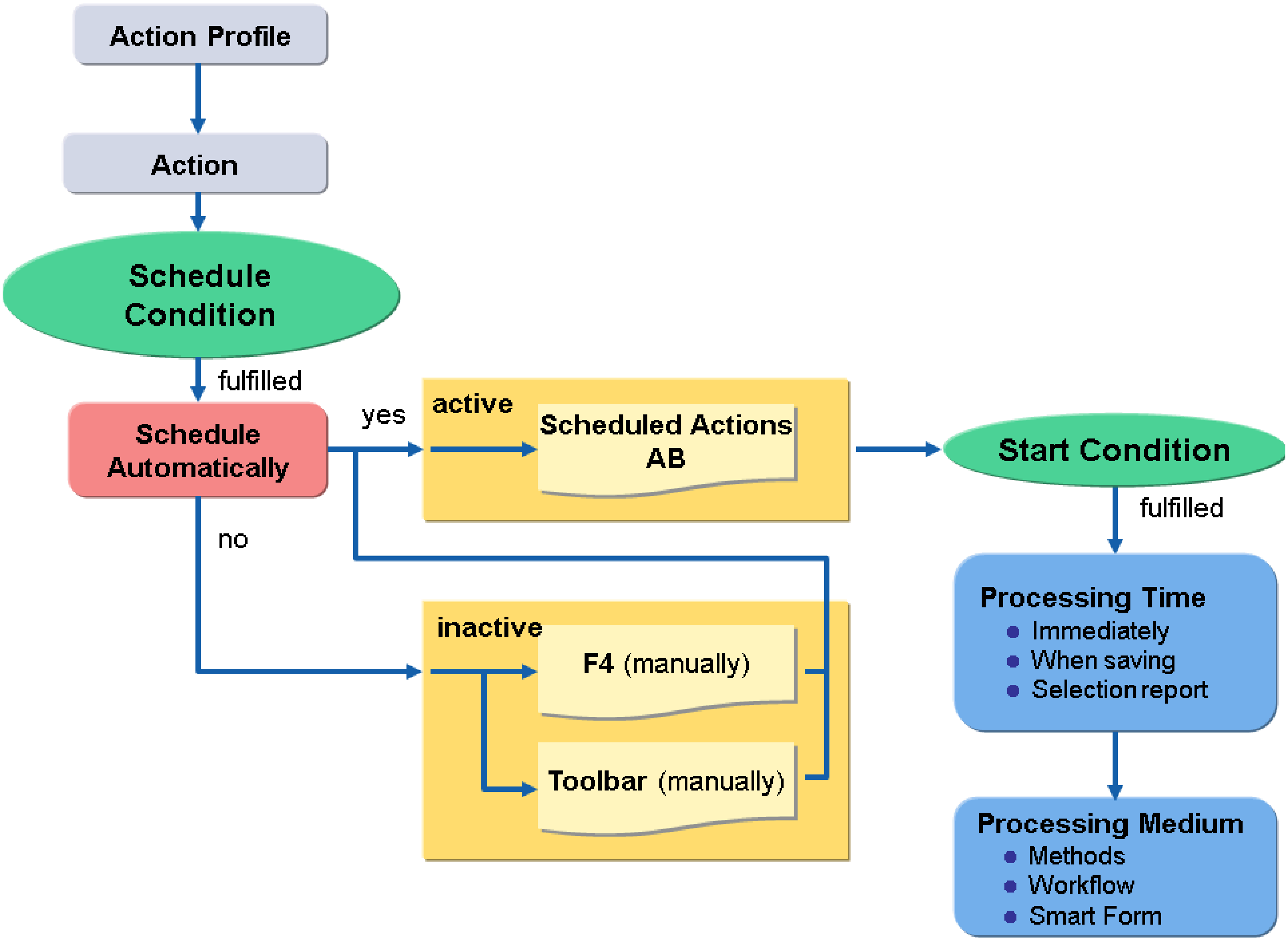
- Created an action profile with action definitions

  You define the attributes of the actions in action definitions, such as the processing time, or whether the action may be changed in the transaction.

- Defined conditions (schedule and start conditions) for action types
- Assigned the action profile to the relevant transaction type

For more information about controlling actions, see controlling actions.

Process



The processing of actions can be structured in three segments:

- Planning Actions
- Starting Actions
- Monitoring Actions

Planning Actions

1. The system uses the action profile assigned to the transaction type to check which actions are possible for the transaction.
2. The system checks whether the schedule conditions are met, and for which actions.

> **i Note**
>
> It is only possible to start scheduled actions.

Starting Actions

When the start conditions are met, or if no start conditions have been defined, the system triggers the action at the [Processing Time] defined in Customizing.

The following possibilities are available for starting:

- Starting automatically

  The system triggers the action automatically when the start condition for a scheduled action is met.

- Starting manually using the action list

  You have to have selected the indicator **Execute in Dialog** in the action definition for this function.

Scheduled Actions

Use

Actions are displayed in transactions that support actions and to which an action profile is assigned. The following information is displayed for actions:

- Status
- Description of a definition of an action

  Here, the description from Customizing that can give comments on the purpose of the action is used.

- Conditions, condition parameters, processing parameters

  The settings for the action definition and the conditions defined in Customizing are displayed here.

- Creator and creation date of the action
- The transaction ID that has been created by the action.

Prerequisites

You have made the settings in Customizing for **Service** under **Basic Functions** > **Actions**.

Activities

Depending on your settings in Customizing, you can execute the following:

- Activate selected action (schedule).

  The actions that are possible for the current business transaction are displayed in the list field next to this symbol.

- Execute selected action directly.

  You can only execute actions that have **Field executable in dialog** selected in their action definition in Customizing. Other actions are processed automatically.

- Repeat processing of the selected action.

  You can only repeat processing for actions that have the **Field changeable in dialog** selected in their action definition in Customizing.

- Deactivate actions selected, that is, remove them from the list.

  The action becomes inactive and can be rescheduled when required. This is only possible for actions that have **Field changeable in dialog** selected in their action definition in Customizing.

- Display details for the selected action.

- Preview actions.

  You can call up a print preview here for actions using Smart Forms.

- Display processing log for the selected action.

  This function is only possible for processed actions.

- Information regarding the definition of actions.

  The description from Customizing is displayed here.

- You can change the conditions and processing parameters of an action.

  This is only possible for actions that have **Field changeable in dialog** selected in their action definition in Customizing.

Processing Actions Using the Application Toolbar (SAP GUI)

Use

When processing transactions using the SAP Graphical User Interface (SAP GUI), you can schedule or start actions directly in the application toolbar.

Prerequisites

You have selected the indicator **Display in Toolbox** in the action definition in Customizing for **Service** under **Basic Functions** > **Actions** > **Actions in ...** > **Change Actions and Conditions** > **Define Action Profiles and Actions**, and you have not selected **Schedule Automatically**.

The schedule condition must have been met for an action to be displayed in the application toolbar.

Activities

1. Open a business transaction (transaction `CRMD_ORDER`).

   You see a list of inactive actions at the header and position level, whose schedule conditions have been fulfilled.

2. Choose an action to activate it.

   The action is scheduled (that is, activated and appears in the action list).

The system starts the action when the start condition has been met, depending on the predefined processing time stored in Customizing. For more information, see Processing Time.

> **i Note**
>
> If you have preset the processing time for an action as **Immediate**, the system triggers the action as soon as it has been activated in the application toolbar, provided that the start condition is met.

Action Monitor

Use

You can monitor and trigger the processing of actions for several documents by using the action monitor.

If, for example, you have set the processing time for an action (for example, output) as **Process Using Selection Report** in Customizing, you can use this program to select the actions and trigger processing manually.

> **i Note**
>
> It can also make sense to process actions with the processing time when saving using the action monitor, if, for example, time-dependent conditions have been fulfilled after a certain time without any changes to the document. Execute the program regularly so that actions are also started in these cases.

If you regularly schedule the program as a background job, the system can automatically start certain actions, whose start conditions have been fulfilled, at a particular time.

Define a variant for this, in which you specify your selection criteria and select the flag **Process without Dialog**.

Integration

The action monitor program `RSPPFPROCESS` (transaction `SPPFP`) originates from the Post Processing Framework (PPF). You can use this program to check processing for all actions (for example, output, follow-up documents).

Features

- Selection

  You select actions based on the application, the action profile, the document number (application key), the status (for example, **Unprocessed**) or the processing time.

  If you select **Process without Dialog**, the system automatically processes the selected actions without outputting a list first.

- Task

  You receive a list of actions according to your selection criteria.

  In the status column, you can see whether the action:

  - Has not been processed (yellow)
  - Has been processed successfully (green)
  - Has been processed incorrectly (red)

You can select individual actions in the list and:

- Trigger processing manually.

  To do this, select **Process**.

- Repeat processing.
- Display output issue.

  To do this, select **Preview**.

- Display a processing log.

  To do this, select **Processing Log**.

You can also display whether the action has been changed, repeated, created manually, or blocked.

Activities

To display actions that have not been processed to date, proceed as follows:

1. Call a transaction (for instance CRM_ORDER).

   The screen **PPF: Selection and Processing Actions** appears.

2. Enter additional selection criteria, for example:

   Action profile CRMB_BILLING

   Creation date to **Today's date**

3. Choose **Execute**.

   You see a list of unprocessed actions that have been created in the system up to the current date.

More Information

Processing Time

Printing Output via the Action Monitor

Actions for Alerts

Use

You can use actions to trigger alerts.

Prerequisites

You have made settings in Customizing for **Service** under **Basic Functions** > **Actions** > ***<choose area>*** > **Change Actions and Conditions** > **Define Action Profiles and Actions**.

Actions for Contracts

Use

To ensure that you retain an existing customer and maintain an excellent customer relationship, you can define certain actions in the system that need to take place during the contract term. To help the sales representative keep track of the contract, these actions are automatically triggered by the system after certain conditions have been met.

Examples of actions that can take place during different stages of the contract term include:

- Periodic follow-up calls to check whether the customer is satisfied with the product or service
- Follow-on quotation to be sent to a customer in time before the contract ends
- Customer satisfaction survey if the contract expires without renewal, or is canceled

Features

Actions can either be scheduled to be triggered automatically at certain points, or you can start planned actions manually from the contract.

You can create a set of actions that are valid for a specific contract type or a particular item category.

The system triggers the actions depending on certain conditions, such as:

- When a condition has been met (less than 80% of target quantity has been released by 30 days before expiry)
- To start on a particular date (call customer every 60 days)

Activities

1. You define actions and action profiles and assign them to the item category or business transaction in Customizing for **Service**, by choosing **Basic Functions** > **Actions.**

   The system displays the relevant actions in the **Scheduled Actions** assignment block.

2. When the condition in the action has been met, the system triggers the action, for example, it creates an appointment for the relevant sales employee. The transactions appear automatically in the sales employee's list of open documents.

You can also trigger planned actions manually from the **Scheduled Actions** assignment block.

More Information

Actions

Printing Transactions Using Actions

Use

You can use actions to print transactions, such as quotations, opportunities, sales and service orders, and sales contracts. Specific smart forms are delivered with the standard configuration. You can view and use them under Customizing for **Service**, by choosing **Basic Functions** > **Actions** > **Actions in the Transaction** > **Change Actions and Conditions** > **Define Action Profiles and Actions**.

Prerequisites

Actions are displayed on the Print pushbutton if the following criteria are met:

- The print action is set as relevant for the toolbar
- A print smart form is defined as the processing type
- The action is set to **Execute immediately**
- The action is inactive, that is, it is set to be executed manually

Process

You can use actions to print transactions during transaction processing.

Survey Tool

Use

A survey is a type of questionnaire that can be used to gather information from specific users.

Surveys are specialized forms, which can be integrated into Web sites or sent as attachments in personalized e-mails to customers. The submitted answers are collected on a server and, depending on the business scenario, the results are either stored in a database or condensed beforehand. Later, the results are evaluated and can be used for making strategic decisions.

Surveys are an important means of gaining specific information from known and unknown users, for example, as part of a customer satisfaction survey to gather feedback on a product already purchased. The results collected from surveys play an important role in making strategic decisions, for example, when planning a new product launch or when making improvements for future products.

Integration

The Survey Suite can be used with different service solution applications. It is currently used in:

- Activity Management
- Lead Management
- Opportunity Management

Features

The Survey Suite consists of the following components:

- Maintenance component to create, maintain, create different versions of, and translate surveys

You can use the Survey Designer to create surveys to:

- Gain information about customer or employee satisfaction
- Determine market potential
- Ascertain opinions
- Check the training and education status of employees

- Evaluate training courses

You can use surveys to reach the following target groups:

- Customers
- Interested parties
- Partners
- Employees
- Internet users

Survey Runtime

Survey runtime functionality forms the basis of many functions within the **Survey Tool**, for example:

- Intranet, Extranet, and Internet scenarios
- Result Evaluation
- Further screen format options, such as printing, handheld, and voice scenarios

  Caution
  Printing, handheld, and voice scenarios are currently only available as project-based functions. If you want to implement these functions, contact your SAP consultant for further information.

- Increased functionality for creating survey versions
- Unicode platforms

Survey Suite

Use

You can use the **Survey Suite** to manage surveys (for example, questionnaires).

Each survey is assigned to a specific application, and can be accessed by expanding the application folder.

Integration

You can also navigate to the style sheets and parameters from the **Survey Suite**.

Features

You can use the **Survey Suite** to:

- Edit and display surveys
- Import and export surveys (XML files)
- Translate surveys

Note that you can only translate a survey once you are logged on in the language to which you want to translate. For example, if you want to translate a survey into Spanish, you must be logged on in language Spanish.

- Display different survey versions and translations
- Test surveys
- Generate survey URLs
- Create screen format
- Maintain survey attributes (technical settings and administrative data)
- Create survey PDF files
- Import and export style sheets (CSS files) and parameters (XML files)

In **Survey Designer**, the survey hierarchy is depicted on the left-hand side of the screen, and editing options for layout, formatting, and so on, appear on the right-hand side.

The hierarchy node you select to edit determines the following:

- Features that you can add to your survey

  These are displayed in the block immediately above **Hierarchy Node**.

- Editing options available

  These are displayed on the right-hand side of the screen.

The table below lists the editing options for each hierarchy node you select, as well as nodes that can be added to the survey under each hierarchy node:

Hierarchy Node	Editing Option	Activity	Possible Additions to Survey:
Title			Section
	Enter a Title for Survey	Enter a title for survey	
Section			• Subsection • Question • Marketing Attribute • Section
	Layout	Select: • List • Table • Inline List	
	Maintain Texts and Attributes	Enter text for section	
Question			• Question • Answer • Marketing Attribute • Subsection
	Maintain Question and Attributes	Enter text for question	
	Rating Factor	Enter a number. The higher the number, the greater the evaluation weighting applied to the question	
Answer			Answer

Hierarchy Node	Editing Option	Activity	Possible Additions to Survey:
	Answer Category	Select how the answer is to be displayed in the survey. This determines the format options available	
	Display Only	Specify whether user input can be made or not	
	Required Field Entry	Select whether an answer is mandatory	
	Text Position	Specify whether answer text is left- or right-justified	
	Visible Field Length	Specify the character length of the answer field that appears in the survey	
	Max. Field Length	Specify the maximum character length of the answer field that appears in the survey. This must be equal to or larger than the visible field length	
	Lines	Specify the number of lines available for a text answer	
	Columns	Specify the number of columns available for a text answer	
Answer Option			
	Maintain Answer and Attributes	Enter text for answer	
	Answer Preassignment	Specify default answer	
	Rating	Enter a number. The higher the number, the greater the evaluation weighting applied to the answer	
	Selected	Specify this answer option as the default answer	
Pushbutton			• Submit Button • Reset Button
Save	Text	Enter a text to appear on the button in the survey	
Reset	Text	Enter a text to appear on the button in the survey	

Activities

You call up the Survey Suite under Customizing for Service > Cross-Application Components > Survey Suite .

Version Management

Use

You can use version management to create, manage, and display different survey versions and, if applicable, their corresponding translations. This enables you to store previous survey versions and translations, which you can then use at a later date. Version management enables you to translate previous survey versions long after they have been created. You can also evaluate and test surveys and their translations directly.

Features

In version management, you can perform the following tasks:

- Adjust the layout of different versions to suit specific marketing campaigns and target groups.

  Example

  You may want to present the same basic survey in tabular form to one target group, but in list form to another.

- Navigate quickly and easily through the structure in the version overview
- Enter a short text for each version you create.
- View all existing survey translations for the corresponding versions, in HTML format.

URL Usage

Use

You can use the **Survey Tool** to make surveys accessible to recipients, either as a URL link or HTML file attached to a (personalized) e-mail, or you can insert surveys into any website. This enables you to use surveys in Intranet, Extranet and Internet scenarios.

Example

You can distribute surveys to a number of recipients as part of a marketing campaign or a customer satisfaction survey. The data entered by the recipients is then submitted and can be evaluated at a later date.

Prerequisites

To make surveys accessible through Intranet, Extranet, or Internet, you first need to create a URL.

You have defined the:

- **Get Option**

  This specifies how the survey is sent to the recipient.

- **Send Option**

  This specifies how the results are later sent back when the recipient submits the survey.

Note

The options you choose are mainly scenario dependent.

Features

Possible scenarios and their corresponding options for making the survey accessible to users include:

- Intranet and Extranet Scenarios

  To make your surveys accessible through the Intranet or Extranet, you can offer the URL to specified users directly, for example, in an e-mail. When the user selects the URL link, a logon page appears, in which he or she is required to enter a user name and password for the corresponding service solution system. This is necessary to allow only authorized and recognized users access to the service solution system.

- Internet Scenarios

  Since Internet users cannot be allowed access to the service solution system for security reasons, surveys have to be made accessible in a different way. You can do this by downloading the required HTML survey to your PC. From here, you can:

  - Copy the HTML survey into any website
  - Send the HTML survey attached to an e-mail

In Internet scenarios, results can be returned by the Mailto method.

XML Parameters

Use

Survey parameter XML files are mainly used to control communication – that is, how surveys are sent and how the results are returned - in the individual survey scenarios. Depending on the individual scenarios, you can define how surveys are sent to end users and how the results are later sent back when the survey is submitted. The same survey can be used in several scenarios, using different survey parameter XML files.

Features

The following three scenarios are currently supported:

- Survey runs integrated into service solution application.
- Survey presentation format (for example, HTML) is generated and published, and, for example, is sent as an attachment in an e-mail or integrated into a Web site.
- Survey URL is generated and published, and, for example, is sent as a hyperlink in an e-mail or integrated into a Web site.

In the first two scenarios above, you need to specify the address (**Get** option) to which you want the results to be sent.

In the third scenario, you must specify the address from which the presentation format (HTML survey) is to be fetched, in addition to the **Send** option.

You can only fetch the presentation format with the **Get** option in Intranet scenarios.

You are required to log onto a service solution server with a valid user name and password. The first time you fetch the presentation format, a logon page appears for the service solution server. You can fetch the presentation format by using http or https.

If you want to make the survey accessible in Internet scenarios, you can download the HTML survey to your PC and either copy into a Web site of your choice or send it as an HTML attachment in an e-mail.

Caution

Currently, it is only possible to use XML parameters for sending surveys and receiving results as a project-based function. If you want to implement this function, contact your SAP consultant for further information.

Survey Printing

Use

You can create surveys as Portable Document Format (PDF) files, enabling you to transfer the contents of a survey to a PDF file. You can then print the survey and use it for reference purposes, for example, in meetings or presentations.

You can use PDF format to send a survey to other users as a read-only attachment in an e-mail. The user then receives the survey in PDF format and can print it directly.

Features

Printing the survey contents as a PDF file offers two main advantages:

- A recognized, standardized document format
- Printed information can be distributed to a wide audience

Integration with SAP Digital Payments Add-On

Service supports the SAP digital payments add-on for processing credit card payments. If you decide to implement this add-on, please note that you require a separate license.

This add-on offers out-of-the-box integration with various payment service providers (PSPs).

A tokenization approach avoids costs and risks: No sensitive payment card information is stored in service solution.

In Service, the SAP digital payments add-on is integrated with service order processing.

In several service processes, users can register their credit cards for one-time use. The cards are first registered through the SAP digital payments add-on by the credit card service provider. Once a card is registered successfully, the SAP system gets a token for this credit card from the service provider for processes that follow, for example credit card authorization and settlement.

If the service processes use the Billing Document Request (BDR) to complete their billing process, this token is transferred to SAP Financial Accounting. In the same way, sales items of service processes are replicated to sales orders.

With service processes, the given credit cards are authorized automatically if some of the service items are billing relevant.

Note

Only service items are billed through the service confirmation.

For more information about the SAP digital payments add-on, see https://help.sap.com/viewer/p/DIGITALPAYMENTS.

Reauthorization Report

Use

Each time you edit and save a business transaction with the payment method **Payment Card**, the system triggers an authorization request. However, transactions may sometimes not be automatically authorized when they are created or changed, for a number of reasons:

- The item delivery date was outside the authorization horizon.
- The validity period of an authorization has expired.
- The authorization has already been used.
- The clearing house rejected the authorization.
- The authorization failed due to technical reasons.

You can manually trigger reauthorization for documents or items that were not authorized automatically when you created or edited them.

Activities

Schedule the report `CRM_PAYCARD_AUTH_BATCH` regularly as a background job. The report collects all documents with the payment method **Payment Card** that have a delivery date within the authorization horizon, and triggers an authorization with the payment service provider.

Pricing

Use

Pricing is used for the calculation of prices. The system determines price information when creating a business transaction such as a service order, service quotation, or service confirmation. For example, the system automatically determines the gross price, discounts, and surcharges that are relevant for a specific customer or a particular date in accordance with the valid price conditions. To determine relevant pricing information from condition records for a business transaction, the system uses the condition technique with usage pricing. In Service, the system uses the pricing functions provided by Sales.

> **Note**
>
> A number of pricing functions are not supported in Service. For more information, see [Pricing Functions Not Supported in Service].

You configure the settings for pricing in Customizing for **Service** under **Basic Functions** > **Pricing**.

> **→ Recommendation**
>
> Before defining condition tables and access sequences, check which fields are supported in service processes in Customizing for **Service** under **Basic Functions** > **Pricing** > **Display Supported Price-Relevant Fields (Standard Delivery)**.

Modular Pricing Procedures

You can use modular pricing procedures for automatic inclusion of price conditions. You can configure the system so that during the pricing process in a service transaction, additional price conditions are included dynamically. You can set up several elements in pricing configuration that interact during the pricing process. For more information, see Modular Pricing Procedures for Automatic Inclusion of Price Conditions.

Related Information

Price Management

Pricing Process Flow

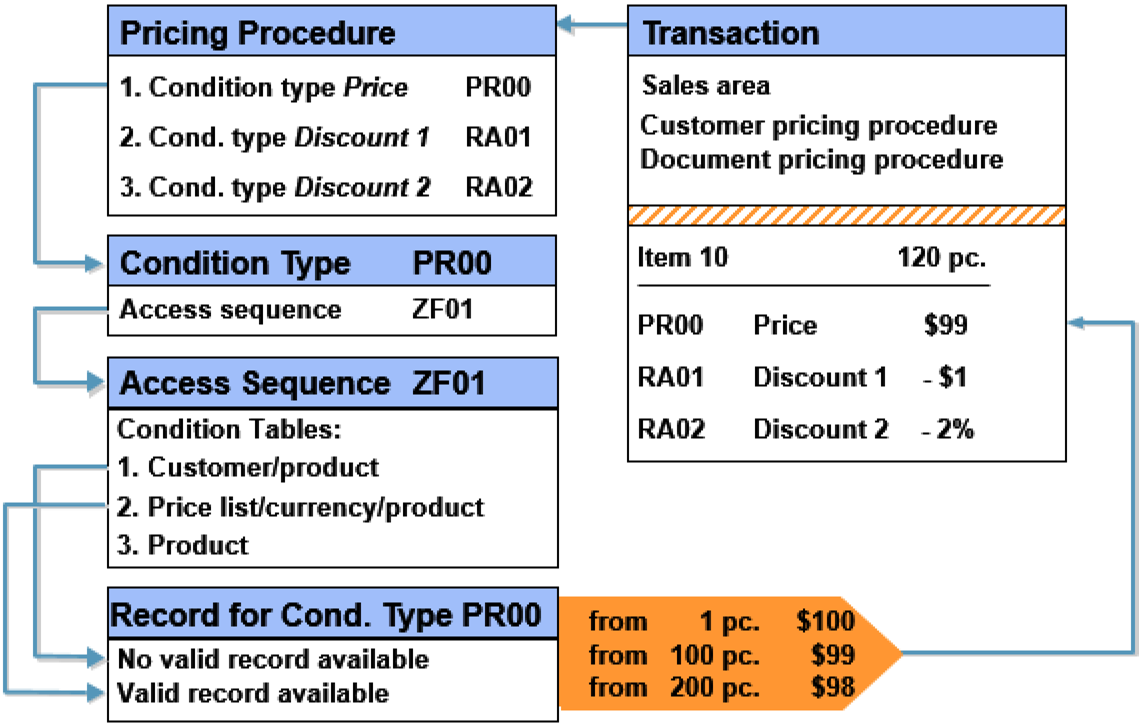
Use

In this process, the system automatically determines prices for a business transaction. To do this, it uses the condition technique. The condition technique processes the price queries by searching for valid values of prices in existing condition records. It does this through the sales area, customer pricing procedure, and document pricing procedure.

Prerequisites

You have defined the settings for pricing in Customizing. You can find these settings in Customizing for **Service** under **Basic Functions** > **Pricing** > **Configure Pricing**.

Process



1. Pricing makes the input values sales area, customer pricing procedure, and document pricing procedure available to the condition technique.

2. The system determines the pricing procedure depending on the sales area, customer pricing procedure, and document pricing procedure.

3. The system reads the first condition type of the pricing procedure and determines the assigned access sequence.

   This step and the following steps are repeated in turn for each condition type in the pricing procedure.

4. The system reads the access sequence by using the condition tables. The sequence of the condition tables forms the search strategy for determining the individual condition records. Each condition table contains the field combinations according to which the system should search in the condition records, for example, customer - product.

5. The system searches for valid condition records for the condition tables. If the system does not find a valid condition record for the first condition table, it continues by searching for a condition record for the next condition table.

6. After the system has found a valid condition record for a condition table, it makes the result available to pricing in the form of prices and discounts.

7. If the pricing procedure contains more than one condition type, the system repeats the search for condition records for each condition type.

Result

The system displays the pricing result in the business transaction for each item as a net value. Additional values (for example, a gross value) are displayed at header level.

Example

You offer a customer a special price for a product and create a special condition record specifically for this customer and this product. In the pricing procedure, the condition type `Price` comes first. The access sequence for this condition type specifies that the system first searches for a customer-specific price (field combination `Customer - Product`). In this way, the system can automatically set the customer-specific price in the business transaction, and calculate the total price using the quantity ordered.

Condition Record

Definition

A condition record defines input and output values for a price determination process.

The condition record is created or changed within condition record maintenance and selected in the price determination process.

The output values are dependent on chosen input values (for example, customer, customer group, and product) and are valid for a certain time period.

Use

The system uses condition records as the basis for the condition technique in pricing. By using the condition technique, the system determines prices in the relevant application.

Structure

You define the structure of a condition record using the assigned condition table in Customizing. The condition table structures the condition record into search fields and results fields.

- The search fields represent the criteria for a determination procedure, for example, customer, customer group, and product.
- The results fields are transferred to the relevant application by the system after the price determination process.

Condition Maintenance

Use

In condition maintenance, you create condition records with concrete values. In doing so, you define, for example, customer-specific prices or discounts. You use Customizing for the condition technique to determine which values you can define for which fields.

A price condition is an agreement on prices, discounts, surcharges, and so on. It is determined according to selected influencing factors (for example, customers, customer groups, and products) and is valid within a specific time period.

Prerequisites

You have made the Customizing settings for the condition technique. For more information, see Customizing under **Service** > **Basic Functions** > **Pricing** > **Configure Pricing**.

Features

Price Condition Key

By selecting a condition type and (if required) a condition table, you define which price conditions are created for which data, for example:

- Customers
- Customer groups
- Products
- Product groups
- Combinations of customers and products
- Combinations of customer groups and products

Calculation Type

The calculation type is specified in Customizing for a condition type. This gives you the option of executing different calculations in the condition record. You can use the following calculation types, for example:

- Fixed amount
- Percentage
- Quantity-based
- Weight-based
- Time-based

Each price condition has a specified validity period.

Scales

You have the option of making a price, a discount or a surcharge dependent on a scale.

For example, you can make every price, discount, or surcharge dependent on a quantity or weight scale.

Maintaining Price Details in Business Transactions

Use

Pricing is used in business transactions (for example, service orders, service confirmations, and service requests), to determine an appropriate price. It consists of the determination and evaluation steps and takes place automatically at item level.

It may be necessary for you, however, to change a pricing result after negotiations with a customer, for example, to supplement it with discounts. In this case, you can make manual changes in the business transaction. This is possible at header level and at item level:

- At header level, you can display the accumulated value for all items for every price element under **Price Details**. You can add price elements, and change or delete them.
- At item level, you can display the price elements under **Price Details**. You can change, add, and delete price elements if your configuration allows it.

> **i Note**
> In transaction processing in Service, the description of the condition types is displayed in the column **Price Element** and the condition value is displayed in column **End Value**.

Manual Price Elements

If you add or change a price element manually, the system automatically calculates the end value and sets the **Manually Changed** indicator for the changed or added price element.

> **i Note**
> You can change, add, and delete header price elements only on header level of a business transaction; you can change, add, and delete item price elements only on item level of a business transaction.

Repricing

You can you perform repricing in a business transaction. With the following actions, you can control how the system treats the price elements:

- Reprice:

  The system carries out a new pricing with determination and evaluation. It copies the manually changed price elements and the manually entered price elements.

- Complete Reprice:

  The system carries out a complete new pricing with determination and evaluation (as if you had created a new document item). It discards your manual changes.

> **i Note**
> If you perform the action on header level, the system performs repricing for all items of the business transaction.
>
> You cannot change or delete price elements for business transactions or its items that have been billed already.

Pricing Functions Not Supported in Service

Information about pricing functions that are not supported in the service solution.

The following pricing functions are not supported in Service:

- Condition exclusion using exclusion indicator KZNEP
- Tracking cumulative values with condition update
- Data sources
- Quantity adjustment
- Agreed conversion factors
- Configurable parameters and formulas in pricing (CPF)
- Commodity pricing

If the configuration of a pricing procedure contains one or more pricing functions that are not supported, and the pricing procedure is used for pricing in the service solution, an error message is triggered if such a condition is determined.

If such a pricing procedure should not be used exclusively for service solution but also for sales documents, and the condition type that uses an unsupported pricing function is not relevant for the service solution, you can suppress the condition type during pricing in the service solution context by assigning an appropriate requirement in the pricing procedure. The requirement could contain a check of field KOMK-KVORG (value C1 is used in the service solution context). This prevents an error message from being triggered in the service solution.

Additional Pricing Functions for Service

Additional pricing functions that are available in Service.

Additional Pricing Functions for Service

Function	Details
Additional subtotal fields	A few subtotal fields that are available in Service pricing are also available in pricing for Sales. These include subtotal fields for the following: • Shipment costs (FREIGHT) • Net value without freight cost (NET_WO_FREIGHT) • Cash discount (SKONTO) The subtotal field for gross value (BRTWR) gets populated if the subtotal value 9 gets assigned accordingly in the pricing procedure.

Fields Used in Pricing Enhancements

Fields that are used in pricing routines and user exits to enhance the pricing functionality.

You can use pricing routines (for example transaction VOFM) and user exits to enhance the pricing functionality. To ensure that the fields of the communication structures KOMK and KOMP are populated with data for executing the pricing routines and user exits during the pricing run, you need to list them in Customizing.

You can maintain the settings in Customizing for **Service** under **Basic Functions** > **Pricing** > **Edit Fields Used in Pricing Enhancements**.

Pricing Analysis

With this function, you can understand how the system determines the pricing conditions for a service transaction such as a service order or service confirmation.

Features

The **pricing analysis** function enables you to do the following:

- Get an overview of pricing procedures for service transactions.
- Display details of accesses for all condition types and check whether a condition record has been found or not for each condition type.
- Obtain more detailed information at the level of the overview tree such as condition record IDs, the input values used to search for a condition table, and in which condition table a condition record has been found.

Use

To use the **pricing analysis** function, you need to set the **Pricing Trace** parameter PRC_TRACE to **X** in the system.

After this step, you can use the **Trace** button on the **Price Details** tab at item level of a service transaction to analyze pricing.

Billing

Get an overview of how Service and Sales Billing work together to facilitate the billing of service transactions.

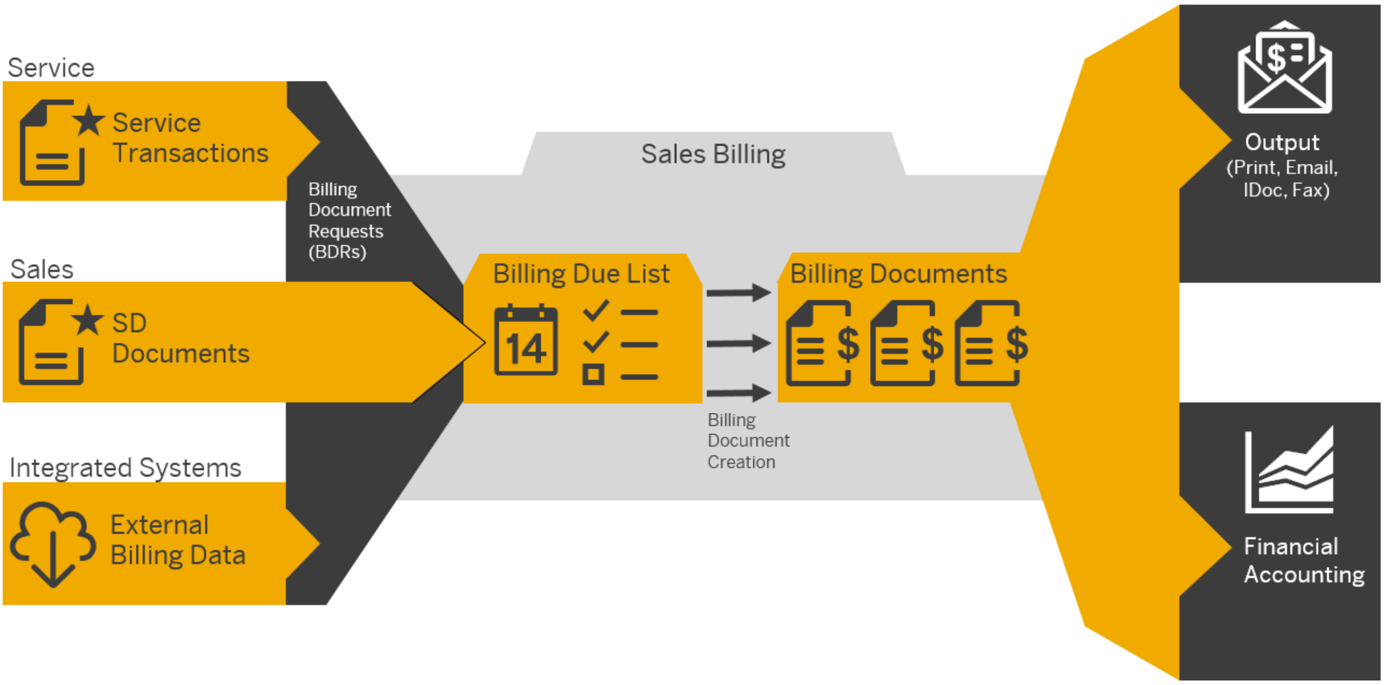
Use

You use the billing functions in Service to initiate billing of the following service transactions:

- Service orders
- Service confirmations
- Service contracts

When service transactions (or, in some cases, individual items) are released for billing, billing-relevant transaction data is transferred to Sales Billing in the form of billing document requests (BDRs), which are added to the billing due list. You can either trigger billing document creation for these billing due list items manually or schedule it to run periodically without user interaction. Both methods create billing documents that can be posted to financial accounting and sent to customers.

The following figure shows the service billing stream in the context of the other two main billing streams that converge in Sales Billing. When it is correctly set up, you can use omnichannel convergent billing to converge all three streams into combined customer invoices.



Prerequisites

You have configured the billing functionality for Service in Customizing under **Service** > **Basic Functions** > **Billing**.

Overview

Triggering the Billing of Service Transactions

Billing in Service is triggered by service transactions. If a service transaction includes sales items, these sales items are replicated to sales orders. These sales orders (along with subsequent deliveries) can then be billed together (converged) with service items and service part items to produce combined customer invoices.

You trigger the billing of service orders and service confirmations by releasing these transactions (or, in some cases, individual completed items in service orders) for billing in the **Release for Billing** app or you can manually schedule jobs by using the report **Generate BDRs for Service Orders** (`CRMS4_SERVORD_BDR_GENERATE`). You can find the report on the SAP Easy Access menu under **Service** > **Service Processes**.

Billing of the condition values in service contracts is triggered once the contract is released.

Creating Billing Documents

When billable service transaction items such as service order items are completed and released for billing, they appear as billing document requests (BDRs) in the billing due list of the **Create Billing Documents** app. Service contracts must simply be released so that a scheduled batch job can create the corresponding BDRs. For more information about billing document requests, see [Billing Document Requests](#).

From the billing due list, you can create billing documents for one or more service transactions manually. For more information about the **Create Billing Documents** app, see [Create Billing Documents](#).

You can also schedule periodic billing creation jobs that automate the billing document creation process. For more information, see [Schedule Billing Creation](#).

As long as key header data (such as business partners, incoterms, and other fields that can cause an invoice split) of the documents that you want to bill are identical, you can use omnichannel convergent billing to converge different document categories into single, combined customer invoices. For more information, see Omnichannel Convergent Billing.

Processing Billing Documents

After billing documents have been created, the system can help you process them easily and effectively. Using the **Manage Billing Documents** app, you can view a list of all billing documents in the system, view each one in detail, cancel them, and post them to financial accounting. You can also view PDF-based previews and output the final billing documents using different media (for example, print, fax, or e-mail). For more information about the app, see Manage Billing Documents.

To automate output for billing documents (for example, to print invoices for customers), you can schedule periodic billing document output jobs using the **Schedule Billing Output** app. For more information about this app, see Schedule Billing Output.

Posting of Billing Data to Financial Accounting

The billing data in billing documents is released to financial accounting by manual or automatic posting of the billing documents. You can choose between manual and automatic posting in the billing settings of the **Manage Billing Documents** app. In addition, you can schedule automated billing release jobs in the **Schedule Billing Release** app. For more information about this app, see Schedule Billing Release.

For more information about billing document posting in general, see Integration with Accounting.

Related Information

Billing of Service Transactions
Periodic Billing Plans for Service Contract Billing
Creating and Processing Billing Documents
Resource-Related Billing

Billing of Service Transactions

Learn the specifics of how billable data from service transactions is transmitted to the billing due list in Sales Billing and get an overview of the different billing procedures.

Use

Service transactions (for example, service orders) can contain one or more of the following item categories, which may be relevant for billing:

- Service items
- Service part items
- Expense items
- Sales items

Note

For the billing of sales items, a sales order is automatically created in the system when the service transaction that contains the sales item is completed. You can use convergent billing to bill this sales order (or the delivery it triggers) together with the corresponding billing document requests (BDRs). This enables the creation of combined customer invoices containing all item categories.

The exact manner in which service transactions are billed depends on which categories of billing-relevant items they contain. The following information describes how service items and service part items are billed. The item category can specify one of the following billing scenarios:

- Not relevant for billing

  The item category is not relevant for billing (for example, a free-of-charge delivery item or quotation item).

- Transaction-related billing

  - Billing after contract release

    The order quantity is billed for the contract once it is released.

  - Billing on completion

    The quantity defined in a transaction (such as a service order or service confirmation) or the fixed price defined for a transaction is billed once the transaction is completed, which is indicated by its header status.

- Partial billing

  Depending on the item category in the service order or service confirmation that is being billed, you can do one of the following:

  - You can bill the quantity specified in the service confirmation, even if the status of the service order at header level isn't completed yet. For this to work, the service confirmation item must have status completed, and the service order must be released.

  - Alternatively, you can bill one or more completed fixed-price service items in a service order, even if the service order isn't yet completed. For this to work, the service order must be released.

→ Remember

All billing-relevant items in service transactions must also be marked as pricing-relevant because billing document requests (BDRs) draw pricing information from the pricing documents associated with these transactions. This also means that prices cannot be recalculated on the fly once an item has been released for billing.

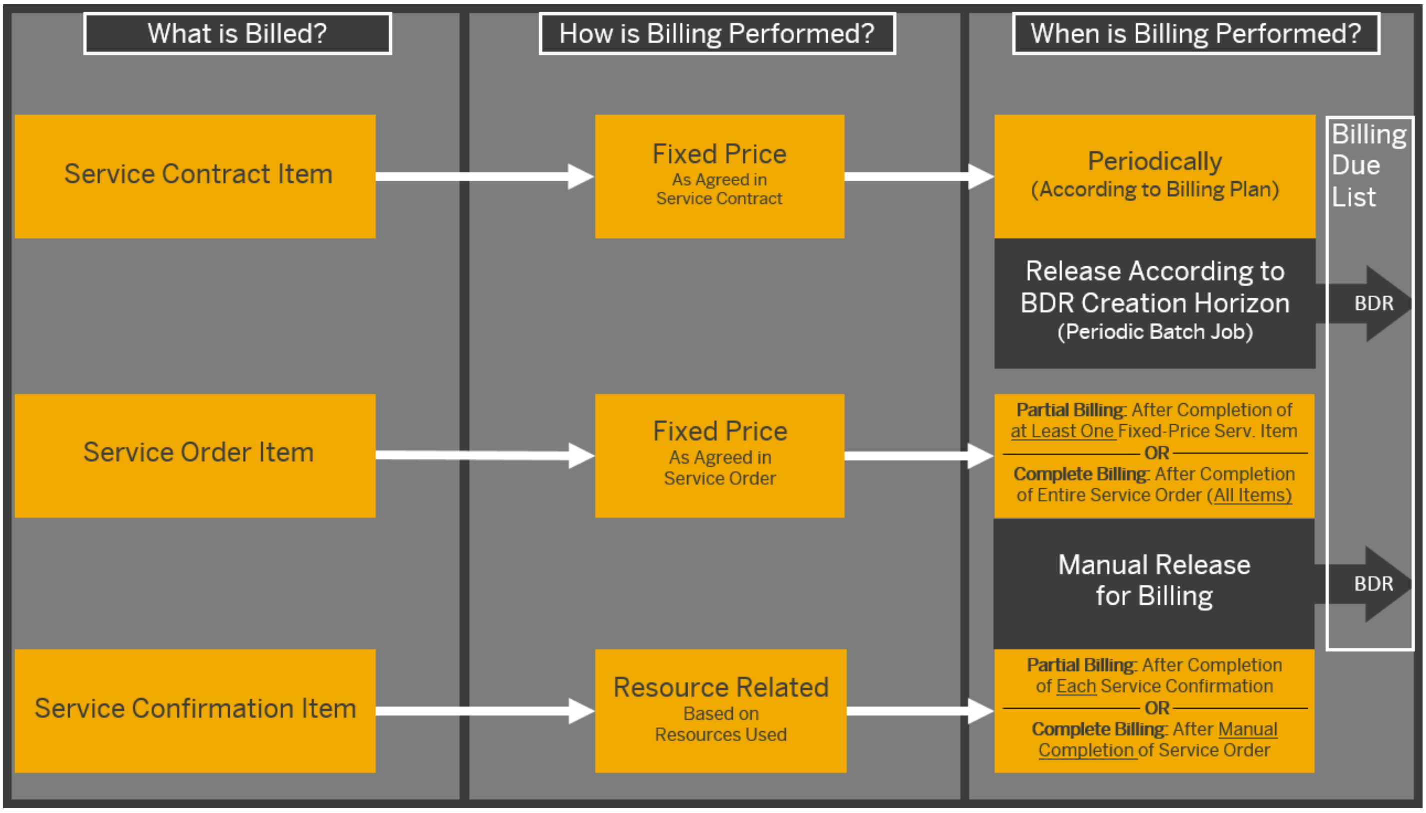
i Note

If billing-relevant items from a single reference service transaction containing electronic payment data (for example, payment card data) lead to a split, meaning that the system creates multiple BDRs, the electronic payment data is only transferred to the first BDR that the system creates. Subsequent BDRs will not contain any reference to the electronic payment data.

In other words, the distribution of electronic payment data from a single reference service transaction across multiple BDRs is not supported.

Billing Procedures for Service Transactions

The following figure shows how service orders, service confirmations, and service contracts can be billed. The figure is followed by an explanation.



- After billable items in service orders or service confirmation have been released for billing in the **Release for Billing** app, they trigger the creation of billing document requests (BDRs), which appear in the billing due list of the **Create Billing Documents** app. From there, a billing clerk can create billing documents, such as invoices.
- In the case of service contracts, the contract must simply be in a released state so that a scheduled BDR creation job can create BDRs for the unbilled service contract items. You can use transaction **SM36** to define jobs that run the BDR creation program **CRMS4_BILLING_BDR_GENERATE** as needed, for example, on specific dates or with a predefined periodicity.

  From the billing due list, a billing clerk can bill the BDRs to create billing documents manually or automatically by respectively using the **Create Billing Documents** or **Schedule Billing Creation** apps.

The amount that is billed and when exactly billing takes place depends on the type of service transaction and other factors. When you release items for billing in the **Release for Billing** app, the system automatically proposes the relevant billing procedure, which are dictated by the item category. For each transaction type, the billing procedures are characterized follows:

Service Contracts

Billing of service contract items takes place periodically. The periodicity and the fixed amount that is billed is specified in the terms of the contract.

Service contract items have billing relevance **Billing After Contract Release**, meaning that billing is triggered according to the billing plan specified in each item of the service contract. BDRs for released (unbilled) items are created periodically by batch jobs that you can schedule to run automatically by using transaction **SM36**.

With this transaction, you define jobs that trigger the BDR creation program **CRMS4_BILLING_BDR_GENERATE** as needed, for example, on specific dates or with a predefined periodicity. For more information about periodic billing plans for service contract billing, see Periodic Billing Plans for Service Contract Billing. For more information about scheduling BDR creation jobs, see the billing plan section under Service Contract Item.

Service Orders

The billing of service orders can differ depending on whether it references a service contract and whether it contains fixed-price service items or not.

- If a service order item refers to a service contract, it is billed according to the price agreements in the service contract. If the contract doesn't contain price agreements, the product prices are applied.
- If a service order item does not refer to a service contract, a fixed amount is billed according to the service agreements and price conditions in the service order.

Both service order items and service confirmation items have the billing relevance **Transaction-Related Billing After Completion**. This means that these items must be manually released for billing on the **Release for Billing** screen after the service transaction has been set to the completed status. This allows finalized transaction data to be checked before billing. When you release the items for billing, the system creates the corresponding BDRs.

Items with the billing relevance **Transaction-Related Billing After Completion** must be manually released for billing in the **Release for Billing** app after the service transaction has been set to the completed status. When you release the items for billing, the system creates the corresponding BDRs.

If you are billing a standard service order (for example, type `SV01`) that contains completed fixed-price items, you can perform **partial billing** for only these items or wait until all items are completed and perform **complete billing**. Whether partial billing is possible depends on the item category that is being billed. The differences between partial billing and complete billing of service orders are as follows:

- Partial billing

  Billing can take place as soon as one or more fixed-price items in the service order are completed and the service order is released. This option results in multiple BDRs and, consequently, enables the creation of multiple invoices per service order.

- Complete billing

  Billing takes place when the entire service order is completed, that is, all items in the service order and the service order itself are completed. This option results in a single BDR and a single invoice for the service order.

For more information about service order types and the item categories that they can contain, see Service Orders.

i Note

A BDR can contain one or more billable items that belong to the same service order or confirmation and were released at the same time. However, a single service transaction can lead to the creation of multiple BDRs if its items are released for billing at different times. For simplicity, it is preferable to release all items in a service transaction together. This minimizes the number of billing due list items that you need to converge to create a combined customer invoice.

Service Confirmations

- If a service confirmation refers to a service order, the amount billed depends on your settings in Customizing under **Service > Basic Functions > Transactions > Define Item Categories**. In this case, you have the following two options:
  - Fixed price (service order-related) billing: Billing of fixed amounts agreed on in the service order

    If you want to bill for fixed amounts, you define the service order item category as billing relevant, but you define the service confirmation item category as not relevant for billing.

  - Resource-related (service confirmation-related) billing: Billing for resources used

    If you want billing to be resource-related depending on the resources confirmed, you define the service confirmation item category as billing relevant, but you define the service order item category as not relevant for billing.
- If you want to bill service confirmations that are not related to a service order, you simply define the service confirmation item category as billing relevant.

i Note

You can only release service confirmations with header status **Completed** for billing. This status signifies that all items in the service confirmation are completed and ready to be billed.

If you have created multiple (that is, partial) service confirmations for a single service order, you must also choose between the following two approaches to billing, which are dictated by the item categories used:

- Complete billing

  Billing takes place when the service order is set to completed manually. This option results in a single invoice for the entire service order. To use this approach, set the billing relevance of the service confirmation item category to **Transaction-Related Billing After Completion**.

- Partial billing

  Billing takes place while the service order as a whole is not completed. Each related service confirmation is billed separately when it is manually set to completed. Note, however, that individual service order items must be completed if they are to be billed. With partial billing, a single service order results in multiple, partial invoices (for example, one invoice for each billed confirmation). To be able to use this approach, you must set the billing relevance of the service confirmation item category in Customizing to **Service Confirmation-Related Billing**.

i Note

When the system creates a BDR with reference to a service transaction, it may copy the value of the 35-character **external reference** field from the service transaction header to the 16-character **reference** field in the BDR header. Whether the system does this or not depends on a corresponding setting in the copying control configuration for billing documents.

When the header copying control setting for reference number is set to A (customer reference), due to the differing field lengths, the system truncates the copied value if it exceeds 16 characters.

Whether the reference number is later copied to the subsequent billing document depends on the copying control settings for the specific billing document type (with reference to the relevant BDR type).

→ Tip

When you trigger billing for a service order item, service confirmation item, or service contract item for which a material sales text has been maintained, this text is copied to the successor items in the subsequently created BDR. When the BDR is invoiced, these texts are copied to the items of the resulting invoice.

Determining Rates for Billing

In service orders and service confirmations, you can determine different rates for billing (for example, for weekend work, overtime, travel time, or depending on the qualifications of the service representative). You do this by specifying a service type or valuation type in the business transaction.

Schedule Creation of BDRs

You can use the report **Generate BDRs for Service Orders** (`CRMS4_SERVORD_BDR_GENERATE`) to schedule jobs to create billing document requests for billable service order items or billable service confirmation items. You can find the report on the **SAP Easy Access** menu under **Service** > **Service Processes**.

Related Information

[Billing](#)
[Creating and Processing Billing Documents](#)

[Periodic Billing Plans for Service Contract Billing](#)

[Service Contract Item](#)

Transfer of Address Data from Service Transactions to Billing Document Requests

Whether the system can pass a specific business partner address from a service transaction to the subsequent billing document request depends on what kind of address it is, but also on whether multiple address handling for SD documents is active in your system or not.

When the system passes billing-related data from service transactions to billing document requests (BDRs), this includes the addresses of the involved parties (such as the sold-to party, bill-to party, and payer). In this context, these business partner (BP) addresses can be divided into three broad categories:

1. Standard address of the business partner (from BP master data)
2. Alternative addresses of the business partner (from BP master data)
3. Document-specific addresses (manually entered into a specific service transaction)

Which of these three categories can be passed from service transaction to BDRs (and subsequently used in invoices) depends on whether multiple address handling for SD documents is active in your system or not, as follows:

- When multiple address handling is **not active**, the system automatically transfers addresses of the following categories from service transactions to the subsequent BDRs:
  - Standard address of the business partner
  - Document-specific addresses for business partners of category Organization or Group
- When multiple address handling is **active**, the system automatically transfers addresses of the following categories from service transactions to the subsequent BDRs:
  - Standard address of the business partner
  - Alternative addresses of the business partner for various partner functions, but excluding partner functions Payer and Employee
  - Document-specific addresses for business partners of category Organization or Group

⚠ **Caution**

When a service transaction contains an address of a category that differs from the address categories listed above, the system issues a warning. Subsequently, when the system creates the BDR, the BP's standard address is passed to the BDR instead of the specified address.

Related Information

[Business Partner Address](#)

[Multiple Address Handling in SD Documents Using SAP Business Partner](#)

Combined Billing of Fixed-Price Items and Time-and-Material Items into One Invoice

Adapt Customizing to prevent invoice splits caused by non-identical pricing procedures of different billing document request (BDR) types representing service orders and service confirmations.

Per default, the respective pricing procedure for each BDR type originating from Service (carrying the billing data of either preceding service orders or service confirmations) is through Customizing. In Customizing, you can set the pricing procedure for a given BDR type by going to **Sales and Distribution** > **Billing** > **Billing Documents** > **Define Billing Types** and maintaining the **Doc. Pricing Procedure** field.

If you want to converge fixed-price items and time-and material (that is, resource-related) items into one combined invoice by billing the corresponding BDRs, the differing pricing procedures (one for service order BDRs and another for service confirmation BDRs) would usually prevent a combined invoice. This is because certain BDR header fields cannot be combined into a single billing document, resulting in a split into two separate invoices.

Such a scenario would occur when, for example, you are billing items resulting from a "mixed" service order that contains both of the following:

- Fixed-price items (service-order related billing)
- Time-and-materal items, whose price is determined according to the resource consumption specified in the subsequent service confirmation items (service confirmation-related billing)

Solution

You can prevent the invoice split by setting the **Doc. Pricing Proc.** field in the **Define Billing Types** Customizing activity to initial (blank).

The empty field instructs the system to carry forward the single pricing procedure from the originating service order, through the service confirmation, to the respective BDR.

The result is that the pricing procedure defined in the service order header is applied to all billing-relevant item categories in the service order and subsequent service confirmations.

Related Information

[Billing Document Split and Convergence]

Resource-Related Intercompany Billing

You can use resource-related intercompany billing for service orders.

Use

You can use resource-related intercompany billing to distribute revenue between company codes based on the services performed by employees belonging to another company code. For example, if a customer orders a service through a service order from a company code 1010 (DE) and another company code 1710 (US) delivers this service by sending a technician to execute the service for the customer, resource-related intercompany billing can be used to handle this situation.

You can use the following items in service orders for resource-related intercompany billing:

- Service Item
- Expense Item
- Execution Order Item (Only labor in execution order items is supported for resource-related intercompany billing.)

i Note

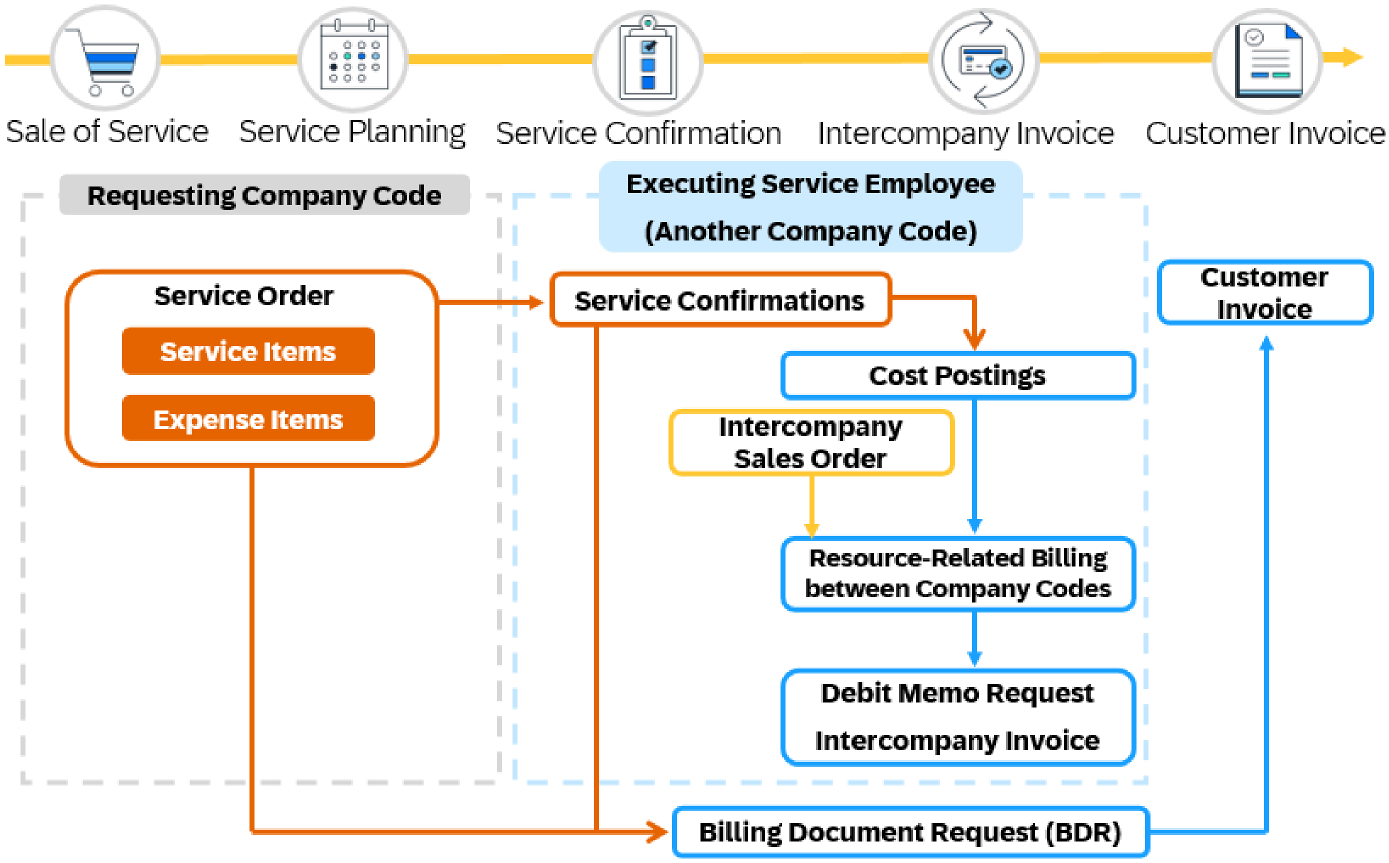
You cannot use service parts for resource-related intercompany billing.

Prerequisites

Before you can use the process for billing between company codes, you need to carry out the settings for the configuration. For more information on the steps, see Configuration.

Process

The following diagram shows you a sample process of how resource-related intercompany billing works using a service order.



For more information about the process, see Billing Between Company Codes.

Release for Billing

You trigger the billing of service orders and service confirmations by releasing these transactions (or, in some cases, individual completed items in service orders) for billing in the **Release for Billing** app or you can manually schedule jobs by using the report **Generate BDRs for Service Orders** (`CRMS4_SERVORD_BDR_GENERATE`).

Generate BDRs for Service Orders

Use

You can use this program to create billing document requests (BDRs) based on the billing request lines found in the billing plans of service order items or create BDRs for billable service order items or billable service confirmation items.

You can start BDR creation immediately or schedule it to run in the background. Note that you can reduce system load and processing times by running BDR creation during periods when system usage is lower (at night, for example).

Selection

Enter your selection criteria as needed. Choose at least one of the following criteria:

- BDR creation horizon: Time period of a number of days into the future (relative to the program run date) that the system uses to determine those billing request lines for which BDRs must be created at program run time. When the program runs, the system creates BDRs for only those billing request lines whose billing dates lie in the period up to the program run date plus the horizon you have defined. The system automatically makes this calculation and processes the billing request lines that it classifies as being due for BDR creation.
- Service Order IDs: One or more IDs of the service order or orders to be included in your BDR creation run.
- Service Confirmation IDs: One or more IDs of the service confirmation or confirmations to be included in your BDR creation run.

You can filter your selection with additional criteria, including:

- Billing date from billing request lines
- Sold-to party
- Sales organization
- Distribution channel
- Division

You need to select at least one of the following checkboxes to include relevant items in the job to create billing document requests (BDRs).

- Items with Billing Plans: Select this checkbox if you want to include items to which you have assigned ad hoc billing plans in the BDR creation run.
- Billable Service Order Items: Select this checkbox if you want to include service order items that can be billed in the BDR creation run.
- Billable Confirmation Items: Select this checkbox if you want to include service confirmation items that can be billed in the BDR creation run.

You can use Simulation Only to quickly test run the job without actually creating any billing document requests (BDRs). The simulation result tells you the number of BDRs that would be created for a certain number of service orders. The simulation result also helps you decide whether to choose Parallel Processing or not.

Parallel Processing improves job performance when dealing with large volumes of data. It enables the job to process billing request lines in multiple parallel tasks. You can specify a maximum number of parallel tasks from two to five.

You can only use Parallel Processing for items with billing plans.

You can use Max. Number of Billable Items to provide the maximum number of billable items for which you want to create billing document requests (BDRs). For example, if a service order or a service confirmation has 20 billable items for which BDRs can be created and you provide a value of 7 in Max. Number of Billable Items, then only 7 items will be processed for BDR creation. If you don't provide any value, all billable items in a service order or a service confirmation will be processed for BDR creation.

Output

You can check program execution in the application logs.

Example

If you enter "3" as the BDR creation horizon and the program runs on 11.26.2018, the system creates BDRs for all billing request lines that have billing dates up to 11.29.2018 (three days after the program run date).

Periodic Billing Plans for Service Contract Billing

Get an overview of how the system generates billing document requests from periodic billing plans to facilitate the billing of service contracts.

Use

You use periodic billing plans to schedule individual dates for the billing of service contracts, independent of the provisioning of the service. Periodic billing plans have a start and end date. They bill fixed (predetermined) amounts at regular intervals, for example, a recurring quarterly maintenance fee in a maintenance contract. You can see information about valid billing plans in the billing plan assignment block of the relevant business transaction, for example, in a service contract item within a service contract. The billing plans discussed here are only valid at service contract item level (and not at header level).

Prerequisites

1. You have made the necessary settings for billing plans in Customizing for Service by choosing **Transactions** > **Basic Settings** > **Billing Plan**.

2. The necessary settings in Customizing for billing request lines are as follows (note that this is also the predelivered configuration):

   An item category is defined to be used for billing request lines.

   You do this under **Transactions** > **Define Item Categories**. The item category must have the object type `BUS2000137` (**CRM Service Contract Item**) and business transaction category `BUS2000115` (**Sales**). In the **Scheduled Bill.** field, select **Billing Request Line Billing** and set the billing relevance to **Billing After Contract Release**.

   The following Customizing settings are predefined by SAP for use with billing request lines:

   - Transaction type `SC`
   - Item category `SCN`, `SCNA`, and `SCNC`

Billing Request Lines and Billing Document Requests

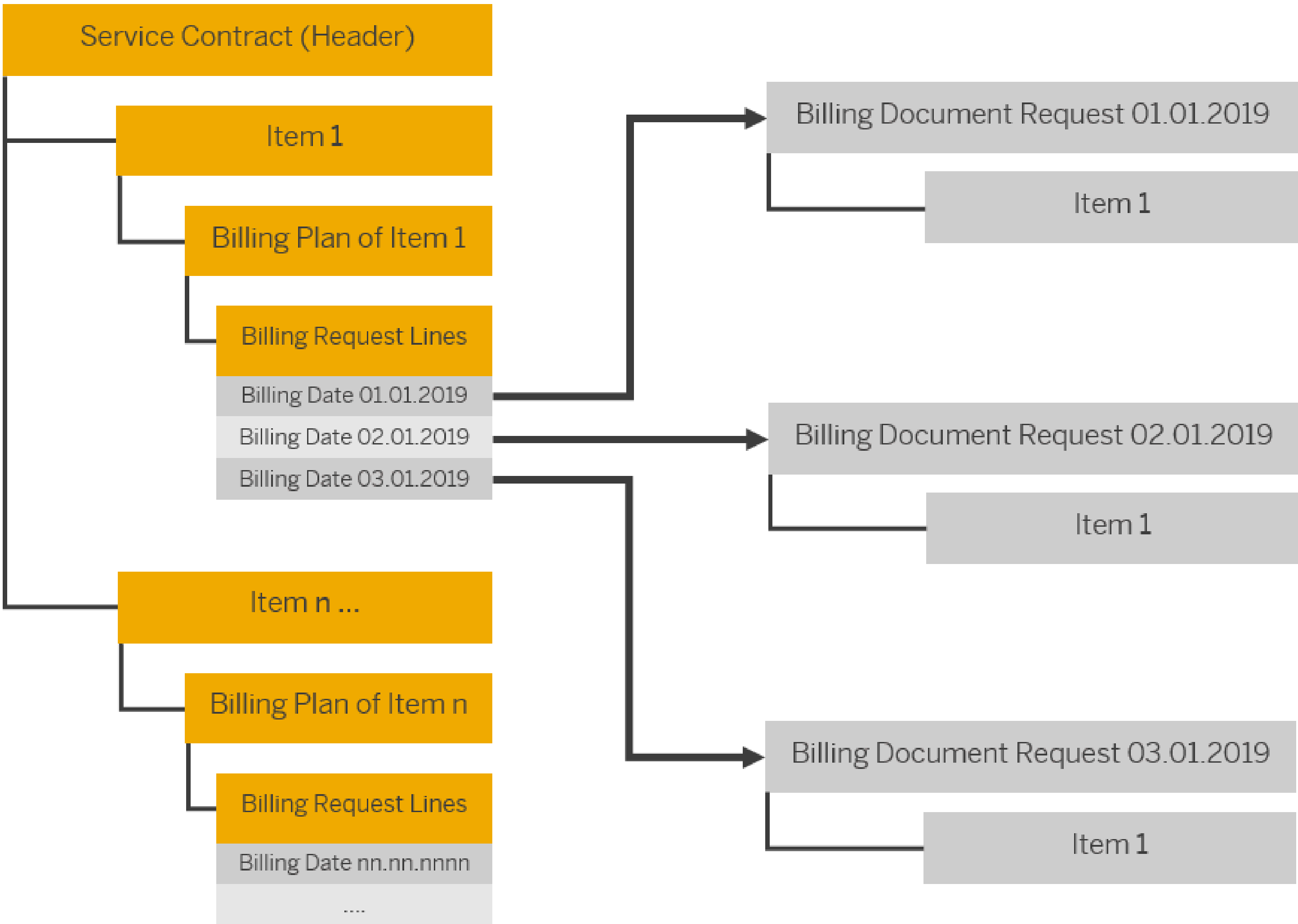
The billing plan contains a list of billing dates within the settlement period on which billing is to occur. For each of these dates, the system generates one billing request line within each service contract item that is to be billed.

To be able to process service transactions in the billing due list, the system must generate billing document requests (BDRs) based on the open billing request lines. A billing request line is open when no corresponding BDR exists for it. You can see the billing status of billing request lines within the service contract item details, in the billing status column. The following table explains each status.

Billing Status of Billing Request Line	Meaning
Not transferred to billing	No BDR has yet been created for the billing request line. This means that the billing request line is not yet represented in the billing due list. This status is also re-applied when a billing clerk uses the **Manage Billing Document Requests** app to delete or reject an existing BDR. The status reversal allows the billing request line to be released for billing again.
Transferred to billing	A BDR has been created for the billing request line, meaning that the billing request line is currently represented in the billing due list.
Fully billed	The BDR that was created for this billing request line has already been billed, meaning that a corresponding billing document has been created. It is therefore no longer represented in the billing due list.
Error in transfer to billing	An error occurred while creating the BDR for the billing request line. For example, if a material (that is, a product) that the contract refers to has not been properly maintained. You can check the application log for additional details. After correcting the root cause, try to create the BDR again. If successful, the status will change to show that the billing request line has been transferred to billing.

A BDR based on a service contract item contains the value that is billed for the item on the relevant date and the relevant price information. In addition, it contains fields that reference the underlying billing request line ID and the contract ID.

The following figure shows the relationship between service contract items, billing request lines, and the resulting BDRs that the system creates for billing plans.



i **Note**

You can use transaction **SM36** to define jobs that run the BDR creation report **CRMS4_BILLING_BDR_GENERATE** as needed, for example, on specific dates or with a predefined periodicity. With this method, it's easy to periodically create BDRs on the basis of open billing request lines. For more information about how to use these transactions, see the billing plan section under Service Contract Item.

You can then bill these BDRs manually by using the **Create Billing Documents** app, or automatically by using the **Schedule Billing Creation** app to schedule periodic jobs. For more information about these next steps, see Creating and Processing Billing Documents.

Please note the following information about how billing request lines and BDRs are created and processed:

- You can make settings in Customizing to block the billing of billing request lines by default. These lines must be manually unblocked so that they appear in the billing due list.
- BDR creation jobs are scheduled based on a creation horizon (a number of days into the future). Every time a job runs, it creates BDRs for all open billing request lines that have a billing date up to the (and including the) designated horizon. If necessary, you can set up several jobs in parallel to meet your business requirements. It makes sense to use a horizon value which ensures that the BDRs required for the next billing document creation run are created in time to be included in the run. You can determine when these runs take place using the **Schedule Billing Creation** app.

  ### Example

  You want all released, unbilled service contract items (represented by open billing request lines) to be billed before the 15th of each month because you want to get the invoices to your customers by the 15th. You therefore schedule a BDR creation job to run on the 11th of each month and set the horizon to four days. On the 11th of each month, the job runs and creates BDRs for all open billing request lines with billing dates up to the 15th of the respective month. Because billing is already triggered on the 11th, a scheduled billing document creation run on the 12th would then still give you three days to send out the created invoices to your customers.

- Per service contract, a BDR with exactly one item is created for each billing date of the billing plan (that is, for each billing request line). Even when different service contract items share the same billing dates, separate BDRs, each containing a single item, are created for each individual billing request line.
- BDRs cannot be changed, only deleted and then replaced. When a service contract item is changed, the data in the billing due list must be updated. How this is accomplished depends on the type of change that occurred.
  - For some types of changes (for example, a changed text on header or item level), the old BDR is deleted and a new BDR is created automatically to replace it.
  - For other types of changes (for example, a settlement rule is changed), the old BDR is deleted and a new BDR is not created until the next BDR creation run.

  If the BDR in question has already been billed to produce a billing document, the corresponding billing request line cannot be changed. If the billing document is canceled, the billing request line can once again be changed.

  ### Note

  It is recommended to create BDRs shortly before they are needed to minimize the number of BDRs that must be replaced because a service contract item has been changed between creation of the BDR and creation of the billing document.

- When an unbilled BDR is deleted or rejected in the **Manage Billing Document Requests** app, the status of the corresponding billing request line in the preceding service contract is updated to indicate that it has not been transferred to billing. This status reversal allows the billing request line to be released for billing again, which leads to the creation of a new BDR.

- When the billing document is created on the basis of the BDR, the pricing conditions are taken from the pricing document that is associated with the service contract.

Billing Status at Service Contract Item Level

The billing status at service contract item level represents the overall status of all the billing request lines that the billing plan of the item contains. The following table explains each status.

Billing Status at Item Level

Billing Status	Meaning
Not billed	Initial status when a billing plan is created. No billing request lines in the billing plan are invoiced.
Partially billed	Some billing request lines in the billing plan have been invoiced.
Completely billed	The cumulative net value of all the invoices is equal to or greater than the net value of the service contract item.

Related Information

Billing of Service Transactions

Creating and Processing Billing Documents

Ad Hoc Billing for Service Contracts

Setting Rules for Billing Plans in Service Contracts

Get an overview of how to set rules for billing plans in service contracts so that the system can calculate the settlement periods, billing dates, and billing values of service contract items.

Use

The system uses various rules to generate periodic billing plans for individual items of service contracts. The following rules control billing plans in service contracts:

- Customized rules for billing plan types in your system. These rules become available as default and alternative billing plan rules that business users can choose for service contract items.
- Settlement rules specified by a business user for a service contract item according to specific business requirements. For more information, see Settlement Rules in Service Contracts.

In both cases, the system automatically generates a billing plan that consists of a number of settlement periods. Each settlement period contains basic billing data, including the settlement start and end dates, billing date, billing value, and more, making up a billing request line for subsequent billing process.

Prerequisites

Most rules for billing plans originate from the date rule settings in date management. SAP predelivers specific date rules in the system. You can also create and set your own date rules. For more information, see Date Management.

Billing Plan Types

You can define billing plan types. To do so, go to the Customizing settings for Service under Transactions > Basic Settings > Billing Plan > Define Billing Plan Types. The following settings for billing plan types are the most important rules for the generation of billing plans.

Origin of General Data

Here, you can define the rules for the generation of the start and end date/time of settlement periods by choosing the rules for start and end dates and the rules for from-time and to-time respectively. These rules are based on the **date rules** that you have set up for date management. These rules convert the start and end date/time of service contract items to the settlement start and end date/time of billing plans.

→ Recommendation

We recommend that you use the same date rules for date and time. For example, use `CONT011` (Start Date) for both the start date and the time from while using `CONT012` (End Date) for both the end date and the time to.

When a business user creates a service contract item, the system uses the predefined rules to determine the settlement start and end dates and then generates a billing plan that covers the entire validity period of the service contract item.

i Note

In billing plans business users can only see dates (but not times). This is due to the integration of billing plans with price management. The from-time and to-time are only used for the back-end calculation of billing plans and are therefore not visible to business users.

For the start and end date/time of service contracts and service contract items, business users can specify both date and time.

SAP predeliveres two sets of settings to fulfill different purposes.

- `CONT006` Contract Start and `CONT007` Contract End

  Use `CONT006` and `CONT007` when you want the settlement start and end date/time of billing plans to exactly match the start and end date/time of service contract items. In this case, if a service contract item ends at 00:00:00 one day, the system doesn't count the day as a business day for which a whole-day's price should apply. This can be expressed as follows:

  - Settlement Start Date and Time = Contract Start Date and Time
  - Settlement End Date and Time = Contract End Date and Time

  The following example explains how the system works.

  ### Example

  Imagine a one-year service contract item that starts on January 1, 2020 at 00:00:00 and ends on January 1, 2021 at 00:00:00. Therefore, in the billing plan, the settlement period also starts on January 1, 2020 at 00:00:00 and ends on January 1, 2021 at 00:00:00.

Contract Start Date and Time	Contract End Date and Time	Settlement Start Date and Time	Settlement End Date and Time
2020.01.01 at 00:00:00	2021.01.01 at 00:00:00	2020.01.01 at 00:00:00	2021.01.01 at 00:00:00

  In this case, if a business user chooses to bill the item monthly, the system automatically generates a billing plan as shown in the following table:

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.01	2020.01.31	100	USD
2020.02.01	2020.02.29	100	USD
2020.03.01	2020.03.31	100	USD
2020.04.01	2020.04.30	100	USD
2020.05.01	2020.05.31	100	USD
2020.06.01	2020.06.30	100	USD
2020.07.01	2020.07.31	100	USD
2020.08.01	2020.08.31	100	USD
2020.09.01	2020.09.30	100	USD
2020.10.01	2020.10.31	100	USD
2020.11.01	2020.11.30	100	USD
2020.12.01	**2021.01.01**	100	USD

- `CONT011` Start Date (Billing Plan) and `CONT012` End Date (Billing Plan)

  Use `CONT011` and `CONT012` if you want a whole day's price not to apply when the item starts at 00:00:00 but a whole day's price to apply no matter whether the item starts at six in the morning or ends at five in the afternoon of the day. The system uses the following formulas:

  - Settlement Start date/time = Date (Contract Start Date) + Time (always at 00:00:00)

    The system uses the date that a business user sets for the start date of the service contract item. The system always sets the time to 00:00:00, no matter at which time the service contract item starts. For example, if the contract start date and time is January 1 at 08:00:00, the settlement start date is January 1 at 00:00:00.

  - Settlement End date/time = Date (Contract End date/time minus one second) + Time (always at 23:59:59)

    The system uses the date minus one second from the contract end date and time. For example, if the contract end date and time is January 1 at 00:00:00, the settlement end date is December 31 at 23:59:59. If the contract end date and time is January 1 at 12:00:00, the settlement end date and time is still January 1 at 23:59:59.

  The following example explains how the system works.

Example

Again, we use a one-year service contract item that starts on January 1, 2020 at 08:00:00 and ends on January 1, 2021 at 00:00:00. Therefore, in the billing plan, the settlement period also starts on January 1, 2020 at 00:00:00 but ends on December 31 at 23:59:59.

Contract Start Date and Time	Contract End Date and Time	Settlement Start Date and Time	Settlement End Date and Time
2020.01.01 at 08:00:00	2021.01.01 at 00:00:00	2020.01.01 at 00:00:00	2020.12.31 at 23:59:59

In this case, if a business user chooses to bill the item monthly, the system automatically generates a billing plan as shown in the following table:

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.01	2020.01.31	100	USD
2020.02.01	2020.02.29	100	USD
2020.03.01	2020.03.31	100	USD
2020.04.01	2020.04.30	100	USD
2020.05.01	2020.05.31	100	USD
2020.06.01	2020.06.30	100	USD
2020.07.01	2020.07.31	100	USD
2020.08.01	2020.08.31	100	USD
2020.09.01	2020.09.30	100	USD
2020.10.01	2020.10.31	100	USD
2020.11.01	2020.11.30	100	USD
2020.12.01	**2020.12.31**	100	USD

Billing Data for Date Proposals

Here, you define the **default** settlement rules for the following three aspects:

- How frequently the periodic billing is scheduled, such as monthly, quarterly, or yearly. This setting also determines how many billing request lines the system creates for the periodic billing plan.
- On which date the billing is to occur, such as on the same day of the settlement start or 14 days after the settlement start.
- On which date a valid pricing condition applies for billing, such as on the same day of the billing date or on the day when the service contract item is created. This setting is only relevant for price adaptation items. For standard service contract items, the default setting is always the item pricing date, which means the day when the service contract item is created.

These rules are based on the date rules that you have set up for date management. When a business user creates a new service contract item, the system uses these default settings to generate the settlement periods, billing dates, and billing values along with other relevant information. A business user can always choose a different rule to launch the system calculation again.

Control Data for Creating Dates

Here, you define **days in month** for monthly fees and **days in year** for yearly fees. The system uses these values to calculate settlement periods, independent of the actual number of calendar days per month or year. These settings impact the calculation of the billing value of each settlement period according to the price condition type that you use for your service contracts. The settings predelivered by SAP are 30 days per month and 360 days per year for price condition type `PSI1` (monthly fee).

→ Recommendation

In the following two special cases, we recommend that you make the respective settings:

- Case 1: When you use a yearly price condition and expect the same price for the yearly billing of both normal years and leap years, apply **empty** to both settings.
- Case 2: When you use a yearly price condition but want prices calculated according to the days contained in the year, use **365** for days in year.

Assigning Date Rules to Settlement Period and Billing Date

You can customize date rules for service contracts. Note that all of these rules, whether they are relevant or not, are displayed as options for determining settlement periods and billing dates in the **Manage Service Contracts** app. Telling the rules apart is only possible based on the explanatory texts, which may cause misunderstanding and inconvenience for your business users. However, you can improve this situation by adding the corresponding date profile, and filtering the date rules in the date profile, using the customizing activity **Service > Transactions > Basic Settings > Billing Plan > Assign Date Rules to Settlement Period and Billing Date**. As a result, only the rules that are assigned to the corresponding purposes are displayed in the app.

Example

- Settlement period rules are the rules that determine the settlement frequency, such as monthly, quarterly, or yearly.
- Billing date rules determine the billing dates within the settlement periods, such as to first of the month, on the settlement start date, or 14 days after settlement start.

Related Information

[Periodic Billing Plans for Service Contract Billing](#)
[Date Management](#)
[Settlement Rules in Service Contracts](#)

Settlement Rules in Service Contracts

Get an overview of the settlement rules that you can specify for your service contract items. The system uses these rules to calculate the settlement periods, billing dates, and billing values.

Use

The system uses various rules to generate periodic billing plans for individual items of service contracts. The following rules control billing plans in service contracts:

- Customized rules for billing plan types in your system. These rules become available as default and alternative billing plan rules that business users can choose for service contract items. For more information, see [Setting Rules for Billing Plans in Service Contracts](#).
- Settlement rules specified by a business user for a service contract item according to specific business requirements.

In both cases, the system automatically generates a billing plan that consists of a number of settlement periods. Each settlement period contains basic billing data, including the settlement start and end dates, billing date, billing value, and more, making up a billing request line for the subsequent billing process.

Prerequisites

1. Most rules are based on the date rule settings for date management. SAP predelivers specific date rules in the system. You can also create and customize your own date rules. For more information, see Date Management.

2. You have made Customizing settings for billing plan types. These settings impact the default behaviors of the system. For more information, see Setting Rules for Billing Plans in Service Contracts.

Settlement Rules

As a business user, you can specify settlement rules to change the default periodic billing plans that the system generates automatically. You can specify three types of settlement rules: settlement period rules, billing date rules, and pricing date rules.

Settlement Period Rules

Settlement period rules determine how frequently the periodic billing is scheduled, such as monthly, quarterly, or yearly. They also determine how many billing request lines the system creates for the service contract item. SAP predelivers several settlement rules to fulfill various business requirements.

- Standard settlement period rules (such as billing monthly, quarterly, half-yearly, and yearly)

  With these rules, the system divides settlement periods into equal intervals by adding one interval to every settlement start date.

  The following example explains how the system works in most cases when the contract start date is earlier than the 28th of a month or equal to the 28th.

  Example

  Contract start date and time: 2020.01.01 at 00:00:00

  Contract end date and time: 2020.12.31 at 23:59:59

  Settlement period rule: Monthly

  The system automatically generates a billing plan as shown in the following table:

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.01	2020.01.31	100	USD
2020.02.01	2020.02.29	100	USD
2020.03.01	2020.03.31	100	USD
2020.04.01	2020.04.30	100	USD
2020.05.01	2020.05.31	100	USD
2020.06.01	2020.06.30	100	USD
2020.07.01	2020.07.31	100	USD
2020.08.01	2020.08.31	100	USD
2020.09.01	2020.09.30	100	USD
2020.10.01	2020.10.31	100	USD
2020.11.01	2020.11.30	100	USD

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.12.01	2020.12.31	100	USD

One special scenario is when the contract start date is later than the 28th of a month, which is likely to place one settlement end date at the end of February. The system sets the settlement end date at one day before the end of February and creates an additional interval at the end of the billing plan. The following example explains how the system works in this case.

Example

Contract start date and time: 2020.01.30 at 00:00:00

Contract end date and time: 2021.01.29 at 23:59:59

Settlement period rule: Monthly

The system automatically generates a billing plan as shown in the following table. Note that the billing value of the first month is less (due to a monthly fee, `PSI1`, set for 30 days in month) and the shortfall is compensated by the last settlement period.

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.30	2020.02.28	**96.67**	USD
2020.02.29	2020.03.28	100	USD
2020.03.29	2020.04.28	100	USD
2020.04.29	2020.05.28	100	USD
2020.05.29	2020.06.28	100	USD
2020.06.29	2020.07.28	100	USD
2020.07.29	2020.08.28	100	USD
2020.08.29	2020.09.28	100	USD
2020.09.29	2020.10.28	100	USD
2020.10.29	2020.11.28	100	USD
2020.11.29	2020.12.28	100	USD
2020.12.29	**2021.01.28**	100	USD
2020.01.29	**2021.01.29**	**3,33**	USD

- Settlement period rules based on calendar months (such as billing monthly, quarterly, half-yearly, and yearly by calendar)

  The system always cuts off settlement periods at the end of a calendar month, until the last interval, which ends at the contract end date. The following example illustrates one of these rules.

Example

Contract start date and time: 2020.01.16 at 00:00:00

Contract end date and time: 2021.01.15 at 23:59:59

Settlement period rule: Monthly by calendar

The system automatically generates a billing plan as shown in the following table:

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.16	**2020.01.31**	**50**	USD
2020.02.01	2020.02.29	100	USD
2020.03.01	2020.03.31	100	USD
2020.04.01	2020.04.30	100	USD
2020.05.01	2020.05.31	100	USD
2020.06.01	2020.06.30	100	USD
2020.07.01	2020.07.31	100	USD
2020.08.01	2020.08.31	100	USD
2020.09.01	2020.09.30	100	USD
2020.10.01	2020.10.31	100	USD
2020.11.01	2020.11.30	100	USD
2020.12.01	2020.12.31	100	USD
2021.01.01	**2021.01.15**	**50**	USD

- Rules for moved settlement periods (such as billing monthly, quarterly, and half-yearly)

  These rules can help you avoid the additional interval and a splitting of billing values generated by the system when your contract start date is later than the 28th of the month. To do so, the system uses the days contained in the **next** interval to calculate settlement end dates, instead of using the days contained in the current interval. We use the term "moved" (from the current interval to the next interval) to describe that system behavior. Note that in case you use quarterly or half-yearly billing, the move always starts from the next month.

  You can also use these rules when your contract start date is earlier than or equal to the 28th of the month. In this case, the system doesn't apply the "move" mechanism.

  The above can be summarized as follows:

  - If the contract start date is after the 28th of the month: Settlement End Date = Settlement Start Date + Days Contained in the Next Interval – 1 Day
  - If contract start date is earlier than and equal to the 28th of the month: Settlement End Date = Settlement Start Date + Days Contained in the Current Interval – 1 Day

  ### i Note

  - The contract start date defines the first settlement start date. Thereafter, the system uses the previous settlement end dates plus one day to determine the next settlement start dates.

- If the interval is monthly, the next interval refers to the next month. If the interval is quarterly or half-yearly, the next interval refers to the quarter or the half year starting from the next month.

The following example illustrates one of these rules.

Example

Contract start date and time: 2020.01.30 at 00:00:00

Contract end date and time: 2021.01.29 at 23:59:59

Settlement period rule: Monthly (Moved Settlement Period)

The system automatically generates a billing plan as shown in the following table:

Settlement Start Date	Settlement End Date	Billing Value	Currency
2020.01.30	**2020.02.27**	**100**	USD
2020.02.28	2020.03.29	100	USD
2020.03.30	2020.04.28	100	USD
2020.04.29	2020.05.29	100	USD
2020.05.30	2020.06.28	100	USD
2020.06.29	2020.07.29	100	USD
2020.07.30	2020.08.29	100	USD
2020.08.30	2020.09.28	100	USD
2020.09.29	2020.10.29	100	USD
2020.10.30	2020.11.28	100	USD
2020.11.29	2020.12.29	100	USD
2020.12.30	2021.01.29	100	USD

Let's check the first two rows together:

1. The first settlement period starts in January. The system uses 29 days because the next month is February and 2020 is a leap year. After subtracting one day, the system determines February 27 as the first settlement end date.

2. The system first adds one day to the previous settlement end date February 27 to get the settlement start date of February 28. The system then uses 31 days because the next month is March. By subtracting one day, the system determines March 29 as the settlement end date.

Billing Date Rules

The system uses the specified billing date rule to schedule the dates on which billing is to occur and the resulting billing document posted to financial accounting. The system offers billing date rules predelivered by SAP and set up through Customizing settings.

Pricing Date Rules

The system uses pricing date rules to determine a pricing date for each settlement period. Subsequently, the system looks for the valid pricing condition on that date so that it can calculate the billing value in each settlement period. Note that you can specify pricing date rules only when the service contract item is for price adaptation. The following table explains the pricing date rules predelivered by SAP:

Pricing Date Rule	Meaning
Item pricing date	Uses the date on which the service contract is created for pricing. This is the default and unchangeable setting for standard service contract items.
Billing date in billing request line	Uses billing dates for pricing.
Settlement start date in billing request line	Uses settlement start dates for pricing.
Settlement end date in billing request line	Uses settlement end dates for pricing.

Related Information

[Setting Rules for Billing Plans in Service Contracts](#)
[Date Management](#)
[Manage Service Contracts](#)
[Price Adaptation for Service Contracts](#)
[Periodic Billing Plans for Service Contract Billing](#)

Ad Hoc Billing for Service Contracts

Get an overview about ad hoc billing.

Use

Ad hoc billing is a structured method of service contract billing that doesn't conform to periodic or milestone billing plan rules, but instead enables you to define billing values and dates freely.

Prerequisites

1. You have made the necessary settings for ad hoc billing plans in Customizing under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Define Billing Plan Types**. In the **BilPlanUse** field, choose **Ad Hoc Billing**.

2. The billing plan type **Ad Hoc Billing** is assigned to the relevant item category. You do this under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Assign Billing Plan Type to Item Category**. The predefined Customizing settings include the transaction type SC and the item category SCNB.

Features

When you create a service contract item, you can choose a product and assign it to the item category **SC Itm (Ad Hoc Bill)** (SCNB), which is intended to be used exclusively for ad hoc billing.

i Note

Be aware that you can't switch to other item categories once you have selected the item category for ad hoc billing.

Ad Hoc Billing Plan

You can use an ad hoc billing plan to define the dates on which billing is to occur and the value or percentage that is to be billed. For each billing date, the system generates one billing request line. In addition, you can optionally specify the settlement period that an individual billing request line covers.

After you create a billing request line for ad hoc billing, processing of billing request lines and the subsequent billing process is the same as for the periodic billing plan.

> **Note**
>
> You must schedule batch jobs to schedule the periodic creation of billing document requests on the basis of open billing request lines. You can do this on the **SAP Easy Access** menu, under **Service** > **Service Processes** > **Generate Billing Document Requests**. For more information, see the section *Billing Request Lines and Billing Document Requests* under [Periodic Billing Plans for Service Contract Billing](#).

You can trigger a warning or error message when the sum of billing values across all billing request lines does not equal the corresponding item net value. This can be achieved by configuring the appropriate **Message Type** in the **Induvidual reporting** for an **Application area** under Customizing **Service** > **Transactions** > **Basic Settings** > **Define Attributes for System Messages**.

Billing Rule

You can use value based billing or percentage based billing to generate the billing request lines for a service contract item.

- In value based billing, you can enter the billing value for a billing request line.
- In percentage based billing, you define the billing request line percentage, which calculates the billing value. When the net value of the service contract item changes, the billing value is recalculated based on the defined percentage. If some lines have already been billed before the net value change, the subsequent lines are recalculated based on the percentage, and the percentage for the already billed lines is adjusted according to the amount billed.

Billing of Remaining Net Value

This feature enables the system to automatically generate a billing request line with all the remaining net value of the service contract item. The billing date of this billing request line is set to the current date by default.

Billing Finalization

Ad hoc billing allows a total billing value higher, lower, or equal to your item net value. Therefore you must set the **Billing Plan Is Finalized** indicator to let the system know that the billing plan is finalized, namely that no more billing value is to be planned. After you set the indicator, the billing status of the ad hoc billing item is set to completely billed when all the billing request lines in the billing plan are invoiced.

Item Fixed Price

Unlike the service contract items, whose prices are calculated based on the contract validity period, ad hoc billing items can be set to a fixed price by using a price condition type of manual item amount (`PMV1`). By doing so, the net value of the ad hoc billing item is always fixed at the manual item amount that you specify, despite the change to its validity period.

Billing Status at Service Contract Item Level

You can always view the billing status at service contract item level. It represents the overall status of all billing request lines that the billing plan of the item contains. The following table explains each status.

Billing Status at Item Level

Billing Status	Meaning
Not billed	Initial status when a billing plan is created. No billing request lines in the billing plan have been invoiced.
Partially billed	Some billing request lines in the billing plan have been invoiced. Note that for ad hoc billing items, the partially billed status is displayed even when all the billing request lines have been invoiced, but the ad hoc billing plan is not finalized.
Completely billed	All billing request lines in the billing plan have been invoiced, under the condition that the ad hoc billing plan is finalized.

Related Information

[Service Contract Item]
[Differences in Item Categories]
[Periodic Billing Plans for Service Contract Billing]

Billing Plan Templates for Ad Hoc Billing Plans in Service Contracts

Billing plan templates allow you to specify reusable sets of data needed to create ad hoc billing plans for service contract items.

Use

You can use a billing plan template to create billing request lines (BRLs) for an ad hoc billing plan in advance so that BRLs can be added to the billing plan automatically. This way, you don't need to create BRLs manually and minimize the time to create an ad hoc billing plan for a service contract item.

Procedure

The following steps describe how to use a billing plan template for an ad hoc billing plan:

1. Create a billing plan template using the Customizing activity **Create Billing Plan Templates** under **Service** > **Transactions** > **Basic Settings** > **Billing Plan**.

2. In the billing plan template, create billing plan template lines and enter data in each billing plan template line that corresponds to the data for a billing request line (BRL).

3. On the **Manage Service Contracts** app, choose **Create from Template** on the **Billing** tab when creating an ad hoc billing plan for a service contract item.

4. From the popup, search for a billing plan template and select the one you want to use for the ad hoc billing plan. This way the data from the billing plan template lines that you have created is copied to the ad hoc billing plan as BRLs.

Note

You can create multiple billing plan templates according to your business requirements.

Related Information

Ad Hoc Billing for Service Orders

You can use ad hoc billing to define billing values and dates freely for fixed price service order items. This feature is particularly helpful if you want to create billing document requests (BDRs) for fixed-price service order items with long duration that you want to be able to bill on specific dates and not just when the service order item is completed.

Use of Ad Hoc Billing Plan

You can use an ad hoc billing plan to define the dates on which billing is to occur and the value that is to be billed. For each billing date, you can enter a billing request line (BRL) manually.

Prerequisites

To use ad hoc billing plan for service order items, you need to set up the appropriate pricing procedure. The billing value of each BRL of a service order item corresponds to the manually entered amount for a fixed price condition type. Therefore, you need to use a condition type in pricing that allows you to manually add a fixed amount for the item condition. You can use the predefined condition type `PMV1` (Manual Item Amount) in the Customizing activity **Set Condition Types for Pricing in Sales** under **Sales and Distribution > Basic Functions > Pricing > Pricing Control > Define Condition Type**.

Note

You can only use positive values for the condition type `PMV1`.

Item Categories for Ad Hoc Billing

For ad hoc billing, you use the following predefined item categories:

- **Service Product Item (Ad Hoc Billing)** (`SBOS`)
- **Expense Item (Ad Hoc Billing)** (`SBOE`)
- **Service Part (Ad Hoc Billing)** (`SBOM`)
- **Intercompany Service Item (Ad Hoc Billing)** (`SVAI`)

Note

- If you want to use the item category **Intercompany Service Item (Ad Hoc Billing)**, you need to activate item-based accounting.
- If you want to change the item category for an item using ad hoc billing plan, you need to delete the data of the billing request line of the item.

Features

Schedule Creation of BDRs

To create the billing document requests for open billing request lines in billing plans, you can manually schedule jobs by using the report **Generate BDRs for Service Orders** (`CRMS4_SERVORD_BDR_GENERATE`). You can find the report on the **SAP Easy Access** menu under **Service** > **Service Processes**.

Billing of Remaining Net Value

If you want to bill the remaining net value of a service order item in one go, you can have the system automatically generate a billing request line with all the remaining net value. The billing date of this billing request line is set to the current date by default.

Billing Status of Service Order Items

You can view the billing status for each service order item on the **Billing** tab. It represents the overall status of all billing request lines in the billing plan of the item. The following table explains each status.

Billing Status at Item Level

Billing Status	Description
Not Billed	Initial status when a billing plan is created. No billing request lines in the billing plan have been invoiced.
Partially Billed	Some billing request lines in the billing plan have been invoiced.
Completely Billed	All billing request lines in the billing plan have been invoiced and the item status is set to **Completed**.

Planned Revenue of Items with Billing Plans

For service order items that have a billing plan assignment, the planned revenues are taken from the billing request lines and not from the item net value. The consequences of this are as follows:

- If the billing request lines are not defined in the billing plan, the ongoing planned revenue will always be 0.
- If the total value of the billing request line is lower or higher than the net value, this might have an impact on the planned revenue calculation and on event-based revenue recognition.
- If you cannot ensure that before a service order item is set to the status **Released**, all billing request lines have been created and defined in the billing plan, SAP advises that you do not use the revenue recognition method "Recognize Revenue on Cost-Based Percentage of Completion (POC) (Method 3)". For more information, see Revenue Recognition Methods for Service Order Items.

i Note

Planned cost and revenue of service order items can only be calculated when item-based accounting is active.

Percentage-Based Billing for Ad Hoc Billing Plans and Billing Plan Templates

In percentage-based billing, you enter the billing request line percentage, which calculates the billing value in an ad hoc billing plan automatically. When the net value of the service order item changes, the billing value is recalculated based on the entered percentage. If some lines have already been billed before the net value change, the subsequent lines are recalculated based on the percentage, and the percentage for the already billed lines is adjusted according to the amount billed. You can use percentage-based billing when you create a billing plan template as well.

Implementation Details

When you define a billing plan type with the billing plan usage **Ad Hoc Billing** under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Define Billing Plan Types**, you can enable the percentage-based billing for billing plans using the **Billing Rule Percentage Based Billing (Ad Hoc)** .

Related Information

[Billing Plan Templates in Service Quotations](#)

[Billing](#)

Billing Plan Templates for Ad Hoc Billing Plans in Service Orders

Billing plan templates allow you to specify reusable sets of data needed to create ad hoc billing plans for service order items.

Use

You can use a billing plan template to create billing request lines (BRLs) for an ad hoc billing plan in advance so that BRLs can be added to the billing plan automatically. This way, you don't need to create BRLs manually and minimize the time to create an ad hoc billing plan for a service order item.

Example

The following example shows you a sample billing plan template and the ad hoc billing plan created from the template:

Billing Plan Template A

Line Number	Description of Billing Plan Template Line	Billing Value Percentage	Billing Block Reason	Date Rule	Description of Date Rule
0010	Bill 20% at start of service	20,00	None	`TODAY`	Today's Date
0020	Bill the rest at the end of service	80,00	Check Terms of Payment	`BILL0006`	Half-Yearly (Settlement Period)

Ad Hoc Billing Plan Based on Billing Plan Template A

Billing Request Line	Description of Billing Request Line	Billing Value Percentage	Billing Block Reason	Date Rule	Description of Date Rule
0010	Bill 20% at start of service	20,00	None	`TODAY`	Today's Date
0020	Bill the rest at the end of service	80,00	Check Terms of Payment	`BILL0006`	Half-Yearly (Settlement Period)

Procedure

The following steps describe how to use a billing plan template for an ad hoc billing plan:

1. Create a billing plan template using the Customizing activity **Create Billing Plan Templates** under **Service > Transactions > Basic Settings > Billing Plan**.
2. In the billing plan template, create billing plan template lines and enter data in each billing plan template line that corresponds to the data for a billing request line (BRL).
3. On the **Manage Service Orders** app, choose **Create from Template** on the **Billing** tab when creating an ad hoc billing plan for a service order item.

4. From the popup, search for a billing plan template and select the one you want to use for the ad hoc billing plan. This way the data from the billing plan template lines that you have created is copied to the ad hoc billing plan as BRLs.

i Note

You can create multiple billing plan templates according to your business requirements.

Date Rule

A date rule is a method for calculating times. In an ad hoc billing plan, the date on which billing is carried out is determined by the date rule that you select for each BRL. A date rule is based on the date types (for example, a day, a week, or a month) that you provide for a date profile. You then assign a date profile to an item category used in a service order. This way, a billing date can be calculated for a service order item that uses ad hoc billing plan.

You can create date rules in Customizing according to your requirements under **Service** > **Basic Functions** > **Date Management** > **Define Date Types, Duration Types, and Date Rules** .

For more information, see Date Management.

Automatic Determination of Billing Plan Templates for Ad Hoc Billing Plans

When you create an ad hoc billing plan for a service order item, you can use the function **Create from Template** to fill the ad hoc billing plan with predefined billing request lines from a billing plan template. Now, you don't need to manually choose **Create from Template** to copy from a billing plan template because a billing plan template can be determined automatically according to specific keys entered for a service order item, for example, a sales area.

Implementation Details

When you define a billing plan type with the billing plan usage **Ad Hoc Billing** under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Define Billing Plan Types**, you can enable the automatic determination of billing plan templates via the checkbox **Determine Billing Plan Template**.

Under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Create Billing Plan Templates** you can now optionally enter an item category and assign it to a billing plan template. It is possible to assign several item categories to one template. It is also possible to assign one particular item category to several templates. However, when you have activated the automatic determination of a billing plan template for billing plan types with the usage "Ad Hoc Billing" for service order items, it is recommended to keep a 1:1 assignment so that the system can then automatically determine uniquely from which template the billing request lines are to be created.

Assign Item Categories to Billing Plan Templates

When you create billing plan templates, you can now assign item categories to billing plan templates. You can choose these item categories in the value help for billing plan templates that is called up for **Create from Template**.

Implementation Details

You can define the assignment of item categories in the new view **Item Categories of Billing Plan Template**. You can find this view under **Service** > **Transactions** > **Basic Settings** > **Billing Plan** > **Create Billing Plan Templates**.

Related Information

Ad Hoc Billing for Service Orders

Billing Plan Templates in Service Quotations

Billing Plan Templates in Service Quotations

Assign billing plan templates to service items, service parts, or expense items in service quotations to predetermine the billing conditions.

Use

Billing plan templates are reusable sets of data that are used to create ad hoc billing plans for service items, service parts, and expense items in service orders. However, you can assign billing plan templates to service quotation items to predetermine the billing conditions. This enables you, for example, to include the billing conditions defined in the billing plan template in the output document for the service quotation. When you create a follow-up service order to the service quotation, the system automatically creates the billing request lines in the follow-up service order items based on the billing plan template.

Prerequisites

Creating Billing Plan Templates

You create billing plan templates in Customizing for **Service** under **Transactions** > **Basic Settings** > **Billing Plan** > **Create Billing Plan Templates**. For detailed information about billing plan templates and ad hoc billing, see Billing Plan Templates for Ad Hoc Billing Plans in Service Orders and Ad Hoc Billing for Service Orders.

Creating Billing Request Lines in Service Order Items

Billing request lines can only be created for service order item categories for which ad hoc billing is enabled in Customizing for **Service** (see Item Categories for Ad Hoc Billing). In addition, define service quotation item categories that correspond to these service order item categories. This enables the system to automatically create billing request lines in service order items based on the billing plan templates for the service quotation items.

Including Information about Billing Plan on Output Documents

If you want to include texts in the output document that have been defined in the billing plan template, add the required fields to your output document.

Example

In the long text for the billing plan template, you can provide a text summarizing the billing conditions that have been defined in the billing plan template.

Processing Billing Plan Templates in Service Quotations

You process billing plan templates in service quotations as follows:

1. Assign the billing plan template in the **Billing** assignment block in service items, service parts, or expense items.

   You can do this if the items have been set to the status **Open** or **In Process**.

2. Send the service quotation to your customer.

   In the output document that is created for the service quotation, you can include the attributes that have been defined for the billing plan template in Customizing.

3. Create a follow-up service order to the service quotation, for example, when accepting the service quotation.

If the service quotation has an item category assigned that enables ad hoc billing in the follow-up service order, the system automatically generates the billing request lines that are displayed in the **Billing Plan** assignment block in the service order item.

4. When you create a requote or a copy of a service quotation, the assigned billing plans are copied over to the new service quotation created.

Related Information

[Billing Plan Templates for Ad Hoc Billing Plans in Service Orders]

[Ad Hoc Billing for Service Orders]

[Service Quotations]

[Creating Requotes]

[Output Management for Service Orders, Service Quotations, and Service Contracts]

Creating and Processing Billing Documents

Get an overview of how the billing documents containing billing data from service transactions can be created and processed further.

Preparing Billing Document Creation for Service Orders and Service Confirmations

After you have completed a service order or service confirmation that you want to bill, you need to release the corresponding items for billing on the **Release for Billing** screen. Releasing triggers the next step, in which the billing data of service items and service part items is added to the billing due list in the form of one or more billing document requests (BDR). You can view the billing due list in the **Create Billing Documents** app. For more information about billing document requests, seeBilling Document Requests.

i Note

If your service transaction includes sales items, these items are automatically added to the billing due list in the form of a sales order or delivery when you complete the service transaction.

Preparing Billing Document Creation for Service Contracts

To bill service contracts, you must create one or more periodic batch jobs. When a job runs, it creates BDRs for all open (billable) service contract items with a billing date up to the specified horizon. For more information, see Periodic Billing Plans for Service Contract Billing. The resulting BDRs can then be processed from the billing due list as described below.

Creating Billing Documents

You can use the **Create Billing Documents** app to manually create individual and collective billing documents from the BDRs, sales orders, deliveries, and other billable documents that appear in the billing due list. As long as key header data (such as business partners and payment terms) is identical for each document, you can use omnichannel convergent billing to converge these different document types into single, combined customer invoices. If one or more header fields are not identical, the resulting invoice split triggers the creation of two or more separate invoices.

Example

You create a service order that contains one service item, one service part item, and one sales item. After completing the service order, the sales item will automatically be added to your billing due list in the form of a sales order. Once you release the service order items for billing on the **Release for Billing** screen, a BDR containing the service items and service part items will also be added to your billing due list. Through omnichannel convergent billing, you can now bill the BDR and the corresponding sales order together to create a single, combined customer invoice.

When a billing document based on a billing document request from Service is created, the billing document number is written to the transaction item's document flow, the billed value is updated, and the fully billed status is set for the item within the relevant service apps.

Note

When billing BDRs based on service transactions in the **Create Billing Documents** app, ensure that the billing setting for creating separate billing documents for each item of the billing due list is set to OFF. When switched ON, this setting prevents BDRs (and other preceding documents) from being converged into combined billing documents.

If you want to schedule billing document creation based on certain criteria (for example, billing type, billing date, and sold-to party) you can use the **Schedule Billing Creation** app to schedule billing creation jobs to run at predefined times or at periodic intervals.

For more information about creating billing documents, see

Processing Billing Documents

After billing documents have been created in the system, you can use a range of apps to help you process them quickly and efficiently. Processing functionality includes the following:

- Display both simplified and detailed overviews of billing documents
- Preview billing document output (PDF format)
- Output billing documents to email, fax, and print (you can also schedule output)
- Post billing documents to financial accounting (including scheduled posting)
- Cancel billing documents

  Note

  When a billing document based on a service billing document request is canceled, the following information is rolled back:

  - The billing document number is removed from the transaction items' document flow
  - The billed value is reversed
  - The fully billed status of items is reset to the open status

  When the billing document is canceled, the preceding BDR is added back into the billing due list and can be billed again to create a new billing document. Note that if you also want to cancel the preceding service transaction, you must first reject the corresponding BDR. You can do this in the **Manage Billing Document Requests** app.

- Add and edit texts and attachments to billing documents

  Note

  The text that you enter into the service transaction (and choose to keep) will be included on the final billing document. Once the billing document has been created, you can decide to keep, edit, or delete this initial text. If you decide to keep it, it will appear on billing output. Payment card data (if applicable) is also transferred from business transactions to billing documents and subsequent output.

For more information about processing billing documents, see Processing Billing Documents.

Prerequisites

- You have ensured that the conditions for copying sales document to billing documents meet your requirements, as described in Customizing under **Sales and Distribution** > **Billing** > **Billing Documents** > **Maintain Copying Control for Billing Documents**.

Creating a Credit Memo Request with Reference to a Service Invoice

Get an overview of why and how to create a credit memo request that references to an invoice originating from service.

A credit memo request is a type of sales document used to request a credit memo for a customer. Creating credit memo requests makes sense when the price that the customer has paid was too high. This could be due to lacking quality of a provided service, a mismatch between billed and delivered quantities, or delivery of an incorrect product. You can create credit memo requests for the following service transaction:

- Service contracts
- Service Orders
- Service Confirmations

Business users with the `SAP_BR_INTERNAL_SALES_REP` business role can create credit memo requests by using the **Manage Credit Memo Requests** app and entering a document type for service (the credit memo request for service `SCR`). You must reference a billing document (invoice only) when you create a credit memo request.

Note

You should never create new items inside a credit memo request because they wouldn't have the cost assignment for service transactions (which are usually copied from the reference document).

After a credit memo request is created, you can find it in the transaction history of the corresponding service transaction. If you cancel the credit memo request that is created in this way, the cancellation is also displayed in the transaction history.

Note

A credit memo is a type of billing document. If you want to create a credit memo directly, see more information in Creating and Processing Billing Documents.

Related Information

Manage Credit Memo Requests

Multilevel Categorization

Use

You can use this function to set up multilevel categorization. Multilevel categorization enables you to flexibly design categorization. You also use it to integrate functions such as auto suggest and content analysis in your applications.

Integration

Multilevel categorization is available in varying scope for the following:

- Service orders
- Service confirmations
- Service requests
- Knowledge articles
- Rule modeler
- E-Mail Response Management System (ERMS)
- Service contracts

Prerequisites

You have set up multilevel categorization in Customizing under **Service** > **Cross-Application Components** > **Multilevel Categorization**.

Features

- Assigning categorization schemas to applications

  Depending on the default system settings and the settings in Customizing, you can assign schemas to applications, as follows:

  - Using a subject profile

    This form of assignment is possible for service orders.

  - Using a combination of transaction type and catalog category

    This form of assignment is possible for service requests and knowledge articles.

  - Using a combination of item category and catalog category

  Depending on the form of assignment, different categorization attributes are available to you. For example, you can use multilevel categorization to influence response profiles and warranty determination, only if the assignment is made using a subject profile.

- Multiple blocks of categorization fields

  When schemas are assigned using transaction types and item categories, you can assign multiple schemas to an application. These are included in the application in separate blocks of categorization fields.

  For service requests, the categorization blocks **Subject** and **Reason** are delivered at header level.

- **Creating categorization schemas**

  You can create schemas that are tailored to your business requirements, and then assign them to applications. You do this in the category modeler in the WebClient UI in Service. For more information, see Category Modeler.

- **Hierarchy categorization or attribute categorization**

  Depending on the business scenario, you can structure the categories in a schema strictly hierarchically, or in the form of value combinations. For more information, see Forms of Categorization.

- **Versioning**

  You can use a combination of various schema statuses and the temporal validity of a schema, to version your categorization schemas.

  You can use versioning to ensure that historical documents are preserved. From the time stamp of a document (created on), you can determine the categorization schema on which the document was based.

  The option of time-based schemas is advantageous in the public sector, for example. If the law changes on a certain date, the validity of a schema may have to be adapted to the validity of certain legislation.

- **Content analysis**

  Content analysis is used in the E-Mail Response Management System (ERMS). Content analysis determines the most suitable categories depending on the content of incoming e-mails, and automatically provides category suggestions. For more information, see Content Analysis.

- **Auto suggest**

  As part of multilevel categorization, the assignment of objects to categories in the category modeler enables automatic object suggestions. In a service order for example, if you select a category to which a service order template is assigned, you can choose to automatically complete the service order, using this template. For more information, see Auto Suggest.

- **Merging and importing categorization schemas**

  You can merge the categories and object links of two schemas. An import function supports cross-client and cross-system schema transports. For more information, see Merge Categorization Schemas and Import Categorization Schemas.

- **Authorizations**

  You can use authorization object CRM_CATEGO to set up authorizations for the category modeler. If you assign the display authorization appropriately, you can also use the authorization concept to configure the layout of the category modeler to be user-dependent. For more information, see Authorizations for the Category Modeler.

Activities

→ Recommendation

Set up a background job to run the shared memory manager overnight at least once a week on the **SAP Easy Access** screen under **Interaction Center** > **E-Mail Response Management System** > **Utilities** > **Shared Memory Manager**.

More Information

Inheritance of Reference Objects and Categorizations in Transactions

Category Modeler

Use

Using this function, you can create categorization schemas and assign these schemas to applications.

In addition, you can use a categorization schema to make additional functions and information available in your applications. For example, you can assign knowledge articles to categories. When you select a corresponding category in a service order, you can choose to add the knowledge articles that are assigned to the category to the service order.

Features

Page Area	Function	Description
Search result list and Categorization Schema	New	You can create categorization schemas. In some cases, you may not be able to change a schema because you do not have the necessary authorization.
	Version	When you create a new version of an existing schema, the original version of the copied schema remains but it is no longer changeable, and also cannot be used in future. Historical documents, such as service orders that refer to the original version of the schema, remain consistent.
	Copy Schema	You can copy an existing schema, and make changes to the copy. When you copy a schema, you create a new schema with a new ID. The original schema remains unchanged.
Search result list	Merge	You can merge categorization schemas.
	Import	You can import categorization schemas.
Categorization Schema	Translation	You can use this feature to translate categorization schemas in the WebClient UI. The translation function includes the following features: • Translation from any source language in which the categorization schema is already available to any target language that is installed in the system • Translation of schema and category descriptions • Copying of the source language entries to the translation fields to facilitate the editing process or to use source language entries in the target language version of the categorization schema Copying over the source language entries does not overwrite existing translations in the target language. • Update of any connected development systems (RFC) with the new translations provided

Page Area	Function	Description
Category Hierarchy > Categorization Schema > General Data	Logical Structure	You can use either hierarchical categorization without multiple usage of identical categories within a schema, or attributive categorization with multiple usage of identical categories within a schema.
	Authorization Mode	If you assign a schema to more than one application ID, you can use the authorization mode to control to which degree the authorizations that were predefined for each application ID, such as change authorization, are valid for this schema.
	Authorization Mode: AND Combination	If you assign a schema to more than one application ID, authorizations such as change authorization must be predefined for all application IDs for them to be valid for this schema.
	Authorization Mode: OR Combination	If you assign a schema to more than one application ID, all authorizations that are predefined for one application ID are transferred to the other application IDs, and are therefore valid without exception for this schema.
	Status: Draft	Newly created schemas, or schemas in editing mode, have Draft status.
	Status: Released	When you have finished editing a schema, set the status to Released.
	Status: Active	The Released schema status is automatically switched to Active when the validity period for the schema begins.
	Status: Deployed	The Deployed schema status is set automatically when the schema's validity period ends. The schema continues to exist so that completed processes that refer to this schema remain consistent.
	Sequencing	Using the sequencing function, you can influence the order in which categories are displayed in the WebClient UI, in the dropdown boxes for categorizing service transactions such as service requests. The UI components `BTCATEGORIES/Categories` and `GSMCDDLB/DropDownList` are supported. You can define the sequence values per category, by entering 4-digit values in the

Page Area	Function	Description
		corresponding column of the categorization schema. You can also activate the sequencing function and define sequences in schemas that are already active and deployed. To do this, you have to open the categorization schema in edit mode (without having to create a new version). If the sequencing function is activated, the following applies: • Categories with a sequence value are displayed at the beginning of the dropdown box and sorted sequentially, as defined. • Categories with identical sequence values are sorted by creation time stamp. • Categories without a sequence value are displayed at the end of the dropdown box and sorted alphabetically by description. If a description is not available, the categories are sorted by category ID. If the sequencing function is not activated, the categories are sorted by creation time stamp, regardless of any sequence values. If the user switches off the application-based sorting in personalization, the following applies: • Choosing Sort by Value sorts the categories alphabetically, by description. • Choosing Sort by Key sorts the categories by GUID. Note that sequencing does not influence the sorting of categories in categorization schemas in the category modeler or in other tree-based visualizations of categorization schemas, such as the input help for categories in the agent inbox search.
Category Hierarchy > Categorization Schema > Application Areas	Application Areas	Using the fields Application ID, Parameter, and Value, you assign the schema to applications, such as service orders.
	Value	For service processes and for the interaction record, choose the subject profile that should be valid for this schema. You can then assign the individual subject

Page Area	Function	Description
		codes for the profile in the schema, at category level.
Category > General Data	Category Name	Enter the category name that you want to be displayed in the category selection fields in sections of the application, such as in the service process.
	Category Description	You can use this field to enter notes. For example, you could use it to differentiate between two categories with similar names. The description is displayed only in the category modeler.
	Subject Code	If you have assigned a schema in the page area Application Areas to the parameter Subject Profile, you must assign a subject code for each category. This is the case in categorization schemas for service orders, for example. Otherwise, you cannot save the categorization in the relevant business transaction, for example in a service order.
Category > Extended Attributes	Validity	You can assign extended attributes, to integrate time-based versioning at category level.
	Local Valid-From Date/Local Valid-To Date	Category values can be defined locally, that is, at the level of the relevant category.
	Inherited Valid-From Date/Inherited Valid-To Date	The validity period of a category is passed down to its lower-level categories.
	Effective Valid-From Date/Effective Valid-To Date	The effective validity period of an individual category is always a combination of the inherited values and the locally defined values. If there is a conflict between two values, the more restrictive value is used.
Category > Various assignment blocks for the assignment of objects to categories	Various functions, such as Service Order Template	The assignment of objects to categories in the category modeler enables the automatic suggestion of objects. In a service order for example, if you select a category to which a service order template is assigned, you can choose to automatically complete the service order, using this template.

Authorizations for the Category Modeler

You can set up authorizations to control access to the various functions of the category modeler.

You create authorizations by defining values for authorization objects in authorization profiles. You then use roles to assign these to users or user groups.

You manage authorization objects in user maintenance or role maintenance. In the SAP Easy Access menu, choose:

Tools > Administration > User Maintenance > Users

and

Tools > Administration > User Maintenance > Role Administration → Roles

The authorization object for the category modeler is CRM_CATEGO.

For more information on authorizations, search for *ABAP Authorization Concept* and *Role Administration* in the product assistance on SAP Help Portal.

Authorization object CRM_CATEGO

You can select any values for the entries for the Application ID .

* > You have to set the full access

{...} > You can select any combination of values.

Application Case	Element	Schema Status	Object Link	Activity
Display application areas Note: The tab page Application Areas is not displayed if there is no authorization to display application areas according to the authorization mode.	Application area	*	*	Display
Change application areas	Application area	*	*	Display, change, add or create
Delete application areas	Application area	*	*	Display, delete
Reset content analysis	Application area	*	*	Release
Display schemas Note: The input help for the Application field in the extended search only displays those applications for which the user has display authorization for schemas. The user cannot search for application areas if he or she does not have	Schema	{Draft, * }	*	Display

Application Case	Element	Schema Status	Object Link	Activity
display authorization for their schemas.				
Change schemas	Schema	{Draft, * }	*	Display, change, add or create
Delete schemas	Schema	{Draft, * }	*	Display, delete
Release schemas	Schema	{Draft, released, * }	*	Release
Import schemas	Schema	*	*	Import
Merge schemas	Schema	*	*	Merge
Display categories Note: The **Categories** tab page is not displayed if the user does not have display authorization for the authorization mode of categories.	Category	{Draft, * }	*	Display
Change categories	Category	{Draft, * }	*	Display, change, add or create
Delete categories	Category	{Draft, * }	*	Display, delete
Display objects Note: The tab page for the relevant object type is not displayed if the user does not have display authorization for the authorization mode of the object.	Object link	{Draft, released, applied}	{A, B, C, ...}	Display
Change objects	Object link	{Draft, released, applied}	{A, B, C, ...}	Display, change, add or create
Delete objects	Object link	{Draft, released, applied}	{A, B, C, ...}	Display, delete

i Note

You cannot change schemas with statuses **Released** and **Applied** in order to prevent data inconsistencies. It is also not possible to change categories that are assigned to schema with these statuses.

Forms of Categorization

When you create a categorization schema in multilevel categorization, you can choose whether you want to arrange the categories hierarchically or by attributes.

Hierarchical categorization

You can choose the hierarchical categorization in the category modeler.

This is a strict hierarchical arrangement of categories in which each category completely describes a subject. Higher-level categories merely represent the context of each subject.

You cannot use any category duplicates in the hierarchical categorization.

Example of a hierarchical categorization:

- Groceries
    - Packaging
        - Stable
        - Unstable
- Taste
    - Delicious
    - Unpleasant

Attribute categorization

You can choose attribute categorization in the category modeler.

Attribute categorization is not strictly hierarchical and allows you to use category duplicates to assign value combinations.

When you use attribute categorization, each category is a characteristic of a subject, without which a subject would not be fully described. A subject is described using a combination of several categories. Such a combination corresponds to a path in the category hierarchy. Shared properties of different subjects are mapped in the form of category duplicates.

Example of an attribute categorization:

Groceries

Packaging

Good

Bad

Taste

Good

Bad

In this example, **Good** and **Bad** are category duplicates, or semantically identical categories. In general, the following subjects are expressed here as combinations of characteristics:

{Groceries, packaging, good}

{Groceries, packaging, bad}

{Groceries, taste, good}

{Groceries, taste, bad}

Auto Suggest

Use

As part of multilevel categorization, the assignment of objects to categories in the category modeler enables automatic object suggestions.

For some business transactions, for example service requests, you can control the automatic item determination using categorization, by assigning service products to categories. Using content analysis, you can display automatic category suggestions in the E-Mail Response Management System (ERMS).The system can also suggest related business transactions, if the same category is assigned to them.

Prerequisites

For information about the prerequisites, see Multilevel Categorization.

Features

- Automatic suggestion of knowledge articles

  You can assign knowledge articles to categories. For example, if you select a category in a service request, you can call the assigned knowledge articles as a list, and add any of them to the request.

  This function is available for the following standard business transactions in Service: service order, service request, and knowledge article.

- Auto complete

  You can assign templates to categories, for example service order templates.

  When you select the appropriate category in the service order, you can choose to transfer the data of the relevant service order template.

  This function is supported for service orders and service requests.

  E-mail standard responses

  This function is available in the e-mail editor of the interaction center (IC).

  You can assign e-mail standard responses to categories.

  When you select a category, the assigned standard responses are available in the e-mail editor, in the dropdown box for selecting standard responses.

- Auto suggest of category using products

  You can assign products to categories. For example, if a corresponding product is selected in the service order as a reference object, the system automatically completes the categorization fields.

- Content analysis

  You can define search queries for content analysis, and assign them to categories. Using search queries, incoming e-mails are analyzed in the ERMS, and default values for categories are determined.

For more information, see Content Analysis.

- Item determination

  In the background, a service product that is required for billing, pricing, and service level agreement, is assigned to service requests.

  If required, the item determination for the service product can be affected by multilevel categorization. To use this function, you need to make the appropriate settings in Customizing.

- Determining related business transactions

  This function is available for service requests. In the WebClient UI, you can use the **More** pushbutton to display related business transactions in which the same category is used. You must activate this function in Customizing.

Content Analysis

Use

This function identifies the categories that best match specific content and makes automatic category suggestions.

Content analysis is used in the E-Mail Response Management System (ERMS) to analyze the content of inbound e-mails. It uses predefined search queries that are assigned to categorization schemas, to determine suitable categories.

Integration

Function	Description
Category Modeler	You define content queries for content analysis in the category modeler. If necessary, you can reset the content analysis in the page area **Application Areas** for each application area. You do this by choosing **Reset Content Analysis**.
E-Mail Response Management System (ERMS) - automated processing	ERMS manages a high volume of inbound e-mails. This system partially automates routing and responding to e-mails. The automated processing of e-mails in the ERMS uses content analysis to evaluate conditions assigned to categories. The following ERMS components relate to content analysis: • Fact gathering • Rule operators **Matches Category** and **Rule Attributes**
Interaction Center	Content analysis provides suggested categories based on the content of e-mails received in the interaction center and selected for processing in the agent inbox. These categories are then displayed in the relevant fields of the following IC-specific business transactions: • Service order • Service request • Interaction record

Prerequisites

You have made content analysis settings in Customizing for **Service** by choosing **Cross-Application Components** ▸ **Multilevel Categorization** ▸ **Define Application Areas for Categorization** .

Features

- Automated categorization

  Category suggestions are made automatically using content queries.

- Language detection

  Content analysis also supports automatic language detection for textual content.

More Information

E-Mail Response Management System

Merge Categorization Schemas

Use

These functions enable you to combine the categories and object links of two schemas with one another. In this type of merge, the object links and category links of a lower-level schema are added to a leading schema. In the process, the leading schema fulfills the function of a template schema. The categories and object links that are identical in both schemas are transferred according to the leading schema – however, the additional categories and objects of the lower-level schema are also transferred. Where a decision is required – for example, when objects can be assigned to a category once only – the settings and objects of the leading schema are transferred when the merge takes place.

This function is available in the category modeler.

Prerequisites

Both schemas you want to merge must be in the same client of a system in Service. If you are working with multiple clients and systems, you must transport schemas beforehand (see Import Categorization Schemas).

Features

Schemas are merged according to the following rules (illustrated by the example in the figure below):

1. Identical and differing categories and object links of both schemas are merged in the leading schema. When categories and object links are identical, the properties (such as descriptions) of the leading schema are retained.

2. When the object link types are identical (here: link **z**) but the objects differ and only a single assignment is permitted for these objects, the objects of the leading schema are retained. The object of the lower-level schema is not taken into account for the merge.

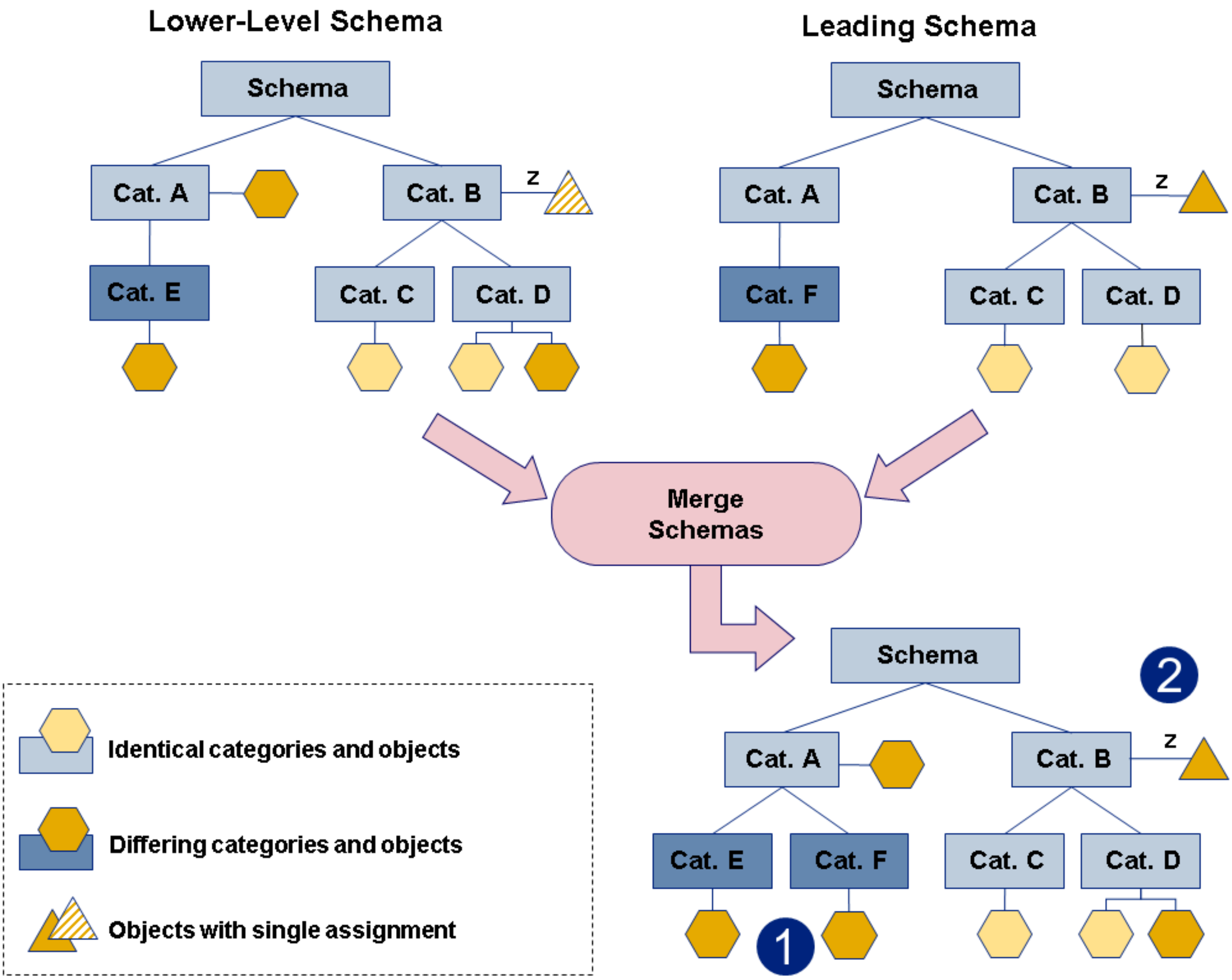


Figure: Merging of Categorization Schemas

The following table provides an overview of the available methods and functions:

Technical Object	Use
Method: IF_CRM_ERMS_CATEGO_ASPECT~MERGE	Merges schemas
Method: IF_CRM_ERMS_CATEGO_ASPECT~CLONE	If you wish the merge to take place in a third schema rather than in the leading schema, this method enables you to create a new schema before the merge.
Method:IF_CRM_ERMS_CATEGO_CATEGORY~MERGE	Merges categories
Business Add-In: CRM_ERMS_CATEGO_OBJECT_BADI	Merges object links. For an overview of the objects that can be merged as standard, see Import Categorization Schemas .
Interface: IF_CRM_CATEGO_OBJECT_BADI	Supports the import and merging of categorization schemas.

For more information about the various functions, see the system documentation for the objects listed above.

i Note

After you have merged two schemas, the new schema has the status **Draft**.

Errors that occur during the merge process are recorded in a merge log.

Import Categorization Schemas

Use

This function enables you to import categorization schemas from a source system into a target system. The import function is supported between two clients of a system in Service and also between two service solution systems.

The import of schemas only accommodates the categories, their hierarchy relationship, the object links, and the object keys. The linked objects themselves are not transferred into the target system by the import function. You must ensure the availability of these objects in the target system using other transport mechanisms, such as the SAP Transport Organizer.

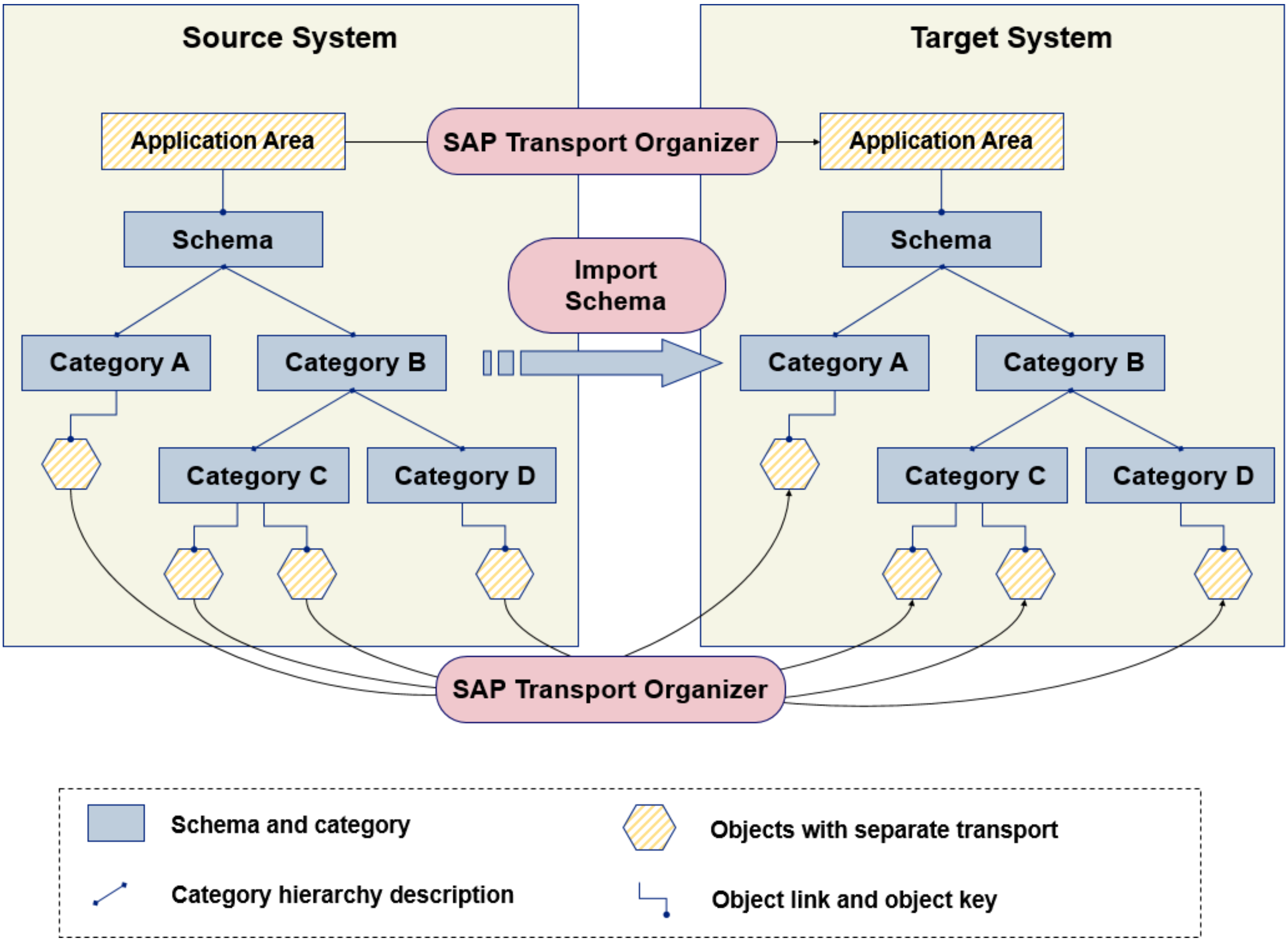


Figure: Importing a Schema

Integration

This function is available in the category modeler. Alternatively, you can also import schemas by running program `CRM_ERMS_CAT_AS_IMPORT` in transaction SE38.

Prerequisites

- You have defined the RFC connections required tor import schemas in Customizing by choosing **Service > Cross-Application Components > Multilevel Categorization > Maintain RFC Destinations for Schema Import**.
- Ensure that the linked objects of a schema exist in the target system before starting the import.

Features

- Before importing categorization schemas, the system checks whether the user has general authorization for import activities (authorization object CRM_CATEGO). In particular, a check is run in the source system to determine whether a user is authorized to export the respective elements.
- The import of schemas is recorded in an import log that you can display in the application log. For more information, search for *Application Log (BC-SRV-BAL)* in the relevant version of SAP NetWeaver on SAP Help Portal.
- Imported schemas are saved with the status **Draft** in the target system.
- The standard implementations of the Business Add-In (BAdI) `CRM_ERMS_CATEGO_OBJECT_BADI` support the import and merging of schemas. The methods `IMPORT_OBJECTS`, `IMPORT_LINK`, `EXPORT_LINK`, and `UPDATE` enable you to structure the category hierarchy and link the objects. BAdI implementations are provided for the following objects in the standard system:

Supported Standard Objects for Import / Merging of Schemas

Object	Import	Merging
Query	x	x
Standard Responses	-	x
Problems and Solutions	-	x
Products	x	x
Accounts	x	x
Enhanced Attributes	x	x
Document Template Profile	x	x
Activity Template	x	x
Subject Code	x	x

> **i Note**
> For technical reasons ‘Queries’ are not imported or merged using the BAdI. A custom enhancement is not supported for this object.

See also:

Merge Categorization Schemas

Inheritance of Reference Objects and Categorizations in Transactions

Use

In the following service transactions, when you create a new item, it automatically inherits the multilevel categorization and the reference objects from the header level.

- Service quotation
- Service order
- Service confirmation

If a reference object or categorization is deleted at header level, it is also deleted at item level. If a reference object or categorization is changed at item level, the inheritance history is lost.

Conversely, categorizations and reference objects that you create for an item do not affect the header level.

Prerequisites

Inheritance of categorization is only possible if the same categorization settings were made at item level (or in the follow-up transaction) and at header level (or in the preceding transaction) as those in standard Customizing.

Features

Inheritance at Creation of Follow-Up Transactions

If you create a follow-up transaction for one of the transactions listed above with categorization and reference objects, the reference objects and the categorization are copied into the follow-up transaction. The inheritance information is available in these follow-up transactions. This means that changes to the reference object or the categorization also follow the rules of behavior outlined above.

Conversely, categorizations and reference objects that you create for an item do not affect the header level.

If a reference object or categorization is deleted at header level, it is also deleted at item level. If a reference object or categorization is changed at item level, the inheritance history is lost.

Rule Modeler

Use

You use the rule modeler to create, maintain, and release rule policies in the WebClient in Service.

Integration

The rule modeler is available for the following areas, known as “contexts”, accessible from the **Process Modeling** area of the WebClient UI in service solution:

- Account and contact management
- Approval management
- Checklist determination
- Checklist step partner determination

- E-mail response management system
- Intent-driven interaction
- Service order/complaint dispatch
- Service request management
- Service transactions

When you select one of the above contexts, you define the area in which you want to create or maintain rules. Rules and rule policies defined in each context are completely independent of one another. The variables that you use in the rule conditions are based on the context.

SAP delivers rules and policies that can be extended with custom coding as needed. Using these objects you create IF — THEN rules.

Prerequisites

You have made the settings in Customizing for **Service** under one of the following paths:

- **E-Mail Response Management System**
- **Basic Functions** > **Rule Modeler**

For example, with activity **Define Repository** you can optionally extend the default attributes, conditions, and actions.

More Information

ERMS Rule Policies

Rule Policies

Rules

Conditions

Actions and Parameters

Rule Policy Authorizations

Conflict Resolution

Creating Rule Policies

Procedure

Creating rule policies can be complex. The procedure outlined below provides the most basic steps for creating a rule policy. Additional information is provided in the related topics for rule modeler.

1. Choose **Rule Policy** under **Create** in the work center page, or choose **Create New Policy** in the search page for rule policies.
2. Select the required context and enter a policy name.

   The system creates a default policy variant.

3. Select the rule policy top node to set authorizations.

4. Select **Draft Rules** and create a rule folder by adding a subnode.

5. Select a rule folder and create a rule by adding a subnode.

   Where permitted, you can add a new rule folder, rule, or subrule as a subnode, or at the same level as the current position in the hierarchy. You do this by selecting either **Insert as Subnode** or **Insert at Same Level**.

6. Create rules and subrules as required.

7. After you have created your rules and checked them to ensure correct logic and syntax, you can release and save the rules.

More Information

Rule Policies

Rule Policy Authorizations

Rule Policies

Definition

A collection of logically grouped rules, for example, by region or country.

Use

You create rule policies to invoke specific actions automatically as soon as predefined conditions are met.

Policy Variants

Rule policies contain rules, which are organized in folders under one or more policy variants. A policy variant is a version of a rule policy. The rules in a policy variant can be evaluated separately and at different times.

Example

You need different policy variants to create time-dependent versions of a rule policy. One collection of rules could apply from November to December and contain a greeting referring to the holiday season. A second collection of rules could be triggered after the holiday period in December, containing many of the same actions as the first collection, but with a different greeting.

A rule policy can have many policy variants, but only one can be active at a time. A policy variant is active when the current date and time coincides with the policy start and end dates and times. When a rule policy is invoked, the system selects the variant that is currently active.

When you create a rule policy, the system creates a default policy variant. As long as there is only one variant for a rule policy, you cannot change the date and time. When there is more than one variant for a rule policy, you can change the start date and time, but not the end date and time. The end date is calculated by the system to ensure uninterrupted rule evaluation.

While policy variants can contain both draft and released rules, only released rules are used in rule evaluation.

To prevent your rule policy from becoming cumbersome, you can split rules into several policies and use the **Invoke Policy** function. This makes it possible to call one rule policy from another.

Rule Policy Relationships

You can view a graphical representation of relationships between rule policies that call other rule policies. The flowchart-style display provides the following:

- Zoom slider for the canvas
- Option to toggle the grid on or off in the canvas
- Option to toggle the **Details** window on or off in the canvas

This window provides information on the rule policy that is currently selected.

- Option to toggle the **Navigation** window on or off in the canvas

This window provides a visual overview of the policy relationships, which can be useful for more complicated scenarios.

- Canvas area to display the rule policies

Enhancing Usability

You can make the most frequently used functions for rule modeler maintenance available as a button or a one-click action. For more information, see Customizing of **Service** under **Basic Functions** > **Rule Modeler** > **Define Profiles for Toolbar and One-Click Actions.**.

More Information

ERMS Rule Policies

Rules

Rules

Definition

A combination of conditions and actions that form a rule policy.

Use

You create and maintain rules in rule folders under existing rule policies in the WebClient UI in Service under **Process Modeling**. The folders are used to group related rules.

Rule Search

You can search the system for rules and rule folders. Subrules are included in rule and rule folder searches, and the system searches all rule policies and policy variants, either active or inactive.

Rule Types

Rules exist as one of the following types:

- Released

  These are rules that you have created and released for use in rule processing.

- Draft

These are rules that you have created and not yet released, or that you have previously released but are currently modifying.

Once rules have been saved and released, you cannot directly change them. This ensures that you do not edit a rule that is currently in use. If you need to change a released rule, choose **Edit Released Rules as Draft** under **More**. The system creates a copy of the released rule and inserts it in **Draft Rules**, which ensures that the released rule is still available for use while you are changing the draft version.

Once you have finished creating or changing your draft rules, you access the function to release the draft rules under **More**.

→ Recommendation

After creating new rules or changing existing rules, choose **Check Draft Rules** under **More**. This ensures that the logic and syntax are correct.

Rule Preview

You can preview the combination of conditions and actions in an “if/then” format to see how rules are applied. In the following example, if an e-mail is received containing the words **Trade Show** or **User Group**, the system sends an invitation automatically, creates an interaction record, and routes the e-mail to the special events queue.

Example

```
If

E-Mail Subject Contains Trade Show or

E-Mail Subject Contains User Group

Then

Send Auto Acknowledgement (Mail Form = Invitation for a Trade Show; Outgoing E-Mail Address

Route E-Mail (Organizational Object = Special Events Queue)
```

Rule Evaluation

Rules are organized in a tree structure in which every rule on a given level is only executed if the upper level rule has evaluated to true.

Every time the system triggers a policy, the rules belonging to the active policy variant are evaluated. Rules are based upon facts, such as the content of an e-mail and business partner information. The system gathers these facts and stores them in a fact base.

More Information

Conditions

Actions and Parameters

Conditions

Use

A given set of circumstances that, if fulfilled, trigger an action. Conditions are one of the defining components of rules when creating rule policies. A rule can have any number of conditions.

Prerequisites

Conditions are defined in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

Features

Conditions consist of the following:

- Attributes

  These can be system variables or facts, such as e-mail content, subject, or business partner attributes such as language, country, and so on. The available attributes depend on the context you are working in. If you do not need certain attributes, you can hide them by deselecting the **Show Attribute** indicator in the **Attributes** step of the **Define Repository** Customizing activity.

- Operators

  Which operators are available depend upon the context you are working in and which attribute you selected. Examples of operators include:

  - **Contains**
  - **Equals**
  - **Is Greater Than**
  - **Greater or Equal**
  - **Less or Equal**
  - **Is Like**: this operator evaluates wildcard expressions.
  - **Is Not Like**

- Values

  The type of values you enter or choose depends upon the combination of the attribute and operator you selected. In some cases, you can only choose entries from a dropdown list box, use an advanced search, or enter values directly in the field. These values are determined in the **Input Support for Conditions** folder of the **Define Repository** Customizing activity.

- Case-sensitivity

  You can define whether the evaluation of a value field should be case-sensitive or not. The Customizing for the attribute defines whether a user can change this field when creating rules, or if this option is a display-only field.

Attributes, operators, and the type of value field you use are all defined in Customizing. You can have more than one condition for a rule. In this case, you must determine whether all conditions should be met to trigger the action or just one. Use **Match All/Match Any** to toggle between the options.

More Information

Rule Policies

Actions and Parameters

Use

Actions are operations that must be triggered if the conditions set in a rule are evaluated to true. Actions are one of the defining components used when creating rule policies. A rule can have any number of actions.

i Note

The available actions for a rule depend on the context in which the rule policy is created. For example, the **Send Auto Acknowledgement** action is available for the **E-Mail Response Management System** context.

The actions that you choose can have corresponding parameters associated to them, which define how the rule policies are created in the WebClient UI in Servicef.

When multiple actions are created for a rule, the actions are performed in their order in the list. You can move actions up and down the list as required.

You define actions and action parameters in Customizing for **Service** under **E-Mail Response Management System** › **Define Repository**.

Features

When you create an action, you must determine the following:

- Action ID

  There are standard delivered actions you can use such as invoking a policy, routing e-mails, sending automatic responses, and creating interaction records.

- Action service ID

  This determines the behavior of the action when executed. You link action service IDs to action IDs in Customizing for **Service** under **E-Mail Response Management System** › **Service Manager** › **Define Services**.

- Action parameters

  These are used to pass values that are to be used in the action, for example, **Mail Form** and **Organizational Object**. The available parameters depend upon the context you are working in, for example **ERMS**, and which action ID you selected.

- Conflict resolution type

  This determines how conflicts are resolved if the same action is invoked more than once during rule execution. Conflict resolution is used primarily for ERMS but is also available for other contexts.

- Time of execution

  You can determine if an action is executed immediately or only after a policy has been evaluated.

i Note

You can use multiple action IDs and action parameters for a single rule.

More Information

Conflict Resolution

Rule Policy Authorizations

Use

You can define authorizations for rule policies to limit who can access the policies in the WebClient UI in Service and to determine whether users can display or maintain them. This is especially helpful in large organizations in order to limit the amount of data that is shown in any rule policy.

Integration

The standard SAP Authorization Concept is used. For more information, search for *ABAP Authorization Concept* in the relevant version of SAP NetWeaver on SAP Help Portal.

Prerequisites

- You have created **Authorization Groups** in Customizing for **Service** under **E-Mail Response Management System** **Define Repository**.
- You have assigned the authorizations groups to authorization object `CRM_ERMS_P` in **Role Maintenance** (transaction `PFCG`).
- You have assigned the authorization groups to policies in the WebClient UI in Service.

  You can see (and in some cases select) authorization groups in the WebClient UI in Service under **Rule Policy Details**, after you select a policy (the top node of the policy structure). The possible values are the entries you maintained in the above Customizing activity.

Conflict Resolution

Use

Conflict resolution determines the outcome if the same action is invoked more than once during rule execution. In some cases, you can have two actions with different content, for example, the **Route To** action is called more than once or two acknowledgments are triggered for an e-mail. The settings made for conflict resolution determine if both actions are triggered or which one has priority. However, in most cases, we do not recommend having two actions with different content.

You can define a priority for an action by using an action parameter. You set the priority when you create rules in the WebClient UI in Service. To display this priority in the user interface, you need to enable the action parameter in the Customizing activity **Define Repository** for the **Conflict Type** field. You must select **Occurrence with Highest Priority is Used** for an action in the ERMS context. Then a new parameter called **Conflict Resolution Priority** appears for the selected action with the default value **500**. If different values are assigned to various instances of the **same** action, then the action with the highest numerical value is used.

Example

You have one rule in your rule policy that states:

If E-mail Contains Christmas, Then Route to Christmas Queue.

The **Route To** action has the Customizing setting **Occurrence with Highest Priority is Used**, and the **Route To** action in the WebClient UI in service solution displays the parameter **Conflict Resolution Priority** with the default value **500**. The company's rule administrator decides to leave the value of this rule at 500.

You also have another rule in the **same** policy that states:

If E-mail Sender Is customer.1@sap.com, Then Route to VIP Customer Care Center Queue.

If an e-mail from customer.1@sap.com arrives containing the keyword “Christmas” in the text, the company's rule administrator raises the priority value from **500** to **1000** in the route action. By increasing the value to 1000, you avoid the possible conflict of whether to route to the Christmas queue because the e-mail contains the keyword “Christmas” or whether to route to the VIP queue because it comes from customer.1@sap.com. The system routes the e-mail to the VIP queue because the value 1000 is higher than 500.

Prerequisites

You have set the conflict resolution indicator for each action in Customizing for **Service** under **E-Mail Response Management System** > **Define Repository**.

Features

The following possibilities are available to determine conflict resolution for actions:

- Any number of occurrences of the action is allowed.
- The first occurrence of the action is used.
- The last occurrence of the action is used.
- The occurrence with the highest priority is used.
- Instead of the above options, a custom ABAP class is used to determine the outcome.

More Information

ERMS Rule Policies

Actions and Parameters

Rule-Based Dispatching of Service Transactions

Use

Rule-based dispatching uses rule policies to forward service transactions to a specific employee or organizational unit for processing. For the technical implementation, the system uses rule policies in the E-Mail Response Management System (ERMS). Rule-based dispatching is available on the UI for the following transaction types:

- Service order
- Service quotation
- Service confirmation

Prerequisites

You have made the required settings for this function as described in Customizing for **Service** under **Transactions** > **Settings for Service Transactions** > **Settings for Rule-Based Dispatching**.

Activities

To start dispatching, choose **More** > **Dispatch**. In the **Parties Involved** assignment block, a new row appears with partner function **Employee Responsible** or **Responsible Group**.

External References for Service Transactions

You can use references to synchronize your service transactions with external systems.

Use

You use external references as identifiers for your service transactions in external systems. For example, you can use a quotation ID in the external system as the external reference for a solution quotation that you synchronize with your SAP S/4HANA or SAP S/4HANA Cloud Private Edition system through an API service.

Features

You provide external references when you create service transactions (and service transaction items) through API services. You can check the assignment block **Additional External References** in the respective user interface (UI) for each service transaction type.

You can use external references in the following service transactions:

- Solution quotation
- Solution quotation item
- Business solution portfolio

Plant and Storage Location

Learn about plant and storage location in service transactions.

Use

A plant is an organizational unit for dividing an enterprise according to production, procurement, maintenance, and materials planning. A storage location is an organizational unit that represents a place used for storing items in a warehouse or other repository.

Plant and storage location are required fields for many integrated functions of various service functions such as:

- Logistics Integration
- Available-to-Promise (ATP)
- Billing
- Pricing

- Finance and Controlling

You can maintain the plant and storage location manually in the items, or the system can determine them automatically. You can also change them after determination.

Determination of Plant and Storage Location

The following item types are supported for the determination of plant and storage location:

- Service item
- Intercompany service item
- Service part
- Expense item
- Tool item
- External service item
- Price item
- Service contract item
- Sales item

Note

Determination of plant and storage location is not supported for execution order items. For more information, see Execution Order Items.

Determination for All Item Types Except Sales Item

The determination for these items depends on the organizational model being used.

- Organizational Management
    1. Service organization, service team, service employee
    2. Service organization, service team
    3. Service organization
- Enterprise Organizational Management

  The determination sequence can be found in Determination Sequence for Plant and Storage Location.

Note

Storage location is only relevant for service part and tool items. For service parts, storage location is used in logistic integration. For example, to create purchase requisitions or purchase orders and for ATP. For tool items there is currently no integration which requires storage location and is therefore used to provide information. The storage location can be changed manually or through the use of a BAdI.

Customizing

You can find more information on the Customizing tables for the above organization models using the following paths:

- Organizational Management (Legacy): Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy)
- Enterprise Organizational Management: Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management

If the determination logic does not meet your requirements, you can implement the respective Business Add-In (BAdI) for Organizational Management or Enterprise Organizational Management.

- Organizational Management (Legacy): Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Organizational Management (Legacy) > Business Add-Ins for Logistics Integration > Business Add-In: Determination of Plant and Storage Location
- Enterprise Organizational Management: Service > Transactions > Settings for Service Transactions > Integration > Logistics Integration > Enterprise Organizational Management > Business Add-Ins for Logistics Integration > Business Add-In: Determination of Plant and Storage Location

Determination for Sales Item

For sales items, the plant is determined from customer material information record, ship-to party, or material master. The determination sequence is as follows:

1. Plant of customer material information record
2. Delivering plant of ship-to party
3. Delivering plant of material

Note

For sales item, the storage location is never automatically determined, but it can be maintained manually if required.

Redetermination of Plant and Storage Location

Changing the product or item category always triggers a redetermination of the plant and storage location.

Redetermination for All Item Types Except Sales Item

If the plant or storage location was not changed manually, changing one of the following fields will trigger a redetermination:

- Sales organization
- Distribution channel
- Division
- Service organization
- Service employee group or service team
- Executing service employee or service employee

Redetermination for Sales Item

Similarly, in a sales item if the plant or storage location is not changed manually, changing one of the following fields will trigger a redetermination:

- Sales organization
- Distribution channel
- Division
- Sold-to party
- Ship-to party

Manual Redetermination

You can also manually trigger a redetermination of plant and storage location by choosing the Redetermine Plant/Location button in the Items assignment block for transactions such as service quotation, service order, service contract, and solution quotation.

Note

Choosing the Redetermine Plant/Location button will also overwrite manual changes.

Constraints

In the following cases the plant and storage location can't be changed:

- If the item status is Completed or Transferred to Billing.
- If a logistical movement such as a goods receipt, goods issue, or delivery was executed or if a reservation was created.
- If for a service confirmation item, the referenced service order item created a reservation.

Copy to Follow-Up Transactions

The plant and storage location are automatically copied to the follow-up transaction. You can influence the copy process in the Define Copying Control for Item Categories Customizing activity using the Copy Plant/Stor Loc field. You can use the following path: Service > Transactions > Basic Setting > Copying Control for Business Transactions > Define Copying Control for Item Categories

You have the following options in the Copy Plant/Stor Loc field:

- Copy from Preceding Transaction

  The system copies the plant and storage location from the preceding transaction.

- Copy Manual Value and Redetermine Automatic Value

  The system copies a manually changed plant or storage location and the others will be redetermined.

- Redetermine All Values

  The system redetermines the plant and storage location when the transaction is copied or created as a follow-up.

Note

During the copy process, if the system determines that any of the determination relevant attributes were automatically changed then it will redetermine the plant and storage location, even if the copy control setting is set to Copy from Preceding Transaction. For example, consider a service order template without a service team. When a follow-up service order is created, the plant and storage location is copied over. But when a service team is now determined in the service order during the copy process, the redetermination of plant and storage location will be triggered for the item. This is because the service team is a relevant attribute for the plant and storage location determination.

i Note

A follow-up from a service contract item to any other items will always lead to a new determination, independent of the copy control setting.

Data Migration

You can use the plant and storage location fields once you upgrade to SAP S/4HANA Cloud Private Edition 2025 FPS0 or SAP S/4HANA 2025 FPS0. The data conversion method, Silent Data Migration (SDMI), will update the plant and storage location fields for all items except execution order items and sales items based on the current Customizing or BAdI implementation for plant and storage location determination. The update is independent of the status of the item, so for example completed items will also be updated. Until the SDMI is executed, the current logic for dynamically determining plant and storage location will continue to be used as a fallback.